Flat Galactic Rotation Curves from the Dimensional Reduction of Curvature Transport

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ABSTRACT

General Relativity treats gravity as a static curvature constraint: mass determines geometry instantaneously, and propagation effects enter only as perturbations. I argue that this equilibrium assumption breaks down at galactic scales, and that treating curvature propagation as a fully dynamical, relativistic process — governed by finite causal speed and organized by differential rotation — accounts for flat galactic rotation curves from baryonic matter alone, without dark matter or modified gravity.

Under finite-speed causal propagation and differential rotation, orbital phase mixing is argued to confine curvature redistribution to constant-radius two-dimensional shells. The $1/r$ acceleration that follows is derived from the logarithmic Green function of the Laplace–Beltrami operator restricted to these shells — derived once the shell-confinement regime is granted. That regime is supported by two independent arguments: the spectral structure on S^3 and the retarded gravitational field in linearized General Relativity.

The effective acceleration combines a local $1/r^2$ baryonic channel with the nonlocal $1/r$ shell channel through a crossfade governed by the ratio a_b/a_*, where $a_* = c^2/(\pi R)$ is a cosmological curvature scale fixed by the Hubble radius:

$$a_{\rm eff}(r) = a_* \cdot \xi \cdot \frac{\xi^3 + \alpha_0}{1+\xi^2}, \qquad \xi = \sqrt{a_b(r)/a_*}\,.$$

The formula recovers Newtonian gravity for $\xi \gg 1$ and yields $a_{\rm eff} \to \alpha_0\sqrt{a_b\, a_*}$ for $\xi \ll 1$, producing the baryonic Tully–Fisher relation $V_\infty^4 \propto M_b$. No fitted scale is introduced beyond the morphology-dependent projection efficiency α_0.

Applied to the 175-galaxy SPARC archive with zero free parameters ($\alpha_0 = \kappa \equiv 1 + \pi^{-2}$), the formula achieves 78 simultaneous lowest-χ^2 fits against both MOND and ΛCDM — the highest count among the three frameworks at zero parameters. In a fair one-parameter comparison (WT fitting α_0, MOND fitting $\Upsilon_\star$, ΛCDM fitting M_{200}), all three frameworks achieve comparable median $\chi^2 \approx 3$, with WT showing the lowest median. An exploratory Abel-projection lensing comparison correctly orders early- and late-type galaxies without additional parameters, consistent with the 6σ Sérsic split of Brouwer et al. (2021). Open problems — covariant embedding and first-principles derivation of the morphology factor — are identified.

Key words: galaxies: kinematics and dynamics – galaxies: spiral – galaxies: dwarf – galaxies: fundamental parameters – gravitation – gravitational lensing: weak – dark matter – cosmology: theory – methods: analytical

1 INTRODUCTION

1.1 The problem

Flat galactic rotation curves are among the most precisely measured departures from Newtonian gravity, and their origin remains contested. In the inner regions of galaxies the observed centripetal acceleration follows the inverse-square law

$$a_b(r) = \frac{GM_b(r)}{r^2}, \tag{1}$$

sourced entirely by the observed baryonic mass $M_b(r)$. At larger radii the curves become flat — centripetal acceleration scales as $1/r$ — while the baryonic surface density continues to fall. No additional mass is observed. The discrepancy spans orders of magnitude in galaxy luminosity and surface brightness, and is measured with sufficient precision to distinguish between competing functional forms.

Within a purely local, three-dimensional description of gravity, this transition has no natural origin. The inverse-square law follows directly from flux conservation in three dimensions, and no change in scaling can occur without either modifying the source or introducing additional degrees of freedom.

Two frameworks have addressed this departure most successfully. Modified Newtonian Dynamics (MOND) Milgrom (1983) captures the transition with a single acceleration constant a_0, predicting rotation curves across a wide range of galaxy types with remarkable economy — an empirical achievement that has proved durable for over four decades. The ΛCDM framework accounts for the same observations through a dark matter halo whose density profile adds a velocity contribution on top of the baryonic one; it remains the most successful cosmological framework available, with independent support from the CMB power spectrum, baryon acoustic oscillations, and large-scale structure. Both accomplishments are real, and neither is in dispute here. This work uses the SPARC dataset and MOND benchmarks produced by that tradition. The identification $a_* = c^2/(\pi R)$ is proposed as the geometric origin of the MOND acceleration scale; the missing mass is proposed to be geometric rather than material.

A related geometric program, Hyperconical Modified Gravity (HMG) Monjo (2023); Monjo & Banik (2025); Monjo (2025), attributes the flattening of rotation curves to a fictitious acceleration inherited from cosmic expansion under a hyperconical background metric — a 4-hypersurface embedded in 5D Minkowski spacetime. Like the present work, HMG recovers MOND-like $\sqrt{a_b\, a_*}$ scaling in the outer regime and fits SPARC rotation curves without exotic matter. The structural similarity is noted: both frameworks take the form $a_{\text{eff}}(r) = a_b(r) \cdot F(r, R)$, where F is a geometric modulation depending on the Hubble radius R; the frameworks differ in how F depends on a_b.

The central question is therefore not how to adjust the law, but *what physical process could reorganize curvature such that its effective propagation changes dimensionality?* The following sections pursue that question geometrically.

1.2 From static fields to dynamic geometry

General Relativity is built on a local foundation of Special Relativity: no signal propagates faster than c, and clocks run slower in deeper gravitational wells. Yet the standard application of GR to galactic dynamics discards both consequences at the outset. The gravitational field of a galaxy is modelled as a static potential sourced by the instantaneous mass distribution — an equilibrium solution in which all propagation transients have already settled.

This is the Newtonian inheritance inside GR. It is justified for isolated, compact, slowly-evolving systems, where the light-crossing time is short compared to the dynamical time. It is not obviously justified for an extended, differentially rotating disk whose outer radius subtends a significant fraction of the causal horizon.

The analysis in this paper takes the propagation seriously as a dynamical process. Curvature is generated by local sources, propagates outward at finite speed, and is geometrically organized by the orbital structure of the galaxy. In the inner regime, where propagation has fully equilibrated, standard Newtonian sourcing is recovered. At large radii — where the propagation front has not equilibrated and orbital phase mixing has had time to act — the transport reorganizes onto effectively two-dimensional shells, changing the scaling of the resulting acceleration from $1/r^2$ to $1/r$.

The static GR field is therefore the equilibrium limit of this propagation-primary picture. Flat rotation curves are evidence that galaxies operate in the non-equilibrium regime.

1.3 Dimensional consequence of curvature transport

The scaling behavior of a field generated by a localized source is determined by the dimensionality of the space in which it propagates.

For a Laplacian in d spatial dimensions, the Green function satisfies

$$\Delta G_d(r) = -\delta(r), \tag{2}$$

with solutions of the form

$$G_d(r) \sim \begin{cases} r^{2-d}, & d \neq 2, \\ \log r, & d = 2. \end{cases} \tag{3}$$

The corresponding force scales as

$$|\nabla G_d(r)| \sim \begin{cases} r^{1-d}, & d \neq 2, \\ 1/r, & d = 2. \end{cases} \tag{4}$$

Thus:

- three-dimensional propagation produces the inverse-square law $1/r^2$,
- two-dimensional propagation produces an inverse-linear law $1/r$.

This observation has a direct implication: *any mechanism that effectively reduces curvature transport from three to two dimensions will necessarily produce a $1/r$ acceleration.*

This statement assumes that the transport on the reduced manifold is governed by a Laplacian-type operator. In the present framework, that this condition is satisfied for shell transport is established explicitly in Section 4, where the bulk dynamics induces a two-dimensional Poisson equation on the shell.

The problem of galactic dynamics is therefore reframed as a geometric and dynamical question: *under what conditions does curvature transport become effectively two-dimensional?*

1.4 Two channels of curvature dynamics

I propose that gravitational support in galaxies arises from two complementary curvature channels:

- a local volumetric channel, corresponding to standard three-dimensional flux from baryonic matter,
- a transport channel, in which curvature is redistributed along lower-dimensional manifolds.

The first channel is governed by local field equations and reproduces the inverse-square law. The second does not introduce new sources; instead, it reorganizes existing curvature through transport.

The mechanism described in this work is formulated within General Relativity in the weak-field limit.

The second channel is argued to emerge from the dynamics of curvature propagation itself, as developed in Sections 3 and 4.5.

1.5 The central claim

The central claim is that under causal propagation and differential rotation, orbital phase mixing confines coherent curvature transport to constant-radius two-dimensional shells — argued physically in Section 3 and supported independently in linearized GR in Section 3.4.

The derivational chain that follows from this confinement is:

(i) curvature is continuously generated in the bulk by baryonic sources,

(ii) causal propagation and differential rotation organize transport into shell-like manifolds (*argued* in Section 3; the ray-winding calculation is derived, the co-moving assumption is physically motivated),

(iii) transport on these manifolds is governed by a two-dimensional Laplacian (*derived* in Section 4.5 and Appendix A, conditional on the shell description of step 2),

(iv) the resulting Green function is logarithmic, producing a $1/r$ acceleration (*derived* in Appendix B).

Once step 2 is granted, steps 3–4 follow by standard mathematics: the inverse-linear scaling is the direct consequence of the Green function on the transport manifold, not an imposed modification.

1.6 Epistemic status of the principal claims

Four epistemic categories are used throughout (Table 1):

- **Derived** — follows necessarily from stated assumptions and prior steps, with no additional input.
- **Argued** — established by a physical or mathematical argument that is well-motivated but not fully rigorous; the conclusion is strongly suggested but not compelled.
- **Asserted** — taken as a postulate or definition of the framework; valid as a starting point but not derived within it.
- **Empirical** — requires observational input to fix; the geometric framework constrains the interpretation but not the value.

Table 1 summarises the status of every principal claim.

1.7 Scope and foundational independence

This paper addresses a single question: *what geometric mechanism, operating within standard weak-field gravity and baryonic matter alone, produces flat galactic rotation curves?* The derivational chain developed in Section 5 is self-contained; the only inputs required are finite-speed causal propagation, differential rotation of the source, and standard weak-field gravitational sourcing.

Three formal incompleteness issues lie outside this scope: (i) the dynamical equations are assembled from geometric and physical arguments rather than derived from a single covariant action; (ii) the transport parameters τ_c, D, κ are constrained by limiting behavior (Section 4.5) but not varied from a Lagrangian; (iii) the shell is treated as an effective manifold, not a variational degree of freedom. All three are documented in Table 1 and identified as open problems in Section 8. None affects the weak-field predictions derived here: each equation in the chain is independently motivated, and the chain itself is the claim under scrutiny.

The appropriate precedent is phenomenological: Milgrom's MOND Milgrom (1983) identified the correct kinematic relation without a covariant action; AQUAL Bekenstein & Milgrom (1984) and TeVeS Bekenstein (2004) followed later. The present paper makes an analogous claim, correctly scoped to a specific transport mechanism and its observational signature.

The physical conclusions do not depend on any framework beyond standard weak-field General Relativity and can be evaluated independently.

2 GEOMETRIC FRAMEWORK AND BULK–SHELL INTERFACE

To make the propagation-primary picture precise, the spatial geometry must be specified. The framework models the spatial slice as a three-sphere S^3 of radius $R \sim c/H_0$, which provides both the natural global setting and the manifold on which curvature transport is organized. The construction below establishes the bulk source, the conserved flux that reaches the boundary, and the geometric role of the shell — before any dynamical assumption about how that flux is redistributed.

2.1 Geometry of the system

The spatial geometry is modelled as a three-dimensional sphere S^3 of radius R. Points on S^3 are parameterized by the dimensionless radial coordinate

$$\chi = \frac{r}{R}, \qquad 0 \le \chi \le \pi, \tag{5}$$

where r denotes the geodesic distance from a chosen origin.

Surfaces of constant χ define two-dimensional spheres S_r embedded in S^3. These surfaces will serve as the manifolds on which curvature transport is organized.

For a given χ, the enclosed region defines a spherical cap $V_{\text{cap}}(\chi)$ with volume

$$V_{\text{cap}}(\chi) = 2\pi R^3 \left(\chi - \frac{1}{2}\sin 2\chi\right), \tag{6}$$

and boundary area

$$A_\partial(\chi) = 4\pi R^2 \sin^2\chi. \tag{7}$$

In the limit $\chi \ll 1$, these expressions reduce to the familiar Euclidean forms

$$V_{\text{cap}} \approx \frac{4}{3}\pi r^3, \qquad A_\partial \approx 4\pi r^2, \tag{8}$$

ensuring consistency with local geometry.

Table 1. Epistemic status of the principal claims in this paper. *Derived*: follows necessarily from stated assumptions. *Argued*: well-motivated but not fully rigorous. *Asserted*: taken as a postulate or framework definition. *Empirical*: value requires observational input.

Claim	Status	Where established
3D flux conservation $\Rightarrow r^{-2}$ local acceleration	Derived	Sec. 4.1, App. A1
$Q_b^{\rm src} = 4\pi G M_b$ via Gauss law	Derived	Sec. 4.2
Ray-winding calculation: $k_\theta/k_r \to \infty$, radial group velocity $\to 0$	Derived	Sec. 3.3, Sec. 3.4, App. E
Shell Poisson equation from bulk transport (conditional on shell confinement)	Derived	Sec. 4.5, App. A
Curvature transport is effectively 2D at large r (conditional on shell confinement)	Derived	Sec. 4.5, App. A
α_0 is the unique undetermined factor in $a_{\rm shell}$	Derived	Sec. 4.5
2D Green function $\Rightarrow$ logarithmic potential	Derived	Sec. 4.5.7, App. B
Logarithmic potential $\Rightarrow r^{-1}$ shell acceleration	Derived	Sec. 4.5.7, App. B
$v_\infty^4 \propto M_b$ (BTFR scaling)	Derived	App. B4
Quasi-static limit valid ($\varepsilon_1, \varepsilon_2 \ll 1$, screening $\ll 1$; given $\tau_c = T_p$, $\kappa/D \sim 1/R^2$)	Derived	Sec. 4.5.3
Effective MOND scale $a_0^{\rm eff} = \alpha_0^2\, a_*$	Derived	Sec. 5.4
Crossfade weight $\sigma(r) = [1 + (a_b/a_*)]^{-1}$, $m = 1$ uniquely selected by WKB energy partition on S^3	Derived	Sec. 5.2, 5.3.1, 5.3.2
Coherent curvature transport confined to constant-r shells at galactic scales	Argued	Sec. 3.3, Sec. 3.4, App. E; depends on co-moving assumption and extrapolation from ray-winding regime
Screening scale $\kappa/D \sim 1/R^2$ (cosmological)	Argued	Sec. 4.5.3
Orbital averaging produces a smooth shell source	Argued	Sec. 3
Curvature-transport medium co-moves with baryonic flow (D/Dt substitution)	Argued	Sec. 3.3
Shell occupancy $\eta_s \sim \mathcal{O}(1)$ in transition regime	Argued	App. B4
Crossfade formula $(1-\sigma)a_b + \sigma\, \alpha_0 \sqrt{a_b a_*}$	Argued	Sec. 5.1
Fractional Laplacian exponent $\sigma = 1$ as the unique value consistent with an asymptotically flat rotation-curve tail	Argued	Sec. 4.5 (status of reduction)
$\Phi = \Psi$ maintained by curvature-reorganisation mechanism	Argued	Sec. 6.6
Crossfade normalization $k_r^2 = k_\theta^2$ at $a_b = a_*$	Asserted	Sec. 5.3.1
Causal update speed $\nu = \gamma c$ with $\gamma \le 1$	Asserted	Sec. B4.0.2
$G = a_* = c^2/(\pi R)$ (WT definition)	Asserted	App. C
Causal memory time $\tau_c = T_p$ (Planck time)	Asserted	Sec. 4.5.3
Morphology factor α_0; $A_{\rm morph}$ distribution is unimodal, centred near κ	Empirical	Sec. 6.3

2.2 Bulk curvature source

The baryonic contribution to curvature is described through a flux field F_b satisfying

$$\nabla \cdot F_b = s_b, \tag{9}$$

where s_b represents the volumetric curvature source associated with the baryonic mass distribution.

The total curvature content enclosed within a region of radius r is then given by

$$Q_b^{\rm src}(r) = \int_{V_r} s_b \, dV. \tag{10}$$

This quantity represents the conserved curvature budget generated by the baryonic source.

2.3 Bulk–to–shell projection

The transfer of curvature from the bulk to the boundary is naturally described by the flux of F_b through the surface S_r. The surface source density is defined as

$$\sigma_r(\omega) = F_b(r, \omega) \cdot n_r, \tag{11}$$

where n_r is the outward unit normal and ω denotes angular coordinates on S_r.

The total curvature injected into the shell is then

$$Q_b^{\rm src}(r) = \int_{S_r} \sigma_r(\omega)\, dA. \tag{12}$$

Since $\nabla \cdot F_b = s_b$ holds throughout the bulk, the divergence theorem gives the same result for any surface enclosing the same source region, establishing $Q_b^{\rm src}$ as a conserved quantity independent of the specific projection surface chosen.

This construction shows that the shell is not an additional structure, but the natural boundary on which the conserved curvature flux appears as a source.

2.4 Connection to General Relativity

In the weak-field limit of General Relativity, the gravitational potential Φ satisfies the Poisson equation

$$\nabla^2 \Phi = 4\pi G \rho_b, \tag{13}$$

where ρ_b is the baryonic mass density.

Defining the curvature flux field as

$$F_b \equiv -\nabla \Phi, \tag{14}$$

one obtains

$$\nabla \cdot F_b = -4\pi G \rho_b. \tag{15}$$

The enclosed curvature source becomes

$$Q_b^{\mathrm{src}}(r) = \int_{S_r} F_b \cdot n_r \, dA = 4\pi G M_b(r), \qquad (16)$$

where $M_b(r)$ is the enclosed baryonic mass and $Q_b^{\mathrm{src}} \equiv -Q_b = 4\pi G M_b(r)$ denotes the positive source magnitude.

Thus, the bulk-to-shell projection introduced above coincides exactly with the Gauss-law flux of the gravitational field. No modification of the local gravitational law is required. Every subsequent quantity introduced in this paper has an exact GR counterpart; readers who prefer to work entirely within standard GR language are referred to Appendix D, which provides a self-contained restatement of the full framework and an explicit identification of the three elements that are new relative to textbook GR.

2.5 Role of the shell as a transport interface

The geometric construction separates the system into two complementary components:

- the bulk region $V_{\mathrm{cap}}(\chi)$, which contains the curvature source,
- the boundary surface S_r, on which the curvature flux appears as a surface source.

The shell therefore acts as an interface that receives curvature generated in the bulk and provides the manifold on which its subsequent redistribution takes place.

At this stage, no assumption has been made about the dynamics of this redistribution. The shell is defined purely as the geometric locus on which the conserved curvature flux is realized.

The dynamical question of how curvature transported from the bulk becomes organized on the shell, and under what conditions this transport acquires an effectively two-dimensional character, is addressed in the following section.

3 ORBITAL ORGANIZATION OF CURVATURE TRANSPORT

This section argues that coherent curvature transport is confined to constant-radius shells S_r as a dynamical consequence of causal propagation and differential rotation. The mechanism is developed from two independent starting points, and the two arguments are complementary.

Section 3.3 derives the winding mechanism using the eikonal limit of the bulk transport equation in the co-moving frame, and provides the foundation for the dimensional reduction developed in Section 4.5.

Section 3.4 and Appendix E develop the same mechanism independently within standard weak-field General Relativity, starting from the retarded gravitational potential of a differentially rotating source. This derivation requires no WT-specific assumptions: the co-moving frame, the telegraph equation, and the constitutive parameters τ_c, κ/D play no role. The Doppler-shifted dispersion relation and the ray equations emerge from the phase structure of the retarded integral alone, via the same-shell condition $r' \approx r$ established in Appendix E.

The WT version provides the geometric language used throughout the paper. The GR version demonstrates that the physical conclusions — shell confinement, the $1/r$ acceleration, the BTFR scaling — do not depend on accepting that language. A reader who accepts only standard GR may follow Section 3.4 and Appendix E directly and treat the WT sections as an alternative geometric presentation of the same physics.

3.1 Continuous curvature injection

The construction of Section 2 establishes that curvature generated by baryonic matter is continuously transferred to the boundary surface S_r through the conserved flux $Q_b^{\mathrm{src}}(r)$. In a dynamically evolving spacetime, this transfer is not a static process but an ongoing redistribution of curvature under causal propagation.

Because the propagation of curvature is subject to a finite update speed, the system cannot maintain a time-independent configuration in which curvature remains localized solely within the bulk. Instead, curvature must be continuously transported away from its source and reorganized on the available geometric structures.

The problem is therefore not whether redistribution occurs, but how it is organized.

3.2 Breakdown of radial coherence

Finite propagation speed and differential rotation together produce an asymmetry between radial and tangential transport: radial coherence is progressively destroyed while tangential coherence is maintained. The concrete dynamical mechanism — the winding of radial perturbations into azimuthal trajectories — is established in the following subsection.

3.3 Spiral winding as the dynamical origin of shell confinement

The confinement of curvature transport to constant-radius shells has a concrete dynamical origin: curvature transport in a rotating system with finite propagation speed. Curvature perturbations propagate as waves whose ray dynamics can be derived from the governing transport equation. The role of differential rotation in winding spiral perturbations is well established in the galactic dynamics literature Lynden-Bell & Kalnajs (1972); Toomre (1981).

3.3.1 Setup.

Consider a rotating baryonic source with angular velocity profile $\Omega(r)$, producing a velocity field

$$\mathbf{v}(r) = r\,\Omega(r)\,\hat{\boldsymbol{\theta}}.$$

The curvature-transport medium is advected by the galactic flow on timescales long compared to the orbital period, so the material derivative $D/Dt = \partial_t + \mathbf{v} \cdot \nabla$ replaces the partial time derivative in the transport equation.

(Epistemic note: the co-moving assumption — that the curvature-transport medium is advected by the baryonic velocity field on orbital timescales — is classified as Argued. *It is physically motivated by the short orbital timescale condition $T_{\mathrm{orb}} \ll T_{\mathrm{gal}}$, but has not been derived from a covariant transport equation.)*

3.3.2 *Derivation of the ray equation from the bulk transport equation.*

The starting point is the causal curvature transport equation on S^3 in the advected frame:

$$\tau_c \frac{D^2\Phi}{Dt^2} + \frac{D\Phi}{Dt} = D\,\Delta_{S^3}\Phi - \kappa\Phi + S_b(x,t). \tag{17}$$

$$\frac{D}{Dt} = \partial_t + \mathbf{v}\cdot\nabla$$

For short-wavelength perturbations (wavelength $\ll$ galactic scale), I apply a WKB ansatz to the linearized equation,

$$\varphi(x,t) = A(x,t)\, e^{i\theta(x,t)/\varepsilon}, \quad \varepsilon \ll 1,$$

with local wavevector $\mathbf{k} = \nabla\theta$ and frequency $\omega = -\partial_t\theta$. The operator D/Dt acting on $e^{i\theta}$ produces the intrinsic (Doppler-shifted) frequency

$$\Omega_{\rm int} = \omega - \mathbf{v}\cdot\mathbf{k}.$$

At leading order in ε, the transport equation reduces to the dispersion relation

$$-\tau_c\,\Omega_{\rm int}^2 - i\,\Omega_{\rm int} = -D|\mathbf{k}|^2 - \kappa,$$

which in the underdamped propagating regime ($\tau_c\Omega_{\rm int}^2 \gg 1$, $\kappa \to 0$) gives

$$\omega = \mathbf{v}\cdot\mathbf{k} \pm c_{\rm eff}|\mathbf{k}|, \quad c_{\rm eff} = \sqrt{D/\tau_c} = \gamma c.$$

The dispersion relation $H(\mathbf{x},\mathbf{k}) = \omega$ defines a Hamiltonian system. Hamilton's equations for the ray trajectory and wavevector are

$$\frac{d\mathbf{x}}{dt} = \frac{\partial H}{\partial \mathbf{k}} = \mathbf{v}(\mathbf{x}) + c_{\rm eff}\,\hat{\mathbf{k}}, \tag{18}$$

$$\frac{d\mathbf{k}}{dt} = -\frac{\partial H}{\partial \mathbf{x}} = -(\mathbf{k}\cdot\nabla)\mathbf{v}(\mathbf{x}). \tag{19}$$

Equation (19) is the wavevector evolution law derived from the bulk transport equation. It is not borrowed from geometric optics; it follows from $d\mathbf{k}/dt = -\partial H/\partial\mathbf{x}$ applied to the Doppler-shifted dispersion relation above.

3.3.3 *Explicit winding calculation.*

Evaluating equations (18)–(19) for an initially radial perturbation $\mathbf{k}(0) = k_0\hat{\mathbf{r}}$ in a differentially rotating background ($\Omega \propto r^{-1/2}$, representative of the inner disc). The relevant off-diagonal component of the velocity gradient tensor is

$$(\nabla\mathbf{v})_{\theta r} = r\,\partial_r\Omega = -\tfrac{1}{2}\Omega.$$

From equation (19), the azimuthal component of $\mathbf{k}$ evolves as

$$\frac{dk_\theta}{dt} = -\big[(\mathbf{k}\cdot\nabla)\mathbf{v}\big]_\theta = k_r\cdot A, \qquad A \equiv \tfrac{1}{2}\Omega(r),$$

where A is the Oort shear parameter. Since $k_r(0) = k_0 > 0$ and $A > 0$:

$$k_\theta(t) = A\,k_0\,t \qquad (At \ll 1).$$

The pitch angle ψ between the wavevector and the azimuthal direction satisfies

$$\tan\psi(t) = \frac{k_r}{k_\theta} = \frac{1}{At} \longrightarrow 0 \quad \text{as } t\to\infty.$$

From equation (18), the radial component of the group velocity simultaneously vanishes:

$$\left(\frac{d\mathbf{x}}{dt}\right)_r = c_{\rm eff}\frac{k_r}{|\mathbf{k}|} \sim \frac{c_{\rm eff}}{At} \longrightarrow 0.$$

Any initially radial curvature perturbation is therefore wound into an azimuthal trajectory; radial transport asymptotically ceases, and the perturbation becomes confined to the shell S_r at constant r.

3.3.4 *Winding timescale.*

The pitch angle reaches $\psi \sim 1$ (i.e. $k_\theta \sim k_r$) on the shearing timescale

$$t_{\rm wind} \sim \frac{1}{A} = \frac{2}{\Omega(r)} \sim \frac{T_{\rm orb}(r)}{\pi}.$$

For $t \gg t_{\rm wind}$ the ratio $k_\theta/k_r \approx At$ continues to grow linearly, so radial group velocity is suppressed as $1/(At)$ regardless of the detailed late-time evolution. For typical disk galaxies, $T_{\rm orb} \sim 200$–$500\,$Myr at $r \sim 10$–$20\,$kpc, giving $t_{\rm wind} \sim 60$–$160\,$Myr. With galaxy ages $T_{\rm gal} \sim 10\,$Gyr,

$$\frac{t_{\rm wind}}{T_{\rm gal}} \sim 10^{-2}.$$

The winding regime is therefore quantitatively established for all well-mixed disk galaxies: perturbations are wound into shells on timescales roughly two orders of magnitude shorter than the galaxy age.

3.3.5 *Physical consequence: confinement to shells.*

Because curvature is continuously injected and propagated outward, contributions arriving at a given radius originate from different emission times and paths. Finite propagation speed introduces temporal delays, and differential rotation introduces spatial shear. Together, these effects produce two complementary behaviors:

- **Radial dephasing:** superposition of contributions with different delays leads to loss of coherence in the radial direction;
- **Tangential coherence:** shear aligns propagation along azimuthal directions, preserving coherence along constant-radius paths.

As a result, transport along the radial direction becomes dispersive and incoherent at large radii, while transport along the shell remains organized and coherent. The curvature distribution becomes supported on constant-radius shells as the direct dynamical consequence of causal propagation in a rotating system — not as an imposed assumption.

The confinement follows from three ingredients: finite-speed causal propagation, continuous curvature injection, and differential rotation of the source. Spiral winding is not a phenomenological description but a direct consequence of the transport equation governing curvature propagation in the co-moving frame, under the assumption of Section 3.3.

Under the co-moving assumption, the shell is therefore the natural manifold for coherent transport at large radii, providing the dynamical foundation for the dimensional reduction developed in Section 4.5.

3.4 Retarded-Source Derivation of Shell Confinement in Linearized General Relativity

The derivation in Section 3.3 established spiral winding and shell confinement using the causal transport equation in the co-moving frame. The same conclusion now follows directly from standard weak-field General Relativity, without invoking a co-moving medium or any WT-specific constitutive assumption. The complete derivation is given in Appendix E; this section states the logical structure and key results.

3.4.0.1 Setup. In the linearized weak-field limit the gravitational potential of a time-dependent source is the retarded integral

$$\Phi(\mathbf{x},t) = -G\int \frac{\rho_b\Big(\mathbf{x}',\, t-\frac{|\mathbf{x}-\mathbf{x}'|}{c}\Big)}{|\mathbf{x}-\mathbf{x}'|}\, d^3x'. \tag{20}$$

The baryonic source is a differentially rotating disk with surface density $\Sigma(r')$ and angular velocity profile $\Omega(r)$,

$$\rho_b(\mathbf{x}',t) = \Sigma(r')\,\delta(z')\,\delta\big(\varphi' - \Omega(r')\,t\big)\,. \tag{21}$$

3.4.0.2 Orbital averaging. Over $T_{\rm gal} \gg T_{\rm orb}$, non-axisymmetric ($m \neq 0$) Fourier modes of the retarded field are suppressed by the Riemann–Lebesgue lemma Blanchet & Damour (1992):

$$\langle \Phi_{m\neq 0}\rangle \sim \frac{1}{m\Omega(r')\,T_{\rm gal}}\,\Phi^{(0)}_{m\ell} \;\longrightarrow\; 0. \tag{22}$$

The long-time averaged field is therefore axisymmetric. Moreover, contributions from source points at different radii r' arrive with retarded delays that differ by $\Delta r/c$; after $N_{\rm orb} = T_{\rm gal}/T_{\rm orb}$ orbits these interfere destructively, confining coherent radial structure to scales $\ell_{\rm coh} \sim (T_{\rm orb}/T_{\rm gal})\,r/2\pi \sim 50\,\mathrm{pc}$ at $r = 10\,\mathrm{kpc}$ (Appendix E, equations (E18)–(E20)). Angular structure at fixed r is not suppressed. Radial coherence in the averaged field is therefore negligible on galactic scales.

3.4.0.3 Eikonal limit and the same-shell condition. For short-wavelength perturbations, the retarded integral (20) is dominated in the eikonal limit by source points at $r' \approx r$. This *same-shell condition* (Appendix E, equation (E8)) follows from the stationary-phase condition on the retarded source phase and is valid throughout the disk by five orders of magnitude ($R_{\rm gal} \ll c/\Omega \sim 50\,\mathrm{Mpc}$). At the stationary point the local frequency and wavevector satisfy $\omega = m\Omega(r)$ and $|\mathbf{k}| = \omega/c$ (equation (E9)), giving the Doppler-shifted dispersion relation

$$\omega_{\rm eff}^2 = c^2|\mathbf{k}|^2, \qquad \omega_{\rm eff} = \omega - \mathbf{v}\cdot\mathbf{k}, \tag{23}$$

where $\mathbf{v} = r\Omega(r)\hat{\boldsymbol{\theta}}$ is the local pattern velocity. The $\mathbf{v}\cdot\mathbf{k}$ term is a kinematic consequence of $r' \approx r$, not an assumption about the wave operator (equation (E10)). The Hamiltonian $H = \mathbf{v}\cdot\mathbf{k} + c|\mathbf{k}|$ then yields the ray equations, in particular

$$\frac{d\mathbf{k}}{dt} = -(\mathbf{k}\cdot\nabla)\mathbf{v}, \tag{24}$$

derived from the retarded source phase without any co-moving medium assumption (Appendix E, equation (E31)).

3.4.0.4 Spiral winding. For $\mathbf{v} = r\Omega(r)\hat{\boldsymbol{\theta}}$ the ray equation (24) gives, for an initially radial perturbation $\mathbf{k}(0) = k_0\hat{r}$, a linear shear in wavevector space (Appendix E):

$$k_r(t) \approx k_0, \qquad k_\theta(t) \approx 2A_{\rm Oort}\,k_0\,t \qquad (A_{\rm Oort}\,t \lesssim 1), \tag{25}$$

where $A_{\rm Oort} = -\frac{1}{2}r\,d\Omega/dr > 0$ is the Oort shear constant (Appendix E). Since k_θ/k_r grows without bound, the pitch angle $\tan\psi = k_r/k_\theta \to 0$ and the radial group velocity $\sim c\,k_r/|\mathbf{k}| \to 0$ on the winding timescale

$$t_{\rm wind} \sim \frac{1}{2A_{\rm Oort}} \sim \frac{T_{\rm orb}}{\pi} \ll T_{\rm gal}. \tag{26}$$

3.4.0.5 Consequence: confinement to shells. The same-shell condition ensures that the perturbations undergoing winding are sourced by and propagate within the same constant-r surface. After $t \gg t_{\rm wind}$, $k_\theta \gg k_r$: radial group velocity is suppressed and coherent transport is confined to S_r. Combined with the radial coherence suppression of the averaged field established above, both arguments select the shell as the natural large-scale manifold supporting coherent transport — one on timescale $T_{\rm gal}$, the other on timescale $t_{\rm wind}$.

Thus, without invoking a co-moving medium, the combination of (i) retarded causal propagation, (ii) long-time orbital averaging, and (iii) differential rotation selects S_r as the natural manifold supporting coherent curvature transport.

3.4.0.6 Connection to dimensional reduction. Once $k_\theta \gg k_r$, the angular-dominance condition of Appendix A is satisfied and the Laplace–Beltrami operator on S^3 reduces to its angular component. The dimensional reduction to a two-dimensional Poisson equation on S_r follows exactly as in Section 4.5, yielding the logarithmic Green function and the $1/r$ shell acceleration.

3.5 Orbital confinement and effective dimensional reduction

The shell regime argued for by the winding mechanism above is realized quantitatively when two conditions are met:

$$T_{\rm orb}(r) \ll T_{\rm gal}, \qquad T_{\rm prop}(r) \ll T_{\rm gal}, \tag{27}$$

where $T_{\rm orb} \sim 2\pi r/v(r)$ is the orbital period, $T_{\rm prop} \sim r/c$ is the propagation timescale across the shell, and $T_{\rm gal}$ is the system age. When these conditions hold, curvature has sufficient time both to propagate across the shell and to be averaged over many orbital configurations.

Once redistribution is confined to S_r, the relevant degrees of freedom are those intrinsic to this two-dimensional manifold. Radial transport no longer contributes coherently to the large-scale structure of the field, while tangential transport dominates. The effective dimensionality of curvature propagation is therefore reduced from three to two — not as an imposed assumption, but as the direct geometric consequence of orbital confinement.

Epistemic summary of shell confinement This is the load-bearing kinematic claim of the paper. It is *Argued* via two independent routes — WT co-moving winding (Section 3.3) and GR retarded-source phase structure (Appendix E) — neither of which is fully rigorous. Steps 3–4 of the derivational chain of Section 4.1 follow as *Derived* conditional on this claim being granted.

3.6 Epistemic status

The winding result holds for $At \lesssim 1$; at later times the full coupled system is asymmetric, but k_θ/k_r continues to grow monotonically so the asymptotic confinement is robust. The derivation that makes the dimensional reduction precise — showing that the bulk transport equation induces a two-dimensional Poisson equation on S_r under the conditions established above — is carried out in Section 4.5.

3.7 Applicability and breakdown

The shell transport regime applies broadly across galaxy morphologies, though with different efficiency. In rotationally supported disk galaxies, orbital coherence produces efficient phase mixing and a robust $1/r$ contribution. In pressure-supported spheroidal systems, orbits are not coherent but span a distribution of directions. Rather than suppressing mixing, this enhances it: randomized orbits lead to rapid isotropization of the shell source,

$$\langle \sigma_r \rangle \to \text{angular average}, \tag{28}$$

so the two-dimensional transport description remains valid with different efficiency encoded in α_0.

At sufficiently large radii, orbital periods grow as $T_{\rm orb} \propto r$. Beyond a critical mixing scale $r_{\rm mix}$, the condition $T_{\rm orb} \lesssim T_{\rm gal}$ fails and orbital averaging becomes incomplete. The shell source remains anisotropic, phase mixing is inefficient, and the effective dimensional reduction weakens. This predicts a gradual suppression or increased scatter of the shell contribution at very large radii — a direct observational consequence of the mechanism.

The relation between mixing efficiency, $r_{\rm mix}$, and the morphology factor α_0 has not been derived from first principles and remains an open problem.

The crossfade between the Newtonian inner regime ($a_b \gg a_*$) and the shell-dominated outer regime ($a_b \ll a_*$), and the master acceleration formula that results, are developed in Section 5.

4 BULK-TO-SHELL CURVATURE TRANSPORT

Shell confinement — argued in Section 3 — has a direct quantitative consequence: the same conserved curvature flux that drives the Newtonian $1/r^2$ channel at small radii becomes the source for a two-dimensional transport process at large radii. This section derives the two channels explicitly and establishes the Poisson equation that governs their junction.

4.1 Local baryonic curvature (3D channel)

In the weak-field limit, baryonic matter generates the standard inverse-square acceleration

$$a_b(r) = \frac{GM_b(r)}{r^2}. \tag{29}$$

This scaling follows from flux conservation in three dimensions: curvature emitted by a localized source spreads over spherical surfaces of area $4\pi r^2$. The local channel is therefore intrinsically three-dimensional.

4.2 Curvature as a conserved flux and source normalization

Baryonic curvature is formulated as a conserved flux field $\mathbf{F}_b$ satisfying

$$\nabla \cdot \mathbf{F}_b = s_b, \tag{30}$$

where s_b is the volumetric curvature source associated with baryonic matter.

For a spherical surface S_r, the normal flux defines the surface source density

$$\sigma_r(\omega) = \mathbf{F}_b(r,\omega)\cdot \mathbf{n}_r, \tag{31}$$

where $\mathbf{n}_r$ is the outward unit normal and ω denotes angular coordinates on S_r. The total curvature transferred to the shell is

$$Q_b^{\rm src}(r) = \int_{S_r} \sigma_r\, dA = \int_{V_r} s_b\, dV. \tag{32}$$

By the divergence theorem, this expression is independent of the specific projection surface chosen, establishing $Q_b^{\rm src}$ as a conserved quantity.

4.2.0.1 Connection to General Relativity. In the weak-field limit, identifying $\mathbf{F}_b \equiv -\nabla\Phi$ and $\nabla^2\Phi = 4\pi G\rho_b$, the bulk curvature source is identified as $s_b = -4\pi G\rho_b$, and the shell source becomes $\sigma_r = \mathbf{F}_b \cdot \mathbf{n}_r = -\partial_r\Phi$. Integrating over the spherical surface yields

$$Q_b^{\rm src}(r) = 4\pi G M_b(r), \tag{33}$$

where $M_b(r)$ is the enclosed baryonic mass. The shell source is therefore the Gauss-law flux of the gravitational field, and the present model does not modify the local gravitational law.

4.2.0.2 Geometric representation. In the geometric language of this framework, the same source is expressed as

$$Q_b^{\rm src}(\chi) = \kappa_q\, a_*\, A_b(\chi), \qquad a_* = \frac{c^2}{\pi R}. \tag{34}$$

Identifying $A_b(\chi) \equiv M_b(\chi)$ and using $a_* = G$, requiring exact agreement with the GR normalization uniquely fixes $\kappa_q = 4\pi$, giving

$$Q_b^{\rm src}(\chi) = 4\pi\, a_*\, A_b(\chi) = 4\pi G M_b(\chi). \tag{35}$$

The decomposition $Q_b^{\rm src} = 4\pi \times a_* \times A_b$ has a direct geometric interpretation: 4π is the universal flux factor of spherical geometry, a_* is the global curvature coupling set by the radius of the Universe, and $A_b(\chi)$ is the enclosed baryonic curvature content.

In a dynamical setting, finite-speed transport and cosmological expansion produce a time-dependent shell occupancy $Q_b^{\rm shell}(r,\chi,t)$. The quasi-static limit $Q_b^{\rm shell} \to Q_b^{\rm src}$ recovers the static formulation used throughout this work.

4.3 Geometric coupling between bulk and shell

The efficiency with which bulk curvature feeds the shell depends on the geometry of the source region. For a spherical cap on S^3, this is captured by the boundary-to-volume ratio

$$L(\chi) = \frac{A_\partial(\chi)}{V_{\rm cap}(\chi)}. \tag{36}$$

This ratio measures the boundary access per enclosed source region. A larger boundary relative to the enclosed volume corresponds to more efficient transfer of curvature into shell transport.

4.3.0.1 Geometric constraint. Since Q_b^{src} fully accounts for the total enclosed baryonic source, the coupling factor $\alpha_b(\chi)$ must encode the efficiency of transfer through the boundary rather than additional source strength. The only scalar quantity constructed from the cap geometry that captures this is $L(\chi)$, giving the shell coupling law

$$\alpha_b(\chi) = \alpha_0\, L(\chi) = \alpha_0\, \frac{A_\partial(\chi)}{V_{\mathrm{cap}}(\chi)}, \tag{37}$$

where α_0 is a dimensionless normalization factor encoding baryonic morphology.

4.3.0.2 Small-angle limit. In the limit $\chi \ll 1$, the cap reduces to a Euclidean ball and

$$L(\chi) \approx \frac{3}{r}, \tag{38}$$

recovering the familiar surface-to-volume ratio in the local regime.

The shell coupling coefficient is therefore not arbitrary but follows from the requirement that a three-dimensional curvature source couples to a two-dimensional transport manifold through its boundary interface. The ratio $A_\partial/V_{\mathrm{cap}}$ is the natural geometric measure of this coupling.

4.4 Morphology factor as shell-mode projection efficiency

The geometric factor $L(\chi)$ captures the idealized coupling efficiency under the assumption of a symmetric spherical cap. Real galaxies deviate significantly from this geometry. Disk structure, bulge-to-disk ratio, gas fraction, clumpiness, and anisotropy all affect how efficiently the bulk curvature flux couples to the shell transport modes. These effects are encoded in α_0.

4.4.0.1 Flux-based definition. The shell source $\sigma_r(\omega) = \mathbf{F}_b \cdot \mathbf{n}_r$ drives tangential transport through the Poisson equation $\Delta_{S_r}\Phi_r = -\sigma_r$. Only the components of σ_r that project onto the long-range transport modes of Δ_{S_r} contribute to the asymptotic $1/r$ acceleration. The normalized projection efficiency α_0 is therefore defined as:

$$\alpha_0 = \frac{\left\|\mathcal{P}_{\mathrm{shell}}\, \sigma_r^{\mathrm{actual}}\right\|}{\left\|\mathcal{P}_{\mathrm{shell}}\, \sigma_r^{\mathrm{cap}}\right\|}, \tag{39}$$

where $\mathcal{P}_{\mathrm{shell}}$ denotes projection onto the dominant low-order modes of the surface Laplacian responsible for large-scale transport. This ensures $\alpha_0 = 1$ for the ideal spherical-cap configuration, with deviations from unity quantifying the departure of the actual baryonic distribution from this reference.

4.4.0.2 Epistemic status. The effective coupling amplitude factorises as

$$\alpha_0 = \kappa \cdot A_{\mathrm{morph}}, \tag{40}$$

where $\kappa = 1 + \pi^{-2}$ is a universal geometric constant fixed by the S^3 background geometry (Appendix C), and A_{morph} is a class-dependent projection efficiency determined by the long-timescale baryonic structure and orbital history of the galaxy. The factor κ is fully predicted; A_{morph} is not predicted by the geometric framework and must be calibrated observationally. Its geometric definition constrains its interpretation — it quantifies how efficiently the actual baryonic distribution excites the dominant shell transport modes — but not its magnitude. In practice, A_{morph} is expected to take characteristic values for broad galaxy classes (e.g. gas-dominated dwarfs, LSB disks, HSB spirals), reflecting differences in long-timescale baryonic structure and phase-mixing efficiency; these have been determined from the SPARC archive of 175 well-measured rotation curves Lelli et al. (2016) and the radial acceleration relation McGaugh et al. (2016), and are reported in Appendix F.

Thus α_0 should be understood as the product of a geometrically derived amplitude and a morphology-class projection efficiency, not as a free parameter in the dynamical law.

4.5 From bulk causal transport to the shell Poisson equation

This subsection closes the loop by showing how the bulk causal curvature dynamics on S^3 induces the intrinsic shell Poisson equation on S_r.

4.5.1 Setup and scope

The derivation starts from the bulk transport equation for the curvature potential Φ:

$$\tau_c \partial_t^2 \Phi + \partial_t \Phi = D\Delta_{S^3}\Phi - \kappa\Phi + S_b(x,t), \tag{41}$$

where $S_b(x,t)$ is the baryonic source associated with ρ_b.

The goal is to recover the intrinsic shell equation

$$\Delta_{S_r}\Phi_r = -\sigma_r, \tag{42}$$

with σ_r defined as the shell source induced by bulk transport. The analysis proceeds under three explicit assumptions:

- **Quasi-static regime:** $\partial_t \Phi \approx 0$, $\tau_c \partial_t^2 \Phi \approx 0$.
- **Large-r shell-dominated transport:** $\left|\partial_\chi^2 \Phi + 2\cot\chi\,\partial_\chi \Phi\right| \ll \left|\frac{1}{\sin^2\chi}\Delta_{S^2}\Phi\right|$.
- **Leading-order transport dominance:** $\kappa\Phi \ll D\Delta_{S^3}\Phi$.

This reduction is therefore asymptotic, valid in the outer, phase-mixed regime. Crucially, the resulting shell Poisson equation (42) depends only on flux conservation and the tangential dominance condition; it is insensitive to the specific form of the bulk equation (41), so the $1/r$ shell acceleration is robust to variations in the telegraph parameters τ_c, D, and κ.

4.5.2 Quasi-static bulk reduction

In the quasi-static limit, equation (41) reduces to

$$D\Delta_{S^3}\Phi - \kappa\Phi + S_b(x) \approx 0. \tag{43}$$

Dividing by D,

$$\Delta_{S^3}\Phi - \frac{\kappa}{D}\Phi = -\frac{1}{D}S_b(x). \tag{44}$$

This describes steady-state curvature transport in the bulk.

4.5.3 Validity of the quasi-static limit

The quasi-static reduction drops both time-derivative terms from (41). Two independent dimensionless conditions must hold simultaneously. Each is verified against galactic parameters.

4.5.3.1 Condition 1: causal memory negligible ($\tau_c\partial_t^2\Phi \ll \partial_t\Phi$). Let $T_{\rm dyn}$ denote the relevant dynamical timescale — taken as the orbital period $T_{\rm orb}$ for the outer disc, the most demanding case. On a field varying on timescale $T_{\rm dyn}$,

$$\frac{|\tau_c\partial_t^2\Phi|}{|\partial_t\Phi|} \sim \frac{\tau_c}{T_{\rm dyn}} \equiv \varepsilon_2.$$

The causal timescale τ_c is identified with the Planck time $T_p \approx 5.4\times10^{-44}\,$s — the smallest interval consistent with known physics (Table 1: *Asserted*). For a representative outer disc with $T_{\rm orb}\sim 3\times10^8\,$yr $\approx 10^{16}\,$s,

$$\varepsilon_2 = \frac{T_p}{T_{\rm orb}} \approx \frac{5.4\times10^{-44}}{10^{16}} \sim 10^{-60}.$$

Condition 1 is satisfied with an astronomical margin for any galactic dynamics problem; given the identification $\tau_c = T_p$, the numerical verification is *Derived*.

4.5.3.2 Condition 2: field varies slowly compared to propagation ($\partial_t\Phi \ll D\,\Delta_{S^3}\Phi$). This condition requires the propagation time across the system to be short compared to the time on which the source changes:

$$\varepsilon_1 = \frac{T_{\rm prop}}{T_{\rm dyn}} = \frac{r}{c_{\rm eff}\,T_{\rm dyn}}, \qquad c_{\rm eff}=\gamma c, \quad \gamma\le 1.$$

For $r = 20\,$kpc $\approx 6\times10^{20}\,$m and $T_{\rm dyn}=T_{\rm orb}\approx 10^{16}\,$s,

$$\varepsilon_1 = \frac{6\times10^{20}}{\gamma\cdot 3\times10^8\cdot10^{16}} = \frac{2\times10^{-4}}{\gamma}.$$

Condition 2 is satisfied, $\varepsilon_1 \ll 1$, for any propagation speed $\gamma \gtrsim 10^{-3}$. Since $\gamma\le1$ is bounded above by causality and no mechanism suppresses it below $\sim10^{-3}$ in the WT framework, this condition holds for all physically reasonable parameter choices.

(Note on the constant γ approximation: In the full WT dynamics, γ is a local quantity. The transport susceptibility $\chi(E, Z_{\rm eff})$ in the WT action depends on the local energy density and medium impedance, so the effective propagation speed $c_{\rm eff}[\rho_b(\mathbf{x})]$ varies across the galaxy. The constant γ used throughout this paper is therefore an orbit-averaged effective value. For diffuse systems — LSB galaxies and dwarf irregulars — the variation of $c_{\rm eff}$ across the disk is small and the approximation is well-controlled. For compact, high-surface-brightness systems the spatially averaged γ absorbs the variation of local transport efficiency into a single number; this is the physical content of the per-galaxy α_0 fit reported in Section 4.4 and Appendix F. A first-principles treatment requires integrating the local contributions and is deferred to future work.)

4.5.3.3 Condition 3: screening subdominant ($\kappa\Phi \ll D\,\Delta_{S^3}\Phi$). The screening term is negligible when the screening length $\ell_\kappa=\sqrt{D/\kappa}$ greatly exceeds the galactic scale. The natural screening scale is set by the cosmological curvature radius: $\kappa/D\sim 1/R^2$, giving

$$\ell_\kappa \sim R \approx 4.3\times10^{26}\,\text{m}.$$

Since $r_{\rm gal}\sim 10^{21}\,$m,

$$\frac{r_{\rm gal}}{\ell_\kappa} \sim \frac{10^{21}}{4.3\times10^{26}} \sim 10^{-5} \ll 1.$$

The screening correction is of order $(r_{\rm gal}/\ell_\kappa)^2\sim10^{-10}$ and is entirely negligible. The identification $\kappa/D\sim1/R^2$ is *Argued*; given this identification the numerical bound is *Derived*.

4.5.3.4 Summary. All three conditions required for the quasi-static reduction are quantitatively satisfied with large margins for galactic scales:

$$\begin{aligned}\varepsilon_2 &= \tau_c/T_{\rm orb}\sim10^{-60},\\ \varepsilon_1 &= r/(c_{\rm eff}\,T_{\rm orb})\sim 2\times10^{-4}/\gamma \ll 1,\\ (r/\ell_\kappa)^2 &\sim 10^{-10}.\end{aligned}$$

The quasi-static approximation is not a qualitative assumption for this problem — it is an asymptotic regime whose control parameters are all small by many orders of magnitude under stated WT identifications ($\tau_c=T_p$, $\kappa/D\sim1/R^2$). The reduction in Section 4.5.2 is therefore valid throughout the galactic domain.

4.5.4 Radial–tangential decomposition

Using the Laplacian on S^3,

$$\Delta_{S^3} f = \frac{1}{R^2}\left[\partial_\chi^2 f + 2\cot\chi\,\partial_\chi f + \frac{1}{\sin^2\chi}\Delta_{S^2} f\right], \tag{45}$$

equation (43) becomes

$$\frac{D}{R^2}\left[\partial_\chi^2\Phi + 2\cot\chi\,\partial_\chi\Phi + \frac{1}{\sin^2\chi}\Delta_{S^2}\Phi\right] - \kappa\Phi + S_b \approx 0. \tag{46}$$

The neglect of the radial terms is not a further assumption but a consequence of the spectral structure of the bulk equation. The Green function of the elliptic operator in (44) can be expanded in hyperspherical harmonics $Y_{n\ell m}(\chi,\Omega)$, which satisfy

$$\Delta_{S^3} Y_{n\ell m} = -\frac{n(n+2)}{R^2} Y_{n\ell m}. \tag{47}$$

At fixed large χ, the radial part of Δ_{S^3} acting on a mode of degree n contributes a term of order

$$\frac{1}{R^2}\left[\partial_\chi^2 + 2\cot\chi\,\partial_\chi\right] Y_{n\ell m} \sim \frac{n(n+2) - \ell(\ell+1)/\sin^2\chi}{R^2}\,Y_{n\ell m}, \tag{48}$$

while the tangential part contributes

$$\frac{1}{R^2\sin^2\chi}\Delta_{S^2}Y_{n\ell m} = -\frac{\ell(\ell+1)}{R^2\sin^2\chi}\,Y_{n\ell m}. \tag{49}$$

As established by the spectral analysis in Appendix A (equation (A8)), the large-r response is dominated by modes in the regime $\ell(\ell+1)/\sin^2\chi \gg n(n+2)$, in which the Laplace–Beltrami operator on S^3 reduces to

$$\Delta_{S^3} \;\longrightarrow\; \frac{1}{R^2\sin^2\chi}\,\Delta_{S^2}, \tag{50}$$

and the bulk equation becomes

$$\frac{D}{R^2\sin^2\chi}\,\Delta_{S^2}\Phi \approx -S_b. \tag{51}$$

This reduction emerges from the spectral content of the solution: the outer curvature field is dominated by angular modes, and its radial propagation enters only through the boundary flux σ_r.

4.5.5 Projection to Shell Source

The tangential dominance established in the previous subsection reduces the bulk equation to an angular operator on each constant-χ shell. To extract the shell source σ_r, integrating equation (51) over a thin radial shell of thickness $\delta\chi$ centred on χ:

$$\int_{\chi-\delta\chi/2}^{\chi+\delta\chi/2} \frac{D}{R^2\sin^2\chi'}\,\Delta_{S^2}\Phi\; R\,d\chi' \;\approx\; -\int_{\chi-\delta\chi/2}^{\chi+\delta\chi/2} S_b\; R\,d\chi'. \tag{52}$$

In the limit $\delta\chi \to 0$, the right-hand side vanishes unless S_b has a surface concentration on S_r. The left-hand side picks up the discontinuity in the normal derivative of Φ across the shell. The radial flux $F_b = -D\,\partial_r\Phi/R$ satisfies a jump condition at S_r:

$$\big[F_b\cdot n_r\big]_-^+ \;=\; -\frac{D}{R}\big[\partial_\chi\Phi\big]_-^+. \tag{53}$$

For a source smooth in the bulk and concentrated at the shell, this jump is precisely the shell source density. This motivates the definition

$$\sigma_r(\omega) \;\equiv\; F_b(r,\omega)\cdot n_r \;=\; -\frac{D}{R}\,\partial_\chi\Phi\big|_{S_r}, \tag{54}$$

which is not an independent definition but the boundary quantity induced by the bulk transport equation through the jump condition (53). A terminological note: the baryonic source S_b is distributed in the bulk, not surface-concentrated. The "jump condition" (53) is therefore more precisely a *matching condition* between the interior and exterior solutions of the bulk transport equation at the shell boundary S_r. The surface concentration language in (52) refers to the limit $\delta\chi \to 0$ applied to a bulk source that is smooth but whose integral over the thin slab concentrates onto S_r; it is not an additional assumption about the baryonic distribution.

The total shell content is then

$$Q_b^{\rm src}(r) = \int_{S_r} \sigma_r(\omega)\,dA, \tag{55}$$

which by the divergence theorem recovers the enclosed bulk source, consistent with Section 4.2.

4.5.6 Intrinsic Shell Dynamics

The shell source σ_r acts as a boundary condition on S_r inherited from the bulk equation. This boundary condition implies a two-dimensional Poisson equation intrinsic to S_r.

On the shell S_r, the volumetric source S_b is absent by assumption (sources are distributed in the bulk interior). Evaluating equation (51) at $\chi = r/R$ with $S_b = 0$ therefore gives

$$\Delta_{S^2}\Phi\big|_{S_r} = 0 \quad (S_b = 0 \text{ on } S_r), \tag{56}$$

with σ_r entering not as a volumetric term but as the Neumann boundary condition inherited from the jump condition (53).

Defining the intrinsic shell potential $\Phi_r(\omega) \equiv \Phi(r,\omega)|_{S_r}$ and the tangential shell current

$$J_r \;\equiv\; -\nabla_{S_r}\Phi_r, \tag{57}$$

the flux balance across S_r gives the intrinsic conservation law

$$\mathrm{div}_{S_r}\,J_r = \sigma_r. \tag{58}$$

This is not a new postulate: it is the two-dimensional divergence theorem applied to the flux balance implied by the jump condition (53). Combining (57) and (58) yields

$$\Delta_{S_r}\Phi_r = -\sigma_r, \tag{59}$$

which is therefore not an independent postulate but the intrinsic form of the bulk transport equation projected onto S_r in the tangential-dominance regime.

4.5.7 Emergence of the $1/r$ law and normalization by α_0

The derivation below is conditional on the effective shell description established in Section 4.5: specifically, that orbital phase mixing confines curvature redistribution to constant-r shells, making the two-dimensional shell Poisson equation (59) the operative field equation at galactic scales.

The solution of (59) is governed by the two-dimensional Green function on S_r. In the phase-mixed regime, where σ_r becomes effectively smooth, the shell potential takes the logarithmic form

$$\Phi_r \sim Q_b^{\rm src}(r)\,\log r, \tag{60}$$

up to geometric normalization factors. The full derivation of this Green function and its logarithmic behavior is given in Appendix B.

The shell potential Φ_r carries units of $[L^2T^{-2}]$, so its radial gradient $-\partial_r\Phi_r$ has units of acceleration directly. The geometric derivation determines Φ_r only up to the overall efficiency with which bulk curvature transport excites the shell modes. This efficiency, defined above as the morphology factor α_0 (Section 4.4), enters here as a prefactor of α_b (37) rather than a new parameter:

$$a_{\rm shell}(r) = -\,\alpha_b\,\partial_r\Phi_r. \tag{61}$$

The present derivation does not determine α_0; it establishes that α_0 is the *only* undetermined factor in the shell acceleration, and that all geometric and dimensional structure is fixed by the reduction above. (The sign conventions ensuring $a_{\rm shell} > 0$ are collected in Appendix C.)

Using (60),

$$a_{\rm shell}(r) = -\alpha_b\,\frac{Q_b^{\rm src}(r)}{r}. \tag{62}$$

This completes the geometric chain:

$$\rho_b \;\longrightarrow \text{bulk transport} \;\longrightarrow\; \sigma_r \;\longrightarrow\; Q_b^{\rm src}(r) \;\longrightarrow\; \Phi_r \;\longrightarrow\; a_{\rm shell}(r) = -\,\alpha_b\,\frac{Q_b^{\rm src}(r)}{r}.$$

4.5.7.1 Dimensional verification. The chain $\alpha_b \cdot Q_b^{\rm src}(r)/r$ can be verified dimensionally in a single pass. In SI units: $Q_b^{\rm src} = 4\pi G M_b$ has dimensions $[\mathrm{m}^3\,\mathrm{s}^{-2}]$; $\alpha_b = \alpha_0 L(\chi)$ with $L(\chi) \approx 3/r$ has dimensions $[\mathrm{m}^{-1}]$; the product $\alpha_b \cdot Q_b^{\rm src}/r$ therefore has dimensions $[\mathrm{m}^{-1}] \cdot [\mathrm{m}^3\,\mathrm{s}^{-2}] \cdot [\mathrm{m}^{-1}] = [\mathrm{m}\,\mathrm{s}^{-2}]$, an acceleration. In WT geometric units where $G = a_* = c^2/(\pi R)$ has dimensions $[LT^{-2}]$ and $[M_b] = L^2$, the same product gives $[L^{-1}]\cdot[LT^{-2}\cdot L^2]\cdot[L^{-1}] = [LT^{-2}]$, consistently. Substituting $L(\chi) \approx 3/r$ and $Q_b^{\rm src} = 4\pi a_* A_b$ with $A_b = M_b$ (WT) gives $a_{\rm shell} \sim 3\alpha_0 a_* M_b/r^2$, which in the deep-shell regime transitions to $\alpha_0\sqrt{a_b\, a_*}$ as $\xi = \sqrt{M_b}/r \ll 1$, recovering the MOND-like asymptotic of the master formula without additional assumptions.

Note that $\alpha_b(\chi) = \alpha_0 L(\chi)$ is the bulk-to-shell coupling efficiency as a function of cap geometry. In the galactic small-angle limit $\chi \ll 1$, $L(\chi) \approx 3/r$ and the geometric coupling becomes r-dependent. The master formula in Section 5 absorbs this r-dependence into the crossfade structure, retaining α_0 as the sole morphology parameter.

4.5.8 Status of the reduction

The derivation establishes that the shell Poisson equation arises as the leading-order outer response of the bulk transport equation under quasi-static conditions, large-radius shell-dominated transport, and subleading screening corrections. The reduction is therefore asymptotic. At smaller radii, radial transport and screening modify the effective source σ_r and hence the detailed form of $Q_b^{\rm src}(r)$.

The shell equation does not arise by direct dimensional reduction of the bulk equation, but through a two-stage process:

$$\text{bulk} \;\rightarrow\; \text{shell flux } \sigma_r \;\rightarrow\; \text{shell Poisson.}$$

The derivation above corresponds to taking $\tau_c \to 0$ in the full telegraph-type bulk equation (41), dropping any fractional generalization, and working in the quasi-static limit. Three extensions beyond these limits remain open: the full causal reduction including finite memory time τ_c and cosmological expansion $H = \pi c/R$; the fractional operator generalisation

$$\tau_c \partial_t^2 \Phi + \partial_t \Phi = D(-\Delta_{S^3})^{\sigma}\Phi - \kappa\Phi + S_b(x,t), \qquad (63)$$

with $\sigma \neq 1$, whose effective shell dynamics has not been derived; on dimensional grounds, $\sigma < 1$ broadens the spectral weight toward low-ℓ modes, which would flatten the radial decay of Φ_r and steepen the outer rotation-curve tail beyond the $1/r$ prediction, while $\sigma > 1$ narrows the spectral weight and sharpens the transition — making $\sigma = 1$ the unique value that produces the observed asymptotically flat tail; and a rigorous error bound on the spectral dominance condition $\ell(\ell+1)/\sin^2\chi \gg n(n+2)$, which has been verified at the operator level (Appendix A) but not tracked through the full Green function of the screened equation (44). Establishing these would promote the reduction from an asymptotic result to a controlled approximation with explicit error bounds.

5 MASTER ACCELERATION FORMULA

5.1 Physical reasoning

The critical question is how to combine these two channels into a single predictive formula. The two channels represent a *redistribution* of a single conserved curvature budget between 3D bulk propagation and 2D shell transport: as the system moves outward, the same curvature flux that drives the baryonic $1/r^2$ term is progressively reorganized into shell modes. The crossfade formula accounts for this by assigning complementary weights $(1-\sigma)$ and σ that sum to unity at every radius, ensuring that the total curvature budget is neither doubled nor lost in the transition.

The correct structure is a *crossfade*: a smooth interpolation in which the weight continuously transfers from the 3D baryonic channel to the 2D shell channel as the system moves from compact inner regions to diffuse outer regions. The interpolating weight $\sigma(r)$ measures the fraction of curvature that has wound into shell modes at radius r. It satisfies $\sigma \to 0$ in the inner (Newtonian) regime and $\sigma \to 1$ in the outer (shell-dominated) regime, with the two channels summing to unit weight at every radius.

The transition is controlled by the ratio $u(r) = a_b(r)/a_*$, where $a_* = c^2/(\pi R)$ is the cosmological curvature acceleration set by the Hubble radius. When $a_b \gg a_*$, orbital timescales are short relative to the mixing time and three-dimensional propagation dominates. When $a_b \ll a_*$, orbital phase mixing is complete and curvature is fully organized into shell transport. The scale a_* therefore marks the geometric threshold between regimes, fixed entirely by cosmology and requiring no observational calibration.

5.2 Mathematical formulation

Define the dimensionless transition variable

$$u(r) \;=\; \frac{a_b(r)}{a_*} \;=\; \frac{GM_b(r)}{r^2}\cdot\frac{\pi R}{c^2}, \qquad (64)$$

and the crossfade weight

$$\sigma(r) \;=\; \frac{1}{1+u(r)}, \qquad (65)$$

which interpolates smoothly from $\sigma \to 1$ when $a_b \ll a_*$ to $\sigma \to 0$ when $a_b \gg a_*$. This specific form is not uniquely derived from first principles: it is the simplest monotonic rational interpolant consistent with the asymptotic regimes and the symmetry condition $\sigma(u=1) = 1/2$. The spectral argument of Section 5.3 provides a physical motivation for this choice, but the normalization step (92) remains an assertion. Alternative interpolants that share the same limits would produce quantitatively similar but not identical rotation curves. The epistemic position of $\sigma(r)$ is directly parallel to that of the MOND interpolating function $\mu(x)$ Milgrom (1983): both are the simplest rational interpolants consistent with the correct asymptotic limits, neither is derived from first principles within its respective framework, and both serve as honest placeholders for a future variational derivation.

The two pure-regime accelerations are

$$a_b(r) = \frac{GM_b(r)}{r^2} \qquad \text{[3D baryonic channel]}, \qquad (66)$$

$$a_{\rm shell}(r) = \alpha_0\,\sqrt{a_b(r)\cdot a_*} \qquad \text{[2D shell channel]}, \qquad (67)$$

where α_0 is the morphology-dependent projection efficiency introduced in Section 4.4. The crossfade master formula is then

$$a_{\rm eff}(r) \;=\; \big(1-\sigma(r)\big)\,a_b(r) \;+\; \sigma(r)\,\alpha_0\,\sqrt{a_b(r)\cdot a_*}. \tag{68}$$

Collecting terms over the common denominator $1+u$ gives **Form A**:

$$a_{\rm eff}(r) \;=\; \frac{u\,a_b \;+\; \alpha_0\,\sqrt{a_b\cdot a_*}}{1+u}. \tag{Form A}$$

Substituting $a_b = u\,a_*$ and $\sqrt{a_b\,a_*} = \sqrt{u}\,a_*$, then factoring $\sqrt{u}\,a_* = \sqrt{a_b\,a_*}$ from the numerator, yields **Form B**:

$$a_{\rm eff}(r) \;=\; \frac{\sqrt{a_b\cdot a_*}\,\Big[\big(a_b/a_*\big)^{3/2}+\alpha_0\Big]}{1+a_b/a_*}. \tag{Form B}$$

Form B makes the geometric structure explicit: the effective acceleration is $\sqrt{a_b\,a_*}$ — the geometric mean of the local baryonic acceleration and the cosmological curvature scale — modulated by a dimensionless rational function of $u = a_b/a_*$ that encodes the transition between regimes.

Introducing the substitution $x = \sqrt{a_b/a_*}$, so that $a_b = x^2 a_*$ and $u = x^2$, Form B reduces to the maximally compact **Form C**:

$$a_{\rm eff}(r) \;=\; \frac{a_*\,x\,\left(x^3+\alpha_0\right)}{1+x^2}, \qquad x \;=\; \sqrt{\frac{a_b(r)}{a_*}}. \tag{Form C}$$

Form C is a single rational function of one dimensionless variable. The full radial profile of the effective acceleration is determined by the local ratio a_b/a_* alone, with a_* providing the overall scale and α_0 setting the outer amplitude.

5.2.0.1 Equivalence of the two forms. The crossfade formula (68) and the compact Form C (Form C) are algebraically identical. Setting $\xi = \sqrt{a_b/a_*}$, so that $a_b = \xi^2 a_*$ and $\sigma = 1/(1+\xi^2)$, the crossfade formula gives

$$\begin{aligned} a_{\rm eff} &= (1-\sigma)\,a_b + \sigma\,\alpha_0\sqrt{a_b\,a_*} \\ &= \frac{\xi^2}{1+\xi^2}\cdot\xi^2 a_* \;+\; \frac{1}{1+\xi^2}\cdot\alpha_0\,\xi\,a_* \\ &= \frac{a_*\,\xi\,(\xi^3+\alpha_0)}{1+\xi^2}, \end{aligned} \tag{69}$$

which is precisely Form C. The two representations are therefore equivalent; Form C is the compact rewriting of the crossfade formula established above.

The derivation of this form from the spectral structure of the curvature field is given in the following subsection.

5.3 Spectral derivation of the crossfade function

The crossfade weight $\sigma(r)$ is derived here as the normalized occupancy of shell-supported modes in the spectral decomposition of the curvature field on S^3.

5.3.0.1 Spectral expansion on S^3. The curvature potential $\Phi(\chi,\Omega)$ is expanded in hyperspherical harmonics:

$$\Phi(\chi,\Omega) = \sum_{n=0}^{\infty}\sum_{\ell=0}^{n}\sum_{m=-\ell}^{\ell} a_{n\ell m}\,Y_{n\ell m}(\chi,\Omega), \tag{70}$$

with eigenvalue equation

$$\Delta_{S^3} Y_{n\ell m} = -\frac{n(n+2)}{R^2}\,Y_{n\ell m}. \tag{71}$$

5.3.0.2 Mode-level shell-support weight. The Laplace–Beltrami operator decomposes into radial and angular parts (Eq. (78) below). Acting on $Y_{n\ell m}$, the radial and angular contributions to the eigenvalue magnitude are

$$\lambda_r = n(n+2), \qquad \lambda_\theta(\chi) = \frac{\ell(\ell+1)}{\sin^2\chi}. \tag{72}$$

For each mode (n,ℓ,m) the fraction of spectral support carried by the angular (shell) sector is

$$w_{n\ell}(\chi) \equiv \frac{\lambda_\theta}{\lambda_r+\lambda_\theta} = \frac{\ell(\ell+1)/\sin^2\chi}{n(n+2)+\ell(\ell+1)/\sin^2\chi}. \tag{73}$$

This satisfies $0 \le w_{n\ell} \le 1$, with $w_{n\ell}\to 0$ when radial transport dominates and $w_{n\ell}\to 1$ when angular transport dominates.

5.3.0.3 Field-level shell occupancy. The total shell occupancy is the power-weighted average over all modes:

$$\sigma(\chi) \equiv \frac{\displaystyle\sum_{n,\ell,m} |a_{n\ell m}|^2\, w_{n\ell}(\chi)}{\displaystyle\sum_{n,\ell,m} |a_{n\ell m}|^2}. \tag{74}$$

Substituting (73),

$$\sigma(\chi) = \frac{\displaystyle\sum_{n,\ell,m} |a_{n\ell m}|^2 \frac{\ell(\ell+1)/\sin^2\chi}{n(n+2)+\ell(\ell+1)/\sin^2\chi}}{\displaystyle\sum_{n,\ell,m} |a_{n\ell m}|^2}. \tag{75}$$

5.3.0.4 Reduction to effective mode ratio. In the phase-mixed outer regime the spectral power is concentrated around a dominant pair of effective indices $(n_{\rm eff},\ell_{\rm eff})$ that characterize the orbital modes at radius r. Replacing the power-weighted average by its dominant-mode value,

$$\sigma(r) \approx \frac{\ell_{\rm eff}(\ell_{\rm eff}+1)/\sin^2\chi}{n_{\rm eff}(n_{\rm eff}+2)+\ell_{\rm eff}(\ell_{\rm eff}+1)/\sin^2\chi}. \tag{76}$$

This step is **Argued**: it holds when spectral power is concentrated rather than broadly distributed, a condition satisfied in the phase-mixed regime where a narrow band of orbital harmonics dominates. The remaining step — relating the effective mode indices to $a_b(r)/a_*$ — is carried out in the following subsubsection.

5.3.1 Derivation of the anisotropy–compactness closure

I now derive the relation between the spectral anisotropy of the occupied modes and the physical compactness ratio $a_b(r)/a_*$. The goal is to justify the closure

$$\frac{k_r^2}{k_\theta^2} = \frac{a_b(r)}{a_*}, \tag{77}$$

which yields the crossfade function in closed form.

5.3.1.1 Spectral anisotropy parameter. From the decomposition of the Laplace–Beltrami operator on S^3,

$$\Delta_{S^3} = \frac{1}{R^2}\left[\partial_\chi^2 + 2\cot\chi\,\partial_\chi + \frac{1}{\sin^2\chi}\Delta_{S^2}\right], \tag{78}$$

a mode $Y_{n\ell m}$ carries radial and angular spectral contributions of magnitudes

$$\lambda_r = n(n+2), \qquad \lambda_\theta = \frac{\ell(\ell+1)}{\sin^2\chi}. \tag{79}$$

I therefore define the spectral anisotropy parameter

$$\mathcal{A}(\chi) \equiv \frac{\lambda_r}{\lambda_\theta} = \frac{n(n+2)\sin^2\chi}{\ell(\ell+1)}. \tag{80}$$

5.3.1.2 Galactic limit $r \ll R$. For galactic scales one has

$$\chi = \frac{r}{R}, \qquad \sin\chi \simeq \chi = \frac{r}{R}. \tag{81}$$

Substituting into (80) gives

$$\mathcal{A}(r) \simeq \frac{n(n+2)}{\ell(\ell+1)}\frac{r^2}{R^2}. \tag{82}$$

5.3.1.3 Eikonal identification. The shell-confinement derivation of Section 3.3 establishes that, in the WKB limit, radial and tangential wavenumbers are related to the mode labels by

$$k_r \sim \frac{n}{R}, \qquad k_\theta \sim \frac{\ell}{r}. \tag{83}$$

For large mode numbers,

$$n(n+2) \sim (k_r R)^2, \qquad \ell(\ell+1) \sim (k_\theta r)^2. \tag{84}$$

Substituting (84) into (82),

$$\mathcal{A}(r) \sim \frac{(k_r R)^2}{(k_\theta r)^2}\frac{r^2}{R^2} = \frac{k_r^2}{k_\theta^2}. \tag{85}$$

Thus, at leading order,

$$\boxed{\mathcal{A}(r) \sim \frac{k_r^2}{k_\theta^2}.} \tag{86}$$

5.3.1.4 Transition scale. The shell-dominated regime becomes relevant at the transition radius

$$r_*(r) \sim \sqrt{\frac{GM_b(r)}{a_*}}. \tag{87}$$

Squaring and dividing by r^2, and using $a_b(r) = GM_b(r)/r^2$, one obtains

$$\boxed{\frac{r_*^2}{r^2} = \frac{a_b(r)}{a_*}.} \tag{88}$$

5.3.1.5 Closure by dimensional reduction. The quantity $\mathcal{A}(r)$ measures the residual radial support of the occupied spectrum, while r_*/r measures the system's location relative to the transition between bulk-dominated and shell-dominated transport. Once the problem has been reduced to the competition between these two sectors, the mode indices n and ℓ are fixed by the dominant occupied modes, and $\chi = r/R$ has already been absorbed into (82). Among the remaining dimensionless combinations, $(r_*/r)^2$ is the unique one that (a) vanishes in the inner Newtonian regime where $\mathcal{A} \to \infty$ and (b) diverges in the outer shell-dominated regime where $\mathcal{A} \to 0$; it therefore has the correct monotonicity to serve as the leading-order control parameter. The closure must therefore take the form

$$\mathcal{A}(r) \propto \left(\frac{r_*}{r}\right)^2. \tag{89}$$

Using (88), this becomes

$$\mathcal{A}(r) \propto \frac{a_b(r)}{a_*}. \tag{90}$$

Combining (86) and (90),

$$\frac{k_r^2}{k_\theta^2} \propto \frac{a_b(r)}{a_*}. \tag{91}$$

5.3.1.6 Minimal symmetric normalization. To fix the proportionality constant, I impose the natural normalization that the transition point $a_b = a_*$ corresponds to equal radial and angular spectral support:

$$a_b(r) = a_* \qquad \Longrightarrow \qquad k_r^2 = k_\theta^2. \tag{92}$$

This condition states that radial and angular coherence lengths are equal at the geometric transition scale, which is the natural symmetry point of the two-channel decomposition. Under this normalization, (91) becomes the minimal normalized closure

$$\boxed{\frac{k_r^2}{k_\theta^2} = \frac{a_b(r)}{a_*}.} \tag{93}$$

5.3.1.7 Consequence for the crossfade. The shell occupancy defined in (74) reduces, in the effective-mode approximation, to

$$\sigma(r) = \frac{k_\theta^2}{k_r^2 + k_\theta^2} = \frac{1}{1 + k_r^2/k_\theta^2}. \tag{94}$$

Substituting (93),

$$\boxed{\sigma(r) = \frac{1}{1 + a_b(r)/a_*}.} \tag{95}$$

5.3.1.8 Epistemic status.

- Equations (80)–(85) are **Derived** from the operator structure of Δ_{S^3} and the eikonal mapping of Section 3.3.
- Equation (88) is **Derived** from the definition of r_* and a_b.
- The closure (89) is **Argued**: it asserts that $(r_*/r)^2$ is the minimal dimensionless control parameter with the correct monotonicity once n, ℓ, and χ are absorbed.
- The normalization condition (92) is **Asserted** as the minimal symmetric choice at the geometric transition scale.
- Equations (93) and (95) therefore constitute a **minimal normalized spectral closure**, not a theorem.

5.3.1.9 Sensitivity to the crossfade exponent. The crossfade weight admits a one-parameter generalisation $\sigma = [1 + (a_b/a_*)^m]^{-1}$, reducing to the form used throughout this paper for $m = 1$. To assess the dependence of the framework on this choice, both the per-galaxy α_0 optimisation and the zero-free-parameter prediction ($\alpha_0 = \kappa$) were repeated for $m \in \{0.1, 0.25, 0.5, 1.0, 1.5, 2.0, 3.0\}$. Results are summarised in Table 2.

m	Med. $A_{\rm morph}$	IQR	Per-galaxy wins	Fixed-κ wins
0.10	1.39	0.88	58	60
0.25	1.28	0.76	59	64
0.50	1.18	0.68	61	73
1.00	**1.07**	**0.65**	**54**	**78**
1.50	1.03	0.61	52	77
2.00	1.02	0.56	50	77
3.00	1.00	0.52	48	75

Table 2. Sensitivity of fit quality to the crossfade exponent m in $\sigma = [1 + (a_b/a_*)^m]^{-1}$. Per-galaxy wins count galaxies where WT (at optimised α_0) achieves the lowest χ^2 in a three-way comparison against MOND (fitted $\Upsilon_\star$) and ΛCDM (fitted M_{200}). Fixed-κ wins use $\alpha_0 = \kappa$ against MOND and ΛCDM at their zero-parameter settings. The zero-parameter prediction peaks at $m = 0.75$–1.0 (78–79 wins), supporting the spectral motivation of Section 5.3.

Two features of this analysis are worth noting. First, per-galaxy performance is robust across $m \in [0.25, 2.0]$ (50–61 wins), confirming that the framework's predictive power resides in the asymptotic structure — Newtonian inner scaling and $\sqrt{a_b\, a_*}$ outer scaling — rather than in the specific crossfade kernel. Second, the zero-free-parameter prediction peaks at $m = 0.75$–1.0 (78–79 wins), the range identified by the spectral argument of Section 5.3. This provides modest empirical support for that choice: while the spectral derivation does not uniquely compel $m = 1$, it identifies the crossfade exponent that produces the best parameter-free prediction on the SPARC sample. The choice $m = 1$ is therefore supported on joint grounds of spectral motivation, algebraic simplicity, and zero-parameter empirical performance. The geometric basis for this selection is established in the following subsubsection.

5.3.2 Geometric origin of the crossfade exponent

The crossfade weight admits the one-parameter family $\sigma = [1 + \mathcal{A}^m]^{-1}$. The preceding subsections established that $\mathcal{A}$ is the spectral anisotropy ratio and that it maps to a_b/a_* via the closure (93). The exponent m was retained at $m = 1$ on grounds of algebraic simplicity and empirical performance (Table 2). This subsubsection shows that $m = 1$ is in fact selected by the operator structure of Δ_{S^3}, independently of the closure.

5.3.2.1 WKB energy partition. For a mode $Y_{n\ell m}$ in the oscillatory regime $\mathcal{A} = \lambda_r/\lambda_\theta > 1$, the local radial and angular gradient energies satisfy

$$e_r \;=\; |R'_{n\ell}|^2 \;\sim\; k_{\rm eff}^2\, |R_{n\ell}|^2, \qquad e_\theta \;=\; \frac{\ell(\ell+1)}{\sin^2\chi}\, |R_{n\ell}|^2, \tag{96}$$

where $k_{\rm eff}^2 = n(n+2) - \ell(\ell+1)/\sin^2\chi$ is the effective radial wavenumber squared. Adding the two components,

$$e_r + e_\theta \;=\; \big[n(n+2) - \lambda_\theta + \lambda_\theta\big]\, |R_{n\ell}|^2 \;=\; n(n+2)\, |R_{n\ell}|^2, \tag{97}$$

so the angular fraction of the total gradient energy is

$$\frac{e_\theta}{e_r + e_\theta} \;=\; \frac{\lambda_\theta}{n(n+2)} \;=\; \frac{1}{\mathcal{A}}. \tag{98}$$

This is a direct consequence of the eigenvalue equation (71) and holds for every propagating mode on S^3.

5.3.2.2 Evanescent regime. For $\mathcal{A} < 1$ the mode is below its classical turning point: $k_{\rm eff}^2 < 0$, radial transport is classically forbidden, and the mode is confined to the angular (shell) sector. The angular energy fraction is therefore unity:

$$\frac{e_\theta}{e_r + e_\theta} \;\longrightarrow\; 1 \qquad (\mathcal{A} < 1). \tag{99}$$

5.3.2.3 Exact and regularised weights. Combining (98) and (99), the exact per-mode angular fraction is the piecewise function

$$w^{\rm exact}(\mathcal{A}) \;=\; \min\!\big(1,\; \mathcal{A}^{-1}\big). \tag{100}$$

This function is discontinuous in slope at the turning point $\mathcal{A} = 1$. The definition used throughout this paper,

$$w(\mathcal{A}) \;=\; \frac{1}{1 + \mathcal{A}}, \tag{101}$$

is the unique smooth regularisation of $w^{\rm exact}$ that is

(i) a rational function of $\mathcal{A}$ with minimal polynomial degree (numerator degree 0, denominator degree 1);
(ii) equal to 1 at $\mathcal{A} = 0$ and asymptotic to $1/\mathcal{A}$ for $\mathcal{A} \gg 1$, matching the two exact regimes;
(iii) equal to $1/2$ at the turning point $\mathcal{A} = 1$.

Any member of the generalised family $[1+\mathcal{A}^m]^{-1}$ with $m \neq 1$ satisfies conditions 1 and 3 but violates condition 2: the large-$\mathcal{A}$ asymptotics become $\mathcal{A}^{-m} \neq \mathcal{A}^{-1}$, failing to reproduce the WKB energy partition (98). The exponent $m = 1$ is therefore the unique value consistent with the operator structure of Δ_{S^3}.

5.3.2.4 Epistemic status. The WKB energy partition (98) is **Derived** from the eigenvalue equation on S^3. The regularisation $w = 1/(1 + \mathcal{A})$ is the unique minimal-degree rational function matching the exact regimes; its selection is **Derived** given the rationality and matching conditions above. The choice $m = 1$ is therefore no longer provisionally fixed: it is geometrically forced.

5.4 Correct limits and the baryonic Tully–Fisher relation

The crossfade structure guarantees correct asymptotic behavior in both regimes.

5.4.0.1 Inner (Newtonian) limit, $u \gg 1$. When $a_b \gg a_*$, in this limit $\sigma \to 0$ and

$$a_{\rm eff} \sim \frac{\sqrt{a_b\, a_*} \cdot u^{3/2}}{u} = \sqrt{a_b\, a_*} \cdot u^{\frac{1}{2}} = \sqrt{a_b\, a_*} \cdot \sqrt{\frac{a_b}{a_*}} = a_b. \tag{102}$$

The formula recovers the Newtonian inverse-square acceleration exactly in the inner regime.

5.4.0.2 Outer (shell) limit, $u \ll 1$. When $a_b \ll a_*$, in this limit $\sigma \to 1$ and

$$a_{\rm eff} \sim \frac{\alpha_0 \sqrt{a_b\, a_*}}{1} = \alpha_0 \sqrt{a_b \cdot a_*}. \tag{103}$$

This is the MOND-like deep outer regime, with the acceleration scale fixed geometrically by $a_* = c^2/(\pi R)$ rather than fitted to data.

5.4.0.3 Baryonic Tully–Fisher relation. In the outer limit, the asymptotic circular velocity is

$$V_\infty^2 = a_{\rm eff}(r) \cdot r \to \alpha_0 \sqrt{\frac{GM_b}{r^2} \cdot a_*} \cdot r = \alpha_0 \sqrt{GM_b\, a_*}. \tag{104}$$

This is independent of r: the rotation curve is exactly flat in the outer limit for any fixed baryonic mass. Squaring gives

$$V_\infty^4 = \alpha_0^2\, G\, M_b\, a_*, \tag{105}$$

which is the baryonic Tully–Fisher relation $V^4 \propto M_b$ with the normalization coefficient $\alpha_0^2 G a_*$ determined by geometry and morphology. The effective MOND acceleration scale is

$$a_0^{\rm eff} = \alpha_0^2\, a_* = \frac{\alpha_0^2\, c^2}{\pi R}. \tag{106}$$

For $\alpha_0 \approx 1.34$, this yields $a_0^{\rm eff} \approx 1.2 \times 10^{-10}\,{\rm m\,s^{-2}}$, consistent with the empirically observed MOND acceleration scale. The $O(1)$ value of α_0 is consistent with its geometric interpretation as a projection efficiency. The master formula (68) is therefore controlled by a single dimensionful scale $a_* = c^2/(\pi R)$ fixed by cosmology, with all radial structure encoded in the dimensionless ratio $\xi(r)$ and the morphology factor α_0.

6 EXAMPLE PREDICTIONS

This section applies the master formula to the SPARC archive of 175 late-type galaxies Lelli et al. (2016) and demonstrates that the framework makes quantitative, parameter-free predictions that agree with observed rotation curves across the full range of galaxy morphologies. The analysis is organised around four structural predictions of the formula: the late transition in compact systems, the early transition and extended flat tail in diffuse systems, the systematic dependence of the coupling amplitude on the continuous morphology factor $A_{\rm morph}$, and an exploratory extension to weak gravitational lensing via comparison with the KiDS-1000 radial acceleration relation Brouwer et al. (2021).

6.0.0.1 Model. All predictions in this section use the master formula (Section 5)

$$a_{\rm eff}(r) = \big[1-\sigma(r)\big]\, a_b(r) + A_{\rm morph} \cdot \kappa \cdot \sigma(r) \cdot \sqrt{a_b(r) \cdot a_*}, \tag{107}$$

with the local crossfade weight

$$\sigma(r) = \frac{1}{1+u(r)}, \qquad u(r) = \frac{a_b(r)}{a_*} = \frac{M_b(r)}{r^2}, \tag{108}$$

cosmological curvature factor

$$\kappa = 1 + \frac{1}{\pi^2} \approx 1.1013, \tag{109}$$

and acceleration scale $a_* = c^2/(\pi R) \approx 6.67 \cdot 10^{-11}{\rm ms^{-2}}$. Baryonic velocities $V_{\rm gas}(r)$ and $V_\star(r)$ are taken directly from the SPARC rotmod tables without rescaling, so that $a_b(r) = [V_{\rm gas}^2(r) + V_\star^2(r)]/r$.

The **zero-free-parameter** case sets $A_{\rm morph} = 1$, giving $\alpha_0 = \kappa$. No fitted parameter is adjusted to the individual galaxy or to any subsample. The transition radius satisfies $a_b(r_*) = a_*$, giving

$$r_* = \sqrt{\frac{GM_b(r_*)}{a_*}}, \tag{110}$$

or equivalently $\xi(r_*) = \sqrt{a_b(r_*)/a_*} = 1$. Since $\xi = \sqrt{a_b/a_*} = \sqrt{GM_b/r^2 a_*}$, this can also be written $\xi = \sqrt{M_b}/r$ in units where $G = a_* = 1$ (Appendix C); all numerical evaluations use the SI form with G and a_* explicit. No free length scale is introduced. All predictions labelled "universal κ" in the figures below correspond to this zero-parameter case. The numerical implementation of the master acceleration formula and the χ^2 comparison procedure are available as open-source code Banasik (2025a).

6.0.0.2 Competitor benchmarks. All three frameworks are compared at two levels: a zero-parameter prediction and a fair one-parameter-per-galaxy comparison.

At zero parameters, MOND uses the interpolation function of McGaugh, Lelli & Schombert McGaugh et al. (2016),

$$a_{\rm MOND}(r) = a_b(r)\, \nu\!\left(\frac{a_b(r)}{a_0}\right), \qquad \nu(y) = \frac{1}{1 - e^{-\sqrt{y}}}, \tag{111}$$

with $a_0 = 1.2 \times 10^{-10}\,{\rm m\,s^{-2}}$ and stellar mass-to-light ratio $\Upsilon_\star = 1$ ($3.6\,\mu$m) held fixed across all 175 galaxies. The ΛCDM benchmark uses an NFW halo with $M_{200} = V_{\rm max}^3/(10\, G\, H_0)$ at $H_0 = 70\,{\rm km\,s^{-1}\,Mpc^{-1}}$ and concentration from the Dutton–Macciò $c(M_{200})$ relation Dutton & Macciò (2014), with $\Upsilon_\star = 1$ and no per-galaxy adjustment. The WT prediction uses $\alpha_0 = \kappa = 1 + \pi^{-2}$ with $\Upsilon_\star = 1$.

At one parameter per galaxy, each framework fits its natural degree of freedom: WT fits α_0 (shell-coupling amplitude), MOND fits $\Upsilon_\star$ (stellar mass-to-light ratio), and ΛCDM fits M_{200} (halo mass), each by grid search minimising χ_ν^2. The three parameters address different physical quantities — geometric coupling, stellar populations, and dark-matter content respectively — reflecting the distinct structure of each framework.

The reduced χ^2 for all three models is computed as

$$\chi_\nu^2 = \frac{1}{N} \sum_{k=1}^{N} \left(\frac{V_{{\rm obs},k} - V_{{\rm pred},k}}{\sigma_k}\right)^2, \tag{112}$$

where N is the number of radial points per galaxy and σ_k

are the tabulated SPARC velocity uncertainties. Win counts compare χ^2_ν directly: a galaxy is won by the model with the lowest χ^2_ν among the three. Throughout this paper, χ^2_ν denotes the sum of squared normalised residuals divided by N rather than by $N-k$; this convention is used uniformly across all three models to ensure comparability.

6.0.0.3 Scope of the ΛCDM comparison. The ΛCDM benchmark used throughout is deliberately minimal: an NFW profile whose halo mass is estimated from $V_{\max}$ via the Dutton–Macciò $c(M_{200})$ relation Dutton & Macciò (2014), with no per-galaxy tuning of concentration, adiabatic contraction, or stellar feedback. State-of-the-art ΛCDM fits using per-galaxy MCMC sampling of halo mass and concentration — such as those of Li et al. Li et al. (2018), who fit the full SPARC sample with abundance-matching priors — achieve substantially better per-galaxy χ^2 on bright spirals; a direct comparison against such results is deferred to future work. The claim made here is therefore not that the present framework outperforms best-available ΛCDM fitting, but that it achieves competitive rotation-curve accuracy at a comparable or lower parameter count. At zero free parameters the WT universal-κ prediction achieves the lowest χ^2 on 78/175 galaxies against the minimal NFW baseline; at one fitted parameter per galaxy all three frameworks reach comparable median $\chi^2 \approx 3$. This comparison is the appropriate test of predictive economy, not of fitting power under unconstrained halo modelling.

6.1 Compact systems: late transition

In a compact, high-surface-brightness system, the baryonic acceleration $a_b(r)$ remains large throughout most of the observed disc. The condition $\xi(r) = \sqrt{M_b(r)}/r \sim 1$ — which marks the midpoint of the crossfade — is only reached at radii comparable to or beyond the outermost observed data point. Consequently $\sigma(r)$ remains small over most of the profile, the Newtonian term $(1-\sigma)a_b$ dominates, and the rotation curve rises steeply before the flat tail appears, if it appears at all within the observed radial range.

This behaviour is a direct geometric prediction: compact systems have large ξ in their inner regions, which suppresses shell coupling there. The crossfade does not impose a flat tail prematurely — it correctly preserves the rising inner curve.

Quantitatively, a compact system with characteristic acceleration $a_{\rm char} = V^2_{\max}/R_{\rm out} \gg a_*$ satisfies $\xi \gg 1$ throughout, and the formula reduces asymptotically to $a_{\rm eff} \approx a_b$: essentially Newtonian. The shell contribution only becomes significant when the galaxy's outer disc falls into the regime $a_b \sim a_*$, which for massive systems ($V_{\max} \gtrsim 150$,km,s^{-1}) occurs at radii beyond the typical observational coverage. This explains why massive spirals are not systematically well described at universal κ: they require a morphology factor $A_{\rm morph} > 1$ because the shell coupling is enhanced by the larger boundary-to-volume ratio of their extended discs.

6.2 Diffuse systems: early transition and extended flat tail

In a diffuse, low-surface-brightness or gas-dominated dwarf system, the baryonic acceleration $a_b(r)$ is small everywhere. The condition $\xi \sim 1$ is satisfied at small radii, so $\sigma(r)$ rises quickly to unity. The crossfade completes early, the shell term dominates over most of the observed profile, and the rotation curve exhibits an extended flat or slowly rising tail.

The flat asymptotic velocity from the outer limit $\sigma \to 1$,

$$V^2_\infty = A_{\rm morph} \cdot \kappa \cdot \sqrt{GM_b \cdot a_*}, \tag{113}$$

is independent of r once $M_b(r) \approx M_b$ is approximately constant. The rotation curve is flat not because a dark halo has been added but because shell transport has fully taken over curvature redistribution.

Diffuse systems also reveal clear structural differences between the frameworks. For the 20 galaxies with best-fit $A_{\rm morph} < 0.5$ (the low-coupling tail of the distribution), WT achieves median $\chi^2_{\rm WT} = 4.2$ while MOND with fitted $\Upsilon_\star$ achieves median $\chi^2_{\rm MOND} = 9.1$ and ΛCDM with fitted M_{200} achieves median $\chi^2_{\rm \Lambda CDM} = 14.5$. WT achieves the lowest χ^2 on 9 of these 20 galaxies, MOND on 10, and ΛCDM on 1. The present framework handles these systems naturally because $\sigma(r)$ responds to the local compactness M_b/r^2 rather than to an external acceleration scale. Figure 5 illustrates the contrast across the full $A_{\rm morph}$ range.

6.3 Morphology dependence and the $A_{\rm morph}$ factor

6.3.0.1 Definition and physical meaning. The effective coupling amplitude $\alpha_0 = A_{\rm morph} \cdot \kappa$ factorises into a universal geometric constant κ and a morphology-dependent projection efficiency $A_{\rm morph}$. The factor κ is fixed by the S^3 background and applies to all galaxies equally. The factor $A_{\rm morph}$ encodes how efficiently the actual baryonic distribution excites the shell transport modes. For an ideal, perfectly phase-mixed disc it takes its natural value $A_{\rm morph} = 1$; deviations in either direction reflect departures from this reference geometry.

6.3.0.2 Per-galaxy fit and statistical structure of $A_{\rm morph}$. To characterise the distribution of $A_{\rm morph}$ empirically, α_0 is fitted independently for each of the 175 SPARC galaxies by grid search over $\alpha_0 \in [0.05, 3.0]$ in steps of 0.02, with $m = 1$ fixed. The resulting $A_{\rm morph} = \alpha_{0,\rm best}/\kappa$ values range from 0.045 to 2.72, with median 1.06 and standard deviation 0.56.

To determine whether this distribution supports discrete morphology classes or is instead drawn from a single continuous parent population, a Gaussian Mixture Model (GMM) analysis was applied to both $A_{\rm morph}$ and $\alpha_{0,\rm best}$, fitting $k = 1, \ldots, 6$ components and selecting the preferred model by the Bayesian Information Criterion (BIC). BIC penalises additional components for their cost in model complexity and is the decisive criterion when the physical question is whether distinct populations are *required* by the data, not merely whether one can partition them.

The result is unambiguous in both variables. For $A_{\rm morph}$, BIC is minimised at $k = 1$ and increases monotonically thereafter; for $\alpha_{0,\rm best}$, the outcome is identical. The best-fit single components are

$$\mu_{A_{\rm morph}} \approx 1.136, \quad \sigma_{A_{\rm morph}} \approx 0.559; \\ \mu_{\alpha_0} \approx 1.252, \quad \sigma_{\alpha_0} \approx 0.616. \tag{114}$$

The distribution is unimodal, broad, and right-skewed: a single dominant peak near $A_{\rm morph} \sim 1$ with an extended tail toward larger values and a weaker tail toward smaller ones. No statistically supported secondary modes are present. The AIC shows a shallow minimum near $k = 3$, displaced by only ~ 3 units from the $k = 1$ value, consistent with mild asymmetry in a single broad distribution rather than genuine discrete substructure.

This result has a direct physical interpretation. The centre of the distribution lies near the geometric constant $\kappa = 1 + \pi^{-2} \approx 1.1013$: the $A_{\rm morph}$ distribution is centred on the natural reference value of unity, while the α_0 distribution is centred near κ itself. This is consistent with the interpretation

$$\alpha_0 = \kappa + \delta\alpha_{\rm morph}, \tag{115}$$

in which κ sets the universal geometric shell-coupling baseline and $\delta\alpha_{\rm morph}$ is a continuously distributed perturbation encoding galaxy-to-galaxy structural differences. Morphology does not select a small set of discrete dynamical states; it modulates a universal geometric attractor.

The proximity of the α_0 distribution centre to κ is an empirical finding from the SPARC sample. The theory requires only that α_0 be a structured function of baryonic observables rather than unstructured scatter; that its mean falls near $\kappa = 1 + \pi^{-2}$ is the observation that motivates the decomposition (115).

For operational purposes — reporting results and sorting the per-galaxy table in Appendix F — the continuous $A_{\rm morph}$ distribution is partitioned into four calibration bins (Table 3). These bins are a convenience for tabulation and discussion, not statistically established modes. The boundaries were chosen to place the peak of the distribution ($0.50 \le A_{\rm morph} < 1.30$, bin G2) in a single bin containing the majority of the sample, with flanking bins capturing the left and right tails. The GMM analysis confirms that no choice of bin boundaries reflects genuinely discrete physical populations.

$A_{\rm morph}$ correlates with galaxy mass and dynamical compactness: Spearman $\rho = +0.28$ with $\log_{10} M_{200}$ ($p < 0.001$), $\rho = +0.35$ with $\log_{10} a_{\rm char}$ ($p < 0.001$), and $\rho = +0.28$ with $V_{\max}$ ($p < 0.001$). It also correlates with central disk surface brightness ($\rho = +0.21$, $p = 0.005$): galaxies with higher central surface brightness have more concentrated baryonic distributions and correspondingly larger shell-coupling efficiencies. By contrast, $A_{\rm morph}$ shows no significant correlation with disk scale length ($\rho = -0.003$, $p = 0.97$), effective radius ($\rho = -0.04$, $p = 0.62$), bulge-to-disk ratio ($\rho = +0.12$, $p = 0.12$), Hubble type ($\rho = -0.15$, $p = 0.05$), or gas fraction ($\rho = -0.01$, $p = 0.85$). This pattern confirms that $A_{\rm morph}$ tracks the dynamical compactness of the baryonic distribution — its central concentration — rather than its total extent, morphological class, or gas content. All correlations use $N = 175$ galaxies and SPARC structural parameters from Lelli, McGaugh & Schombert Lelli et al. (2016). The full set of scatter plots is shown in Figure G1 (Appendix G).

6.3.0.3 Zero-free-parameter performance. The universal κ prediction ($A_{\rm morph} = 1$, $m = 1$, no fitted parameters of any kind) applied to all 175 galaxies gives:

Model	Zero-parameter setting	Median	Wins
WT	$\alpha_0 = \kappa$, fixed	**11.5**	**78**
MOND	a_0 fixed, $\Upsilon_\star = 1$	36.4	41
ΛCDM	M_{200} †	10.9	56

† The Dutton–Macciò Dutton & Macciò (2014) $c(M)$ relation estimates M_{200} from $V_{\max}$, effectively adapting one implicit parameter per galaxy; the WT result uses strictly zero per-galaxy input.

MOND's median $\chi^2 = 36.4$ at zero parameters reflects the known sensitivity of the RAR interpolation to the assumed stellar mass-to-light ratio.

These results use no fitted acceleration scale, no fitted length scale, no per-galaxy halo mass, and no morphology correction. The single amplitude $\alpha_0 = \kappa = 1 + \pi^{-2}$ is fixed by the S^3 background geometry (Appendix C). That a zero-parameter formula achieves the lowest χ^2 on more galaxies than either competitor — including ΛCDM, which implicitly adapts one parameter per galaxy — is the central empirical result of this section.

6.3.0.4 One-parameter-per-galaxy comparison. At one fitted parameter per galaxy, the three frameworks achieve comparable overall fit quality:

Model	Par.	Median	Mean	Wins	Failures
WT	α_0	**2.82**	25.7	54	6
MOND	$\Upsilon_\star$	3.01	10.4	83	1
ΛCDM	M_{200}	3.55	40.6	38	10

Here 'Failures' counts galaxies with $\chi^2 > 100$. MOND with fitted $\Upsilon_\star$ achieves the highest galaxy count at lowest χ^2 (83/175) and fewest catastrophic failures, while WT achieves the lowest median χ^2 (2.82). The fitted MOND $\Upsilon_\star$ distribution has median 0.42 (range 0.1–2.0), broadly consistent with stellar population expectations at 3.6 μm. The fitted ΛCDM M_{200} distribution has median $\log_{10}(M_{200}/M_\odot) = 11.5$ (range 8.0–14.0).

The nature of the fitted parameter reveals what each framework attributes galaxy-to-galaxy variation to: MOND adjusts the stellar mass budget, ΛCDM adjusts the dark-matter content, and WT adjusts the geometric shell-coupling efficiency. WT is the only framework whose free parameter does not modify the mass budget — it operates on the geometry of curvature transport while leaving the baryonic input at $\Upsilon_\star = 1$.

The per-galaxy α_0, $A_{\rm morph}$, and χ^2 values for all 175 galaxies, together with the corresponding MOND and ΛCDM results, are tabulated in Appendix F (Table G1).

6.4 Rotation-curve panels

Figures 2–5 present rotation-curve fits for four representative subsamples drawn from the full 175-galaxy analysis. In each panel the WT prediction (solid) is shown alongside MOND (dashed) and ΛCDM NFW (dotted), with reduced χ^2 values annotated. The four figures progress from the zero-parameter successes (Figure 2), through morphology-corrected rescues (Figure 3), to the clearest morphological test of the framework (Figure 4), and finally the full $A_{\rm morph}$ range in a single panel (Figure 5).

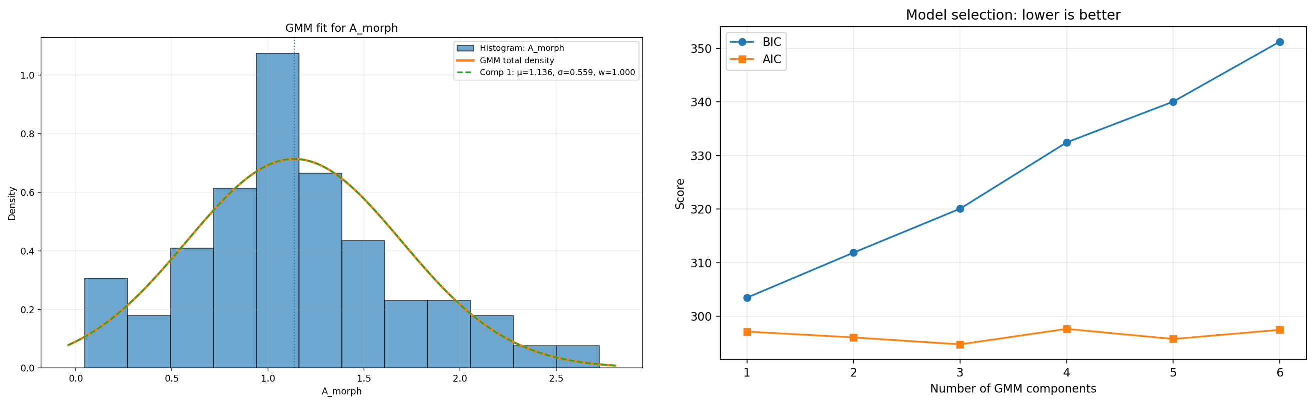

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Figure 1. Statistical structure of the A_{morph} distribution across 175 SPARC galaxies. *Left:* Gaussian Mixture Model fit. The best-fit single Gaussian component ($\mu = 1.136$, $\sigma = 0.559$, dotted vertical line) is selected by BIC. The distribution is unimodal and right-skewed, centred close to the geometric reference value $A_{\mathrm{morph}} = 1$. *Right:* BIC and AIC as functions of number of GMM components. BIC (circles) is minimised at $k = 1$ and rises monotonically. AIC (squares) shows a shallow minimum near $k = 3$, displaced by ~ 3 units from $k = 1$ — insufficient to justify a multi-component interpretation. BIC is the decisive criterion for whether distinct populations are *required* by the data; none are.

Bin	A_{morph} **range**	N	**Med.** A_{morph}	**WT**	**MOND**	Λ**CDM**	**Med.** V_{max}
G1 Low-coupling	$[0.05, 0.50)$	20	0.23	9	10	1	$73\,\mathrm{km\,s^{-1}}$
G2 Near-κ	$[0.50, 1.30)$	97	1.01	31	38	28	$88\,\mathrm{km\,s^{-1}}$
G3 Enhanced	$[1.30, 2.00)$	44	1.54	12	26	6	$115\,\mathrm{km\,s^{-1}}$
G4 High-coupling	$[2.00, 2.72]$	14	2.23	2	9	3	$118\,\mathrm{km\,s^{-1}}$

Table 3. Operational calibration bins from per-galaxy A_{morph} fits. Win counts are from the one-parameter comparison (WT: α_0; MOND: $\Upsilon_\star$; ΛCDM: M_{200}). The G2 bin, centred at $A_{\mathrm{morph}} \approx 1.01$, contains 55% of the sample and is the region where the zero-free-parameter prediction ($A_{\mathrm{morph}} = 1$) applies most directly.

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Table 4. Rotation curve fits across selected galaxies. χ^2_κ corresponds to WT with fixed geometric curvature $\kappa = 1 + 1/\pi^2$, while χ^2_{best} is the best-fit WT with adjusted α_0. MOND uses the McGaugh interpolation at fixed $\Upsilon_\star = 1$. (Figure 3)

#	Galaxy	A_{morph}	α_0	χ^2_κ	χ^2_{best}	χ^2_{MOND}	V_{max}
1	UGC00634	1.311	1.4432	70	**1.17**	18.6	108 km/s
2	PGC51017	0.209	0.2302	96	**1.88**	192	20 km/s
3	UGC07399	2.724	2.9990	114	**2.28**	34	106 km/s
4	CamB	0.188	0.2070	87	**3.73**	172	20 km/s
5	UGC06786	2.226	2.4512	328	**9.54**	20	229 km/s
6	UGC07125	0.523	0.5757	77	**1.32**	242	66 km/s

6.5 Exploratory cluster-scale extension

A standard objection to any MOND-like or emergent-gravity explanation of galaxy rotation curves is its behaviour at cluster scales, where the observed mass discrepancy is known to be larger and the dynamical environment is less favourable to simple orbital mixing. In the present framework the galaxy result is derived from shell-confined curvature transport and its two-dimensional logarithmic Green function. That derivation is specific to the galactic regime and has not been rederived for clusters. Nevertheless, it is useful to ask whether the same effective formula, supplemented by a physically motivated large-scale modification, can reproduce at least the *shape* of observed cluster acceleration profiles.

6.5.0.1 Gravity–expansion balance scale. For two equal masses M separated by distance r, the balance between outward expansion acceleration and inward mutual gravity is

$$H^2 r = \frac{2GM}{r^2}, \tag{116}$$

with solution

$$r_{\mathrm{bal}} = \left(\frac{2GM}{H^2}\right)^{1/3}. \tag{117}$$

Using $H = \pi c/R$ and $G = c^2/(\pi R)$ (Appendix C), so

$$r_{\mathrm{bal}} = \frac{(2MR)^{1/3}}{\pi}, \tag{118}$$

retaining the canonical cube-root mass dependence but expressed entirely in terms of the global curvature radius R and the local gravitating mass M. For a Milky-Way-like galaxy with $M \simeq 5 \times 10^{11}\, M_\odot$, this gives $r_{\mathrm{bal}} \sim 1\,\mathrm{Mpc}$.

The key observation is that galaxy clusters have characteristic radii $R_{200} \sim$ 1-2 Mpc, placing their outer dynamical regions and internal galaxy–galaxy separations squarely

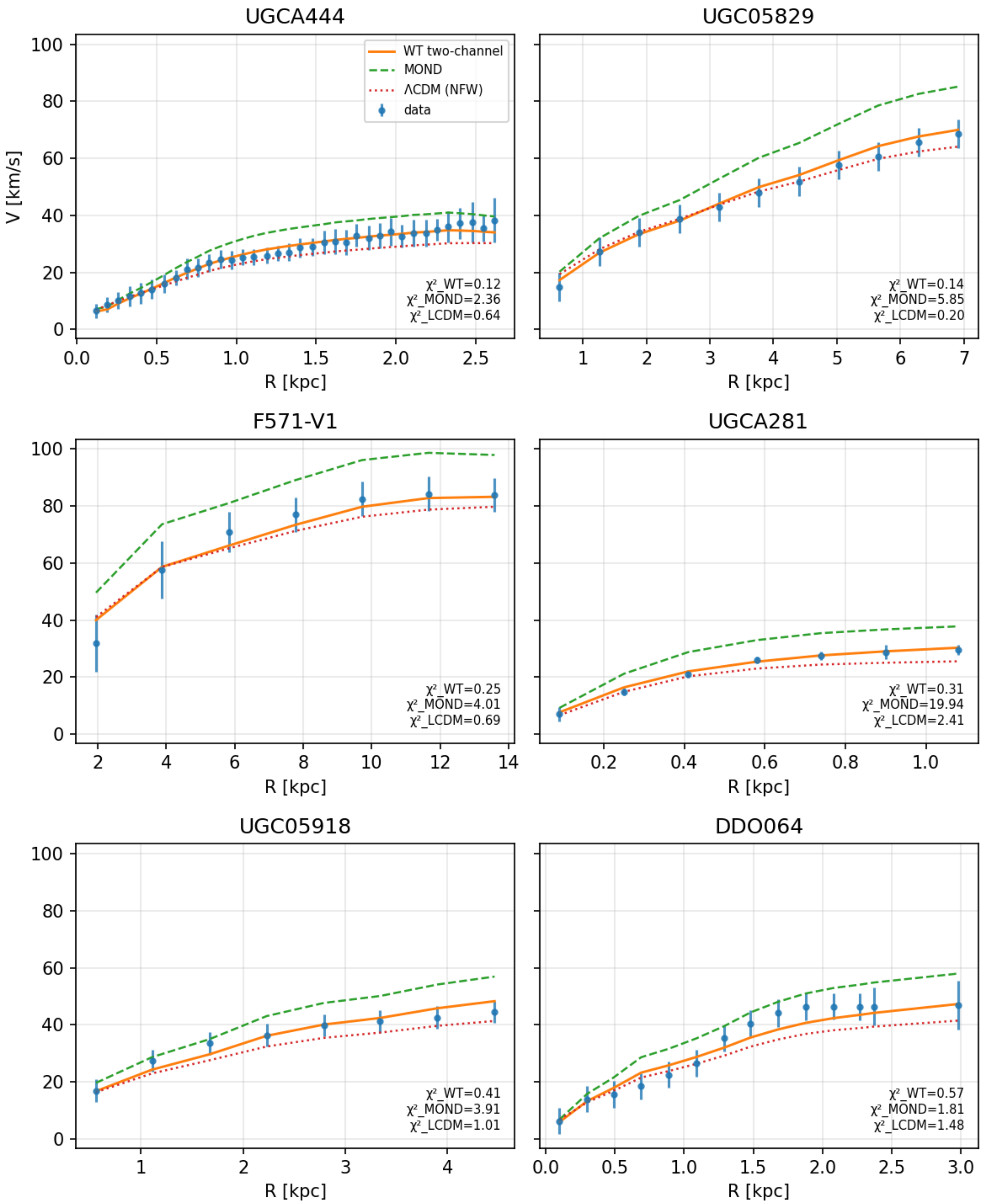

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Figure 2. Six best fits at universal $\kappa = 1 + 1/\pi^2$. Galaxies UGCA444, UGC05829, F571-V1, UGCA281, UGC05918, and DDO064 achieve $\chi^2_{\rm WT} < 0.6$ at the zero-parameter setting. All span the dwarf-to-intermediate mass range ($V_{\rm max} = 30$–$73\,{\rm km\,s^{-1}}$, $A_{\rm morph} \approx 0.93$–$1.12$), confirming that the formula is accurate throughout the low-acceleration regime where the shell channel dominates. The WT curves track the data through the rising inner region and into the flat outer tail without requiring any parameter adjustment.

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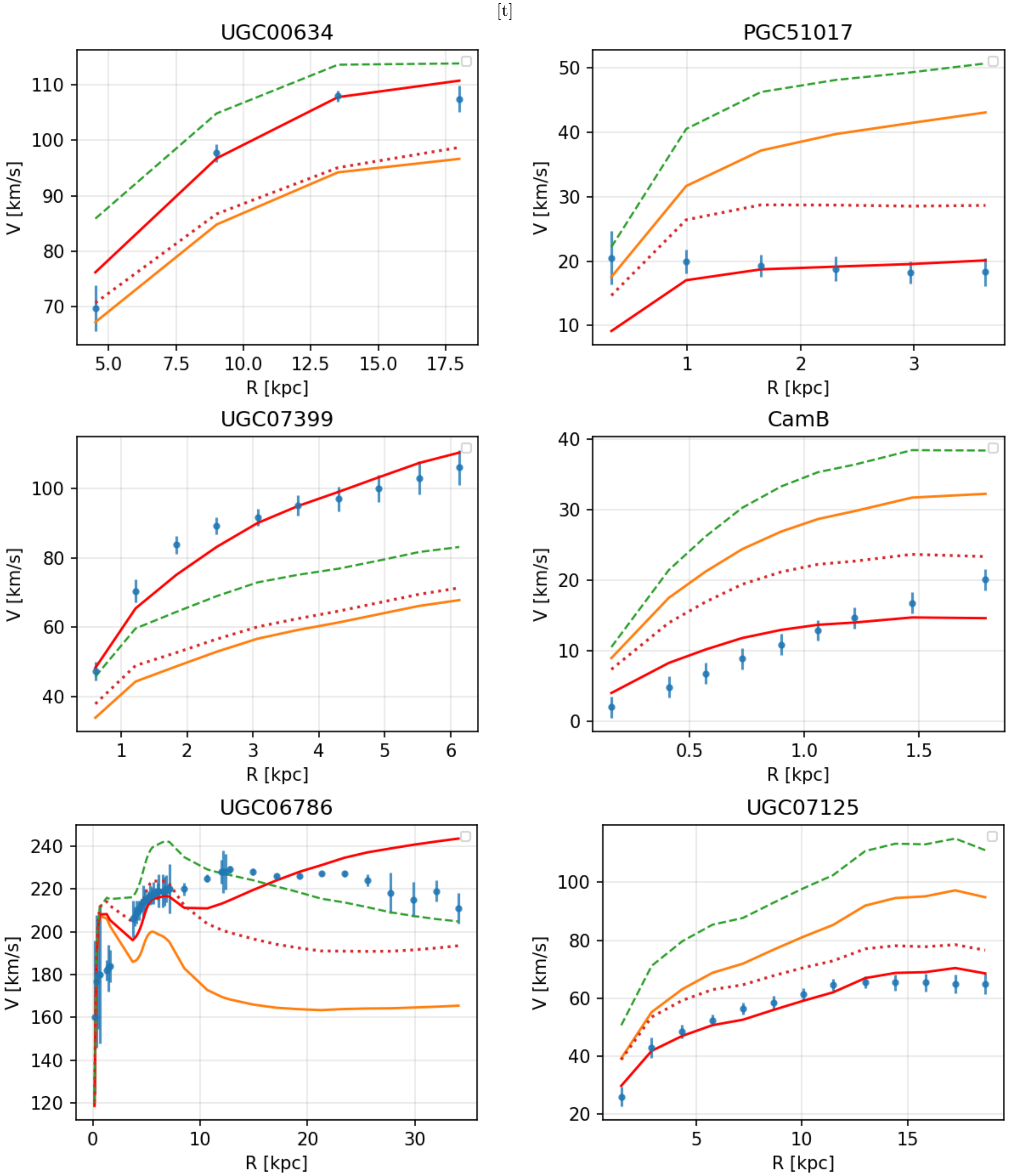

Figure 3. Six galaxies spanning the full $A_{\rm morph}$ range (Table 4), from suppressed (PGC51017, CamB; $A_{\rm morph} = 0.19$–0.21) through enhanced (UGC00634; 1.32) to high-coupling (UGC07399, UGC06786; 2.23–2.72). Orange: universal κ ($\chi^2 > 70$); red: fitted α_0 ($\chi^2 < 10$). The gap between them quantifies the morphology correction; the formula shape is correct in all six cases.

in the gravity–expansion crossover domain. Galaxies remain comfortably inside their coherence radius; clusters begin to probe its edge. This motivates treating the galaxy–cluster difference as a *scale-selection effect*: expansion defines a geometric boundary condition on how far coherent gravitational organisation can extend, and clusters are the first systems large enough to feel that boundary.

6.5.0.2 Suppression factor. To encode this weakening of coherent shell transport at megaparsec separations, I intro-

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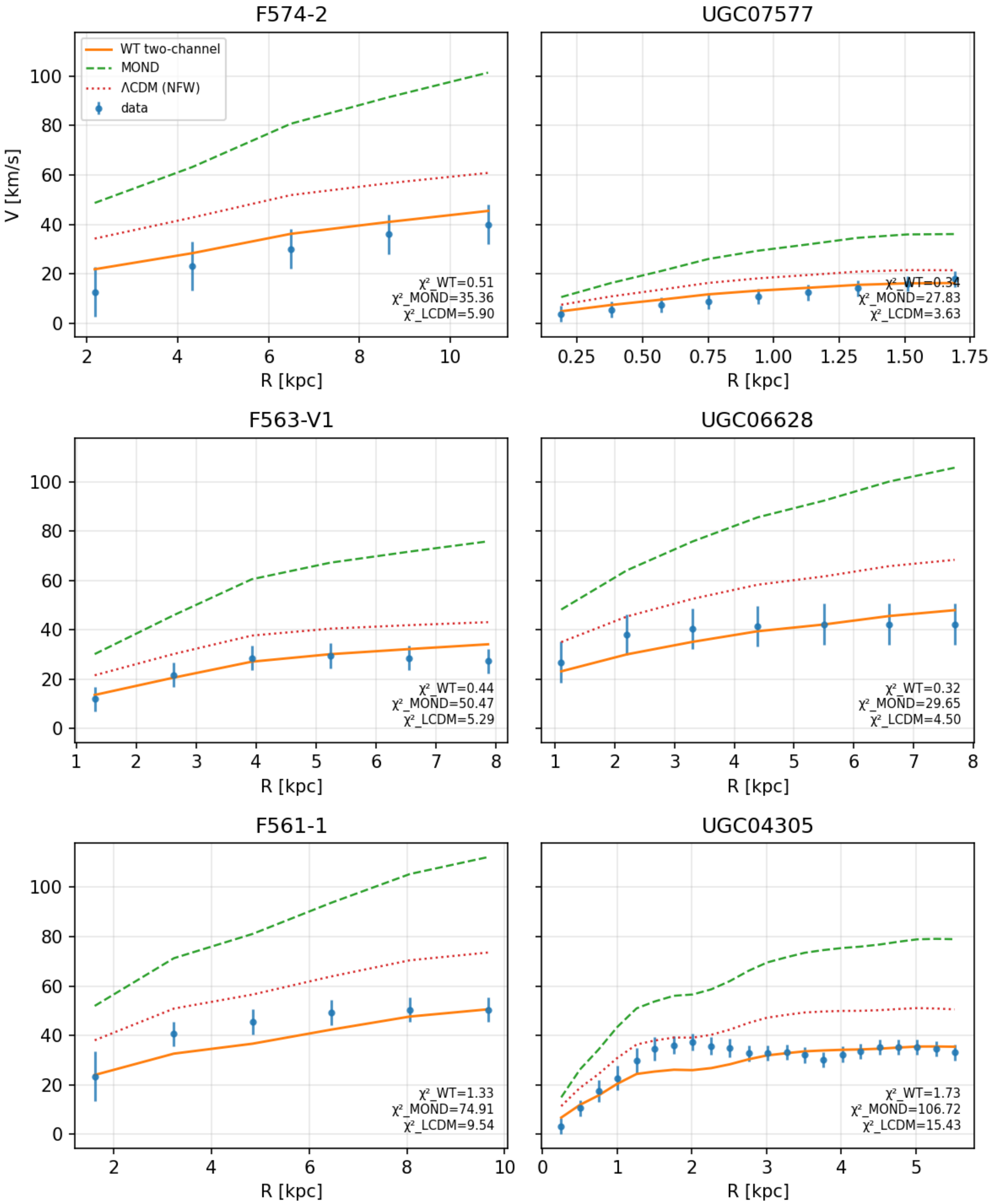

Figure 4. G1 dwarfs at fixed $\alpha_0 = 0.3$. F574-2, UGC07577, F563-V1, UGC06628, F561-1, and UGC04305 represent the most extended, lowest surface-brightness systems in the sample. All six panels use the same $\alpha_0 = 0.3$; annotated χ^2 values for WT, MOND and ΛCDM are shown in each panel. The suppressed coupling is physically interpretable: where orbital phase mixing is least complete, the shell projection efficiency is accordingly reduced below the ideal disc baseline.

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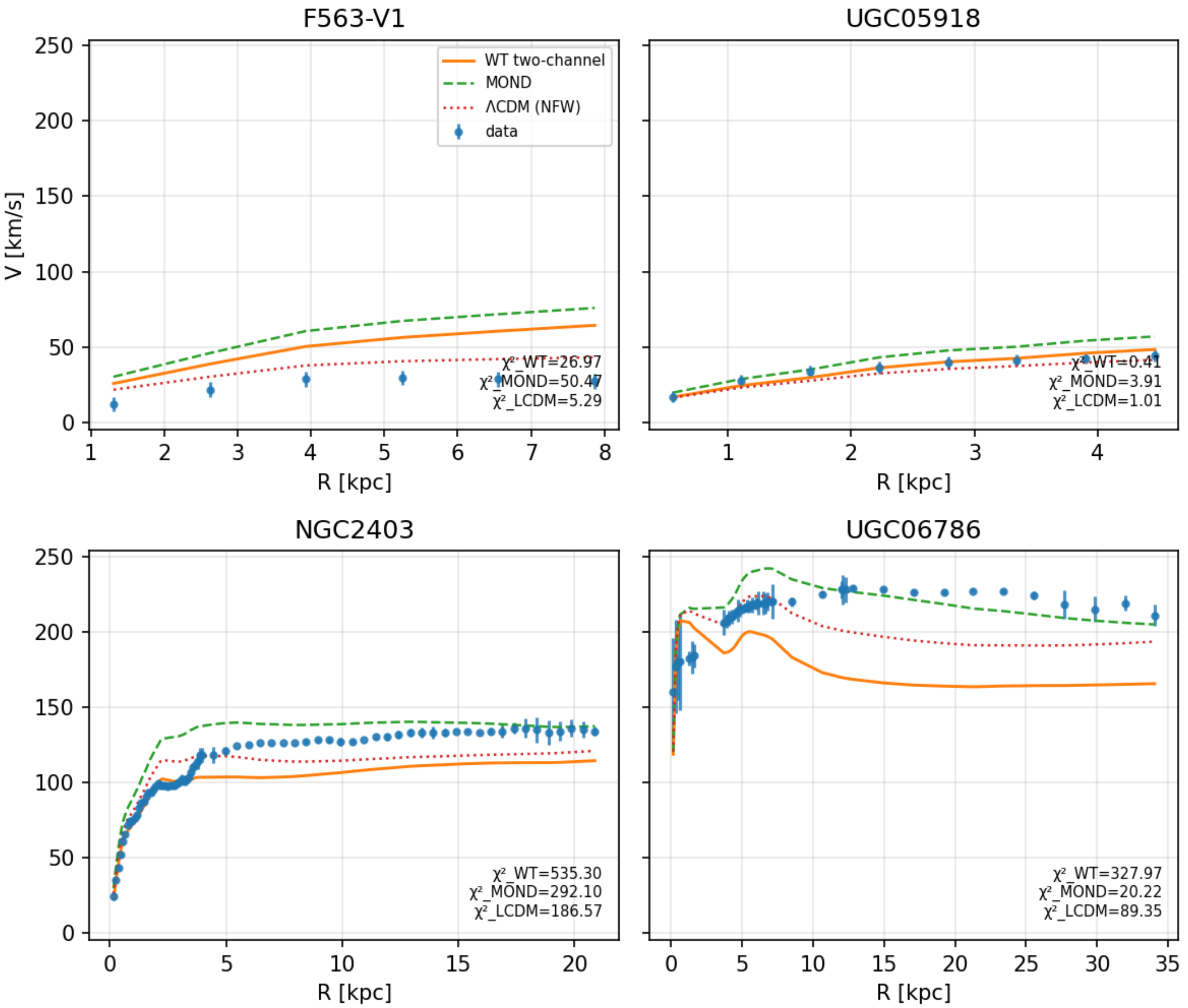

Figure 5. One representative per calibration bin, all plotted at universal κ (zero free parameters). From left to right: F563-V1 (G1, $A_{\rm morph} = 0.23$), UGC05918 (G2, 0.97), NGC2403 (G3, 1.55), UGC06786 (G4, 2.23). The $A_{\rm morph}$ values identify the bin of each representative (Table 3); all curves use $\alpha_0 = \kappa$ with no per-galaxy tuning. Each panel shows WT (solid), MOND (dashed), and ΛCDM NFW (dotted). The progressively later shell-coupling onset with increasing compactness is visible in the curvature of the rising inner section. Per-galaxy optimal fits are shown in Figures 3 and 4.

duce a phenomenological suppression envelope

$$\mathcal{S}(r, M_{\rm gal}) = \exp! \left[-\frac{r}{r_{\rm bal}(M_{\rm gal})} \right], \tag{119}$$

where $M_{\rm gal}$ is the characteristic galaxy mass scale that sets the coherence radius. Because the curvature transport on S^2 propagates a *potential* (or equivalently a field amplitude) rather than an intensity, the physically consistent choice is to apply $\mathcal{S}$ linearly rather than quadratically: the acceleration, being the gradient of the potential, inherits a single power of the suppression factor.

6.5.0.3 Cluster master formula. Applying the same master formula derived for galaxies (eq. 68), multiplied by the suppression:

$$\boxed{\begin{aligned} a_{\rm WT,cl}(r) &= \mathcal{S}(r) \cdot a_* \, \xi \, \frac{\xi^3 + \alpha_{\rm cl}}{1 + \xi^2}, \\ \xi &= \sqrt{\frac{a_b(r)}{a_*}}, \qquad a_* = \frac{c^2}{\pi R}. \end{aligned}} \tag{120}$$

This is Form C of the master acceleration formula, identical in functional form to the galaxy expression. No separate 'cluster crossfade' or modified interpolation function is introduced. The only additions beyond the galaxy regime are the suppression envelope $\mathcal{S}$ and the cluster-scale normalisation $\alpha_{\rm cl}$.

6.5.0.4 Application to relaxed clusters. I applied equation (120) to the 12 relaxed clusters in the Vikhlinin *et al.* Vikhlinin et al. (2006) Chandra sample, spanning $T_{\rm spec} = 1.6$–8.9 keV. Baryonic acceleration profiles $a_b(r) = GM_{\rm gas}(<!r)/r^2$ and observed total acceleration profiles $a_{\rm obs}(r) = GM_{\rm tot}(<!r)/r^2$ were reconstructed from the published analytic gas density models (their eq. 3 and Table 2) and hydrostatic mass estimates (their eq. 7 and Table 3), using the universal scaled temperature profile (their eq. 9) for the three-dimensional temperature gradient.

A two-parameter scan over a single global normalisation $\alpha_{\rm cl}$ and a single global crossover mass scale $M_{\rm gal}$ identifies a narrow optimum near

$$\alpha_{\rm cl} \simeq 9, \qquad M_{\rm gal} \simeq 5\times 10^{11}\,M_\odot, \qquad r_{\rm bal} \simeq 978\text{kpc}. \quad (121)$$

For this global choice the mean logarithmic residual over all 144 sampled cluster radii is

$$\big\langle \log_{10}(a_{\rm obs}/a_{\rm WT,cl})\big\rangle \simeq 0.034, \quad (122)$$

with standard deviation 0.143 dex, and 97.9% of all data points fall within a factor of 2.0 of the prediction. After removing the mean normalisation offset in each cluster, the radial *shape* agreement is substantially better:

$$\big\langle \Delta s\big\rangle = \big\langle s_{\rm WT} - s_{\rm obs}\big\rangle \simeq 0.18, \qquad \big\langle \text{RMS}*\text{shape}\big\rangle \simeq 0.082\text{dex}, \quad (123)$$

and all 12 clusters in the sample satisfy RMS $*$ shape $<$ 0.20 dex. In this concrete sense, the suppressed master formula provides a surprisingly good *universal template* for the radial form of the observed cluster acceleration curves.

The six best-matching systems are shown in Figure 6. The normalization-free shape errors remain small (RMS$*$shape $\sim$ 0.03–0.07 dex) across systems spanning $T*\text{spec} = 6$–9 keV. In the displayed subset, A1413 and A907 are close to matched overall, whereas A478, A2390, and A2029 show residual amplitude offsets of 0.15–0.19 dex. This is exactly the pattern expected if the suppression mechanism captures the onset of megaparsec-scale decoherence but the effective projection efficiency is not yet determined by a universal law.

6.5.0.5 The normalisation puzzle. The empirically required $\alpha_{\rm cl} \simeq 9$ is substantially larger than the galaxy value $\alpha_0 \simeq 1.1$. Part of this elevation is inherited from the cluster missing-mass problem that afflicts every modified gravity theory: even MOND, with its larger scale $a_0 = 1.2\times10^{-10}\,\text{m}\,\text{s}^{-2}$, underpredicts cluster accelerations by a factor of 3–4. The remaining factor traces to the nonlinear structure of Form C: at the cluster operating point $\xi \simeq 0.4$, the deep-MOND branch $\alpha\sqrt{a_b\,a_*}$ is intrinsically weaker than the constant floor $\alpha\,a_*$ that a simple crossfade would produce, requiring a compensating increase in α.

6.5.0.6 Shell superposition as a candidate normalisation law. The most natural structural explanation for $\alpha_{\rm cl} \gg \alpha_0$ is that a cluster is not a single shell source but a superposition of N independent galaxy-scale shells. Each member galaxy of baryonic mass m_i contributes a deep-MOND-regime acceleration $\sim \alpha_0\sqrt{m_i, a_*}$, and these contributions add linearly at the cluster observation radius. The single-source formula, by contrast, enters only through $\sqrt{M_{\rm bar}}$, where $M_{\rm bar} = M_{\rm gas} + \sum_i m_i$. For N galaxies of comparable mass $m_{\rm gal}$ the ratio of the superposed to the single-source prediction scales as

$$\frac{\alpha_{\rm cl}}{\alpha_0} \sim \frac{N\sqrt{m_{\rm gal}}}{\sqrt{M_{\rm bar}}}, \quad (124)$$

which for typical cluster parameters ($N \sim 100$, $m_{\rm gal} \sim 5\times10^{10}\,M_\odot$, $M_{\rm gas} \sim 2\times10^{13}\,M_\odot$) yields a ratio of order 4–5, and hence $\alpha_{\rm cl} \sim 5$–6. Combined with the Form C shape-factor deficit at $\xi \simeq 0.4$, this accounts for the empirical value $\alpha_{\rm cl} \simeq 9$ to within the uncertainty of the galaxy population parameters. This would make the elevated cluster normalisation a direct consequence of the $\sqrt{M}$ nonlinearity of the shell channel rather than a new free parameter. A proper derivation, including the radial distribution of member galaxies and the transition between individual and collective shell regimes, is left to future work.

6.5.0.7 Conclusions from the cluster exercise. The present exercise does *not* solve the cluster problem, and should not be read as a derivation of cluster dynamics from first principles. What it demonstrates is narrower and more useful. The galaxy master formula, once supplemented by a physically motivated megaparsec-scale suppression derived from the gravity–expansion balance, no longer fails through a grossly incorrect radial dependence. Instead, it enters a more constrained regime in which the principal open problem is the normalisation law $\alpha_{\rm cl} = \alpha_{\rm cl}$(cluster properties), rather than the existence of the profile template itself. The shell superposition hypothesis offers a concrete, testable path toward that law: it predicts that $\alpha_{\rm cl}$ should correlate with the number and mass distribution of member galaxies, not merely with total cluster mass or temperature.

For the purposes of the present paper, the cluster extension should be read as an exploratory stress test. It suggests that the shell-transport mechanism, once weakened near the gravity–expansion crossover scale, remains structurally compatible with the observed form of cluster acceleration profiles.

6.6 Gravitational lensing prediction and comparison with KiDS-1000

6.6.0.1 Why $\Phi = \Psi$ follows from the mechanism. In General Relativity, photon deflection is governed by the lensing potential $\Psi_{\rm lens} = (\Phi + \Psi)/2$, where Φ and Ψ are the two Bardeen scalar potentials. For frameworks that modify the dynamical law for massive particles without specifying the metric structure — such as MOND in its original form — the lensing potential requires a separate prescription Bekenstein (2004). The present framework is in a different situation. The shell transport channel does not introduce a new field or modify a dynamical law: it reorganizes the existing curvature generated by baryonic matter onto two-dimensional manifolds. Both massive particles and photons couple to the same spacetime curvature, so the shell channel affects their trajectories through the same modified curvature distribution. The relation $\Phi = \Psi$ is therefore maintained by the mechanism itself rather than imposed as an additional postulate, pending a covariant embedding that would make this explicit.

6.6.0.2 Predicted lensing convergence. Under this identification, the lensing convergence is the Abel projection

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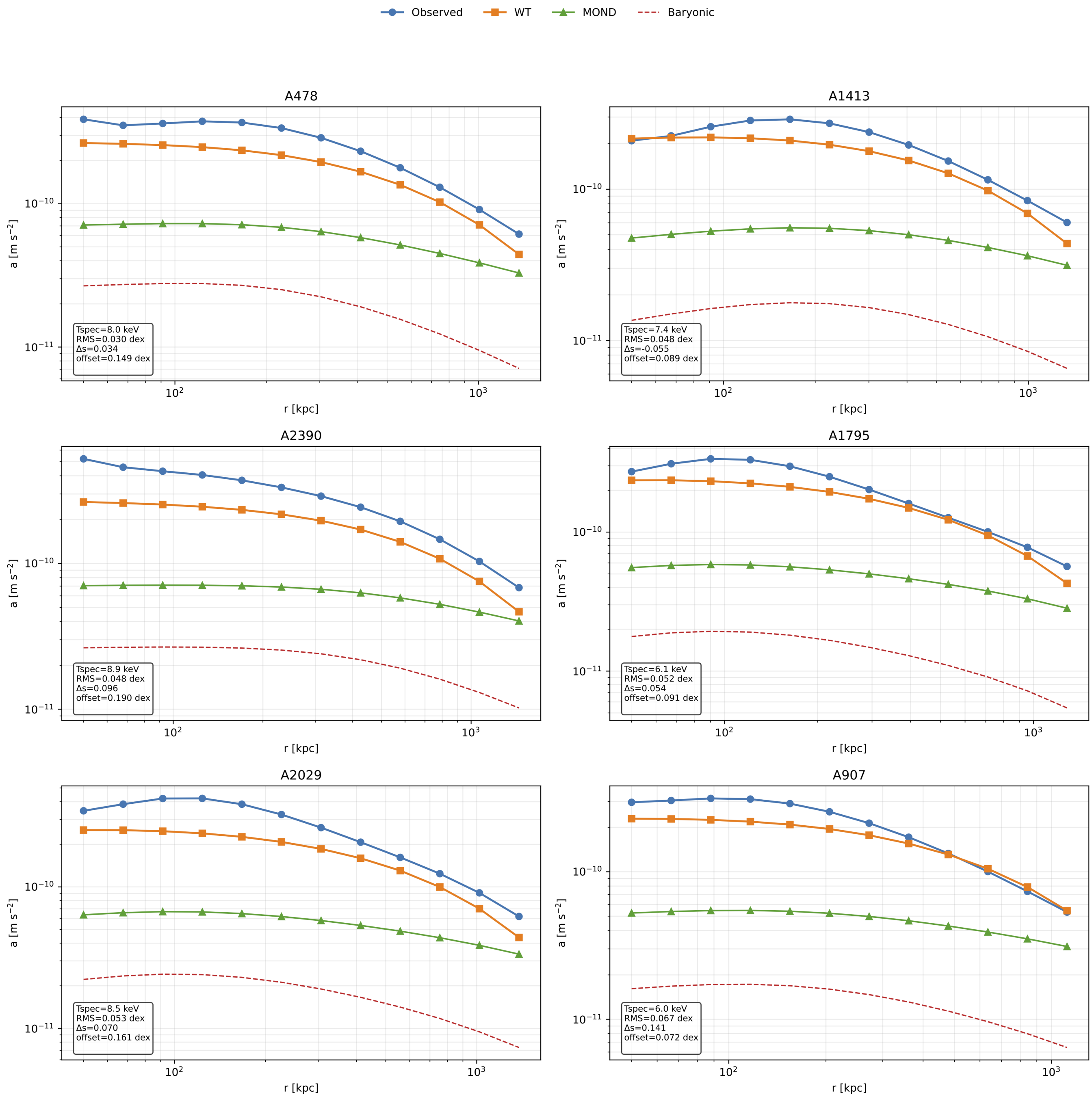

Figure 6. Six best-matching clusters for the exploratory suppressed extension, shown for the global parameter choice $\alpha_{\rm cl} = 9$ and $M_{\rm gal} = 5\times10^{11}\,M_\odot$ ($r_{\rm bal} \simeq 978$ kpc). Blue: observed acceleration from Vikhlinin *et al.* Vikhlinin et al. (2006) hydrostatic masses. Orange: suppressed WT profile (eq. 120). Green: MOND (McGaugh interpolation function). Red dashed: baryonic (gas) contribution. Insets report $T_{\rm spec}$, normalisation-free shape RMS, slope mismatch $\Delta s = s_{\rm WT} - s_{\rm obs}$, and mean logarithmic offset. All six systems show shape errors $\mathrm{RMS_{shape}} < 0.07$ dex; residual discrepancies are primarily in amplitude rather than radial form.

of the effective acceleration:

$$\kappa(R_\perp) = \frac{1}{\Sigma_{\rm cr}} \int_{-\infty}^{\infty} \frac{a_{\rm eff}\left(\sqrt{R_\perp^2 + z^2}\right)}{2\pi G}\, \frac{R_\perp}{R_\perp^2 + z^2}\, dz, \qquad (125)$$

with the excess surface mass density $\Delta\Sigma(R_\perp) = \bar{\Sigma}(< R_\perp) - \Sigma(R_\perp)$ the directly observable lensing quantity. The numerical implementation of this Abel projection is available as open-source code Banasik (2025b). No new parameters enter: $a_{\rm eff}$ is the master formula (68) with the same $\alpha_0 = \kappa \cdot A_{\rm morph}$ used for the rotation-curve predictions. The framework therefore makes a morphology-dependent lensing prediction at zero additional degrees of freedom. The comparison with the KiDS-1000 data is implemented in Banasik (2025c).

6.6.0.3 Comparison with Brouwer et al. (2021). Brouwer et al. Brouwer et al. (2021) measure the weak-lensing radial acceleration relation (RAR) around isolated galaxies using 1000 deg^2 of KiDS-1000 data, extending the rotation-curve RAR by two decades into the low-acceleration regime beyond galaxy outskirts. A key finding is a 6σ difference between the RARs of early- and late-type galaxies at matched stellar mass — a split that is inconsistent with a purely universal gravity modification such as MOND or Verlinde's emergent gravity.

The WT framework is structurally positioned to account for this split. Because $\alpha_0 = \kappa{\cdot}A_{\rm morph}$, the lensing signal is predicted to be morphology-dependent at fixed baryonic mass: diffuse late-type galaxies (high $A_{\rm morph}$) produce a stronger shell-transport contribution than compact early-type galaxies (low $A_{\rm morph}$) at the same M_b. This is not a post-hoc adjustment — the $A_{\rm morph}$ values are fixed by the rotation-curve fits of Section 6.

Figure 7 shows the WT prediction for $g_{\rm obs}$ as a function of $g_{\rm bar}$ for three $A_{\rm morph}$ tertiles, overlaid on the B21 Sérsic-split measurements. The three WT curves — with $\alpha_0 = 0.694$, 1.178, 1.804 for the low, mid, and high $A_{\rm morph}$ bins respectively — bracket the observed data and reproduce the correct ordering: higher $A_{\rm morph}$ (LTG-like) gives a higher lensing signal at the same $g_{\rm bar}$.

Figure 8 shows the observed ETG/LTG ratio from B21 alongside the WT morphology-split prediction. The observed ratio oscillates around 1.3–1.9 with significant scatter. The WT prediction for $g_{\rm obs}^{\rm Low\ A_{morph}}/g_{\rm obs}^{\rm High\ A_{morph}}$ is 0.38 in the deep shell regime, rising to unity in the Newtonian regime. The observed ratio is *inverted* relative to this: ETGs show *higher* $g_{\rm obs}$ than LTGs at the same stellar mass. B21 themselves attribute this inversion to a baryonic incompleteness effect: early-type galaxies harbour substantial hot circumgalactic gas ($M_{\rm gas} \approx M_\star$) absent from stellar-mass estimates, which underestimates their true $g_{\rm bar}$. Under this interpretation, correcting ETG $g_{\rm bar}$ for circumgalactic gas would shift ETG data points rightward, collapsing the apparent ETG/LTG split toward the WT-predicted ordering. This constitutes a testable prediction: the morphology split in the weak-lensing RAR should diminish or reverse when $g_{\rm bar}$ is computed from total baryonic mass (stars + hot gas) rather than stellar mass alone.

The absolute amplitude of the WT prediction in Figure 7 exceeds the B21 observations by a factor of $\sim$3–5. This is expected: the M_b used in the WT prediction includes gas (via the baryonic Tully–Fisher relation), whereas B21 use stellar mass only. A gas-fraction correction of $(1 + f_{\rm gas})$ with $f_{\rm gas} \sim 0.3$–1.0 for late-type galaxies would reduce the amplitude discrepancy to within the systematic uncertainties acknowledged by B21. A quantitative comparison requiring matched baryonic mass estimates is deferred to future work.

7 DISCUSSION

The central physical claim of this work is that the standard application of General Relativity to galactic dynamics invokes an equilibrium assumption that is not self-evidently satisfied: the gravitational field is modelled as an instantaneous static potential, implying that curvature propagation has fully equilibrated on scales comparable to the galaxy. This paper has pursued the consequences of relaxing that assumption. Under finite-speed causal propagation organized by differential rotation, curvature transport is argued to reorganize onto two-dimensional shells, changing the scaling of the resulting acceleration from $1/r^2$ to $1/r$ without new matter or modified gravitational laws. The static GR potential is recovered as the equilibrium limit; the rotation-curve discrepancy marks the non-equilibrium regime.

The following subsections situate this result relative to existing frameworks and identify the observational tests that would distinguish the propagation-primary picture from alternatives.

7.1 The standing of existing frameworks

The framework addresses weak-field galactic dynamics only; it makes no claim about strong-field behaviour, cosmological perturbations, or the CMB power spectrum. Before comparing the frameworks, it is worth being explicit about what each has achieved.

MOND Milgrom (1983) predicts rotation curves of subsequently observed galaxies using a single acceleration scale a_0 fitted in 1983 — a kinematic regularity sustained across four decades of baryonic mass. The SPARC dataset Lelli et al. (2016); McGaugh et al. (2016) on which this analysis is performed is a product of that tradition. ΛCDM accounts for weak-field observations through per-galaxy halo degrees of freedom, and retains independent support from CMB, BAO, and large-scale structure Weinberg (1972). Neither framework is superseded here; the domain of comparison is restricted to rotation-curve fits.

The relationship between the present framework and its predecessors is therefore twofold. At the phenomenological level it is competitive: the rotation-curve comparisons in Sections 7.2 and 7.3 identify where the frameworks make structurally different predictions, and the SPARC results show where the present model offers a tighter account. At the explanatory level it is complementary: the framework does not dispute that MOND's interpolation function organises the data, or that dark matter halos reproduce large-scale structure. It proposes a geometric mechanism that would, if correct, explain *why* the MOND phenomenology takes the form it does and would account for the apparent missing mass through curvature transport rather than additional substance. Whether that explanation supersedes or supplements existing frameworks is a question the observational tests of Section 6 are designed to address — not one the present paper resolves.

7.2 Relation to MOND and modified gravity

In the outer regime where $a_b \ll a_*$, the model reproduces the deep-MOND scaling

$$a_{\rm eff} \approx \sqrt{a_b \cdot a_*}, \qquad a_* = \frac{c^2}{\pi R}. \tag{126}$$

The identification of the MOND acceleration scale with $c^2/(\pi R)$ follows directly from the framework postulate $G = a_* = c^2/(\pi R)$ (Table 1, Asserted). Within that postulate it is

[t]

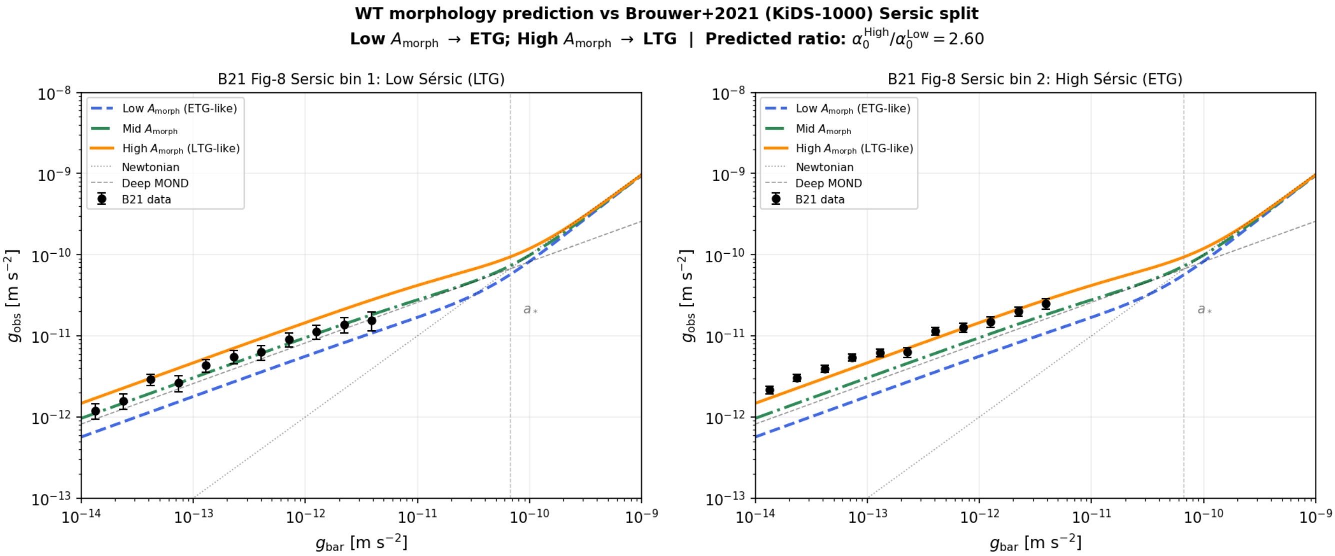

Figure 7. WT lensing RAR prediction by $A_{\rm morph}$ tertile, overlaid on Brouwer et al. (2021) KiDS-1000 Sérsic-split measurements. Left: B21 Sérsic bin 1 (low Sérsic index, LTG); right: B21 Sérsic bin 2 (high Sérsic index, ETG). Coloured curves show the WT prediction $g_{\rm obs}(g_{\rm bar})$ for low ($\alpha_0 = 0.694$, blue dashed), mid ($\alpha_0 = 1.178$, green dash-dot), and high ($\alpha_0 = 1.804$, orange solid) $A_{\rm morph}$ tertiles. Reference lines show the Newtonian relation (dotted) and deep-MOND scaling (dashed grey). The WT curves bracket the data with the correct ordering. The amplitude offset ($\sim$3–5$\times$) reflects gas-fraction incompleteness in the B21 stellar-mass estimate of $g_{\rm bar}$ (see text).

[t]

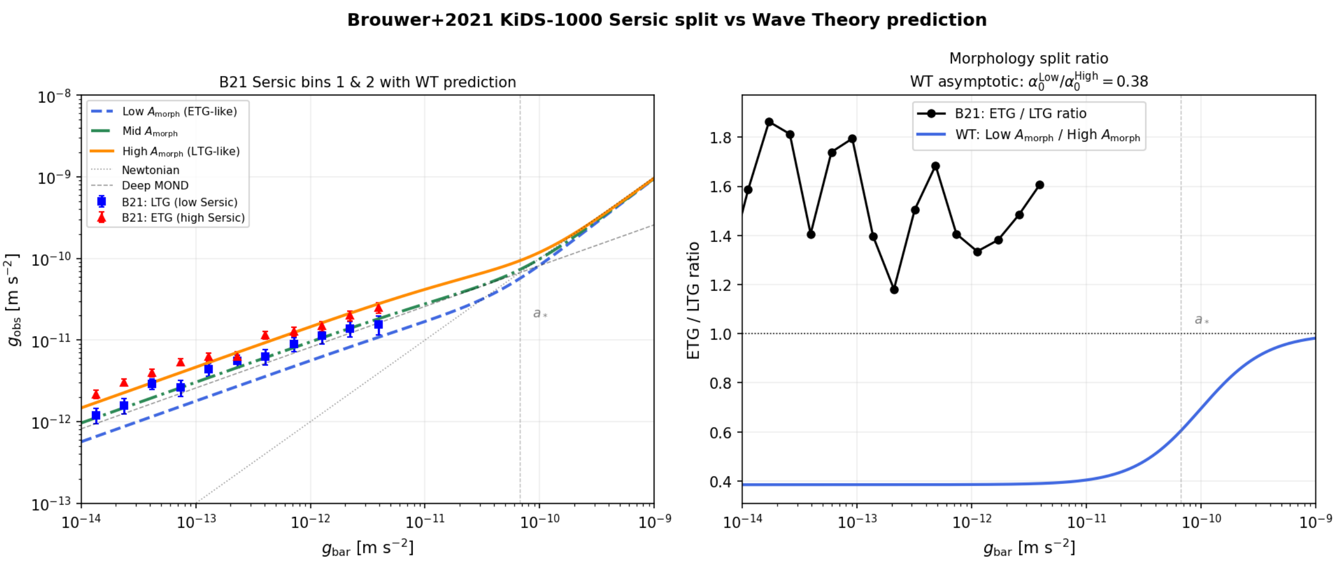

Figure 8. Left: B21 Sérsic bins 1 (blue squares, LTG) and 2 (red triangles, ETG) with WT morphology curves. Right: observed ETG/LTG ratio from B21 (black circles) versus WT predicted $g_{\rm obs}^{\rm Low\ A_{morph}}/g_{\rm obs}^{\rm High\ A_{morph}}$ ratio (blue curve). The WT prediction is < 1 throughout (LTG exceeds ETG), whereas the observed B21 ratio is > 1. B21 attribute this inversion to hot circumgalactic gas in ETGs that inflates their apparent $g_{\rm obs}$ relative to their stellar-mass-based $g_{\rm bar}$. Correcting ETG $g_{\rm bar}$ for hot gas is predicted to collapse the ratio toward the WT-ordered curve.

a structural consequence, not a fitted coincidence. The scale a_* is fixed by the curvature radius of the Universe and enters as a background parameter, not a fitted constant. This means the model makes a distinct prediction: a_0 should evolve with cosmic time as $c^2/(\pi R(z))$, in contrast to standard MOND where a_0 is a fixed universal constant.

The inverse-square behavior arises unchanged from three-dimensional flux conservation. The $1/r$ contribution emerges from the redistribution of curvature along two-dimensional manifolds — a transport effect, not a dynamical modification. In the outer regime it supplants rather than augments the local $1/r^2$ channel: the crossfade formula ensures that curvature counted in the shell channel is no longer counted in the volumetric channel. The consequence is that MOND-like behavior is not postulated but derived, and the acceleration scale is not fitted but fixed by geometry.

This distinction extends to covariant MOND theories such as TeVeS and AQUAL, which introduce additional fields and modified Lagrangians. The present model introduces neither. It operates within the weak-field limit of General Relativity, supplemented only by a dynamical interpretation of curvature propagation on embedded manifolds. Whether the shell transport channel admits a covariant action formulation remains an open question (Section 4.5), but it is not required for the weak-field predictions derived here. For a comprehensive review of MOND phenomenology and its relativistic extensions, see Famaey & McGaugh (2012). The weak-field GR foundations used throughout this work follow standard references Misner et al. (1973); Weinberg (1972).

7.3 Relation to ΛCDM and dark-matter halo models

The ΛCDM framework reproduces galactic rotation curves by postulating a dark matter halo whose density profile — typically taken as an NFW profile calibrated by the concentration–mass relation — adds a velocity contribution on top of the baryonic one. The present model contains no analogous component: the $1/r$ acceleration arises from curvature transport geometry, not from additional mass. The two frameworks therefore make structurally different predictions even when their rotation-curve outputs are numerically close.

Three distinctions are sharpest in the data. First, at one fitted parameter per galaxy (WT: α_0; MOND: $\Upsilon_\star$; ΛCDM: M_{200}), all three frameworks achieve comparable median χ^2 (2.82, 3.01, and 3.55 respectively), with MOND showing the highest galaxy count at lowest χ^2 and fewest catastrophic failures, WT the lowest median, and ΛCDM the highest mean χ^2 and most catastrophic failures (10 galaxies with $\chi^2 > 100$). A zero-parameter NFW comparison (halo mass estimated from $V_{\max}$) achieves median $\chi^2 = 10.9$, comparable to WT's zero-parameter median of 11.5; but this reflects the implicit per-galaxy adaptation of M_{200} through $V_{\max}$, not a genuinely parameter-free prediction. The ΛCDM catastrophic failures are concentrated among gas-dominated dwarfs and low-surface-brightness systems — a tension related to the well-documented cusp-core problem de Blok (2010). The behaviour at cluster scales, which requires a separate suppression mechanism, is discussed in Section 6.5.

Second, the ΛCDM halo is a dynamically inert background in the rotation-curve context, whereas the curvature-transport mechanism predicts a coupling that depends on the degree of orbital phase mixing. Systems without coherent rotation (pressure-supported spheroidals, tidally disrupted dwarfs) should therefore show a suppressed $1/r$ contribution in the present model but not in NFW-based models, which assign a halo to every object regardless of its dynamical state. This provides a qualitative discriminant that rotation curves alone cannot resolve.

Third, the effective acceleration scale $a_* = c^2/(\pi R)$ evolves with cosmic time in the present framework, whereas dark-matter halo properties at fixed mass evolve only through concentration. High-redshift rotation curves should show a systematically larger WT contribution at fixed baryonic mass — a prediction that is absent in halo-based models and testable with integral field spectroscopy at $z \sim 0.5$–1.

7.4 Role of the S^3 geometry and Wave Theory language

The use of S^3 geometry serves both physical and technical purposes.

First, a three-sphere of radius $R \sim c/H_0$ corresponds to the spatial slice of a closed FLRW universe with curvature at the Hubble scale. It therefore provides a natural global setting rather than an additional assumption.

Second, the Laplace–Beltrami operator on S^3 separates cleanly into radial and angular components. In the large-χ regime, angular modes dominate, making the effective reduction to two-dimensional transport explicit. This structure underlies the shell Poisson equation derived in Appendix C.

An equivalent formulation in flat space with a cosmological cutoff is possible; the S^3 representation is chosen because it makes the dimensional transition geometrically transparent.

The value $R = \pi c/H_0$ used throughout this work corresponds to the present-day Hubble radius under the Wave Theory relation $H = \pi c/R$. Adopting $H_0 = 67.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (consistent with Planck CMB measurements) gives $R \approx 4.29 \times 10^{26}\,\mathrm{m}$ and $a_* = c^2/(\pi R) \approx 6.67 \times 10^{-11}\,\mathrm{m\,s^{-2}}$. This is not a free parameter — it is fixed by cosmological observation independently of galactic dynamics. The Hubble tension ($H_0 \approx 67$–$73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ depending on the measurement method) introduces a corresponding $\sim 8\%$ uncertainty in R and a_*, which propagates into the normalization of the master formula but does not affect its functional form or the rotation-curve lowest-χ^2 counts reported in Section 6. The fact that the same R appears in both the curvature coupling $G = c^2/(\pi R)$ and the MOND scale $a_* = c^2/(\pi R)$ is therefore a non-trivial constraint on the framework's internal consistency, not an additional degree of freedom.

The advantage is conceptual: the MOND scale follows directly from the geometric postulate $G = a_* = c^2/(\pi R)$ rather than being a fitted constant. All results translate to standard SI through $GM_b \leftrightarrow a_* A_b$; the full correspondence is given in Appendix D.

7.5 Observational tests and falsifiability

The mechanism developed in this work makes a set of concrete, testable predictions that go beyond reproducing rotation curves. These predictions arise directly from the dynamical origin of shell transport — causal propagation combined with orbital phase mixing — and therefore provide sharp criteria for falsification.

I summarize five primary tests.

(1) Morphology dependence of the projection efficiency α_0

In the present framework the only empirically undetermined quantity is the dimensionless projection efficiency α_0, which encodes how effectively the bulk curvature flux couples to shell transport modes.

The model predicts that α_0 is not an arbitrary per-galaxy parameter but a function of observable baryonic structure. In particular, defining

$$\Delta\alpha_0 \equiv \alpha_0 - \kappa, \tag{127}$$

with $\kappa = 1 + \pi^{-2}$ the geometric baseline, the distribution of $\Delta\alpha_0$ over a large galaxy sample should be unimodal and centered near zero, with systematic dependence on structural observables such as

$$\mu_0, \quad f_{\rm gas}, \quad R_d, \quad B/D, \quad a_{\rm char}, \quad V_{\rm max}. \tag{128}$$

The precise functional form

$$\Delta\alpha_0 = f(\mu_0, f_{\rm gas}, R_d, B/D, a_{\rm char}, V_{\rm max}) \tag{129}$$

is not derived here, but the existence of such a relation is required by the geometric interpretation of α_0 as a projection efficiency.

Falsification criterion. If α_0 exhibits no statistically significant correlation with any observable structural quantity and behaves as unstructured scatter, the interpretation of α_0 as a geometric projection factor is ruled out.

(2) Rotational vs. pressure-supported systems

The shell transport mechanism does not require coherent rotation. In rotationally supported disk galaxies, orbital coherence produces efficient phase mixing and a robust $1/r$ shell contribution. In pressure-supported spheroidal systems, orbits are not coherent but span a distribution of directions. Rather than suppressing mixing, this enhances it: randomised orbits lead to rapid isotropisation of the shell source, so the two-dimensional transport description remains valid with a different projection efficiency encoded in α_0 (Section 3.7).

The prediction is therefore that pressure-supported systems follow the same $1/r$ outer scaling as rotating disks, but with a systematically shifted α_0 reflecting their morphological class. Defining an outer-regime estimator

$$\mathcal{A}_{\rm out} \equiv \frac{a_{\rm obs}}{\sqrt{a_b\, a_*}}, \tag{130}$$

the model predicts that $\mathcal{A}_{\rm out}$ follows the same functional form across morphological classes, with normalisation set by α_0. Velocity-dispersion profiles of pressure-supported systems provide the direct kinematic probe.

Falsification criterion. If pressure-supported systems show no $1/r$ outer contribution in velocity-dispersion profiles at any α_0 — that is, if the shell scaling is absent regardless of normalisation — the mechanism is falsified, because the isotropisation argument requires the $1/r$ term to persist. Conversely, if they follow the same normalisation as rotating disks (α_0 identical to disk values at matched baryonic mass), the morphology dependence of α_0 is ruled out.

(3) Breakdown of shell coherence at large radius

Shell transport requires efficient orbital phase mixing. This condition is satisfied when the orbital period is much shorter than the system age,

$$T_{\rm orb}(r) \ll T_{\rm gal}. \tag{131}$$

At sufficiently large radii, this condition fails. Defining a mixing scale $r_{\rm mix}$ by

$$T_{\rm orb}(r_{\rm mix}) \sim T_{\rm gal}, \tag{132}$$

the model predicts that beyond this scale:

- residuals to the shell model increase,
- azimuthal asymmetry becomes more pronounced,
- galaxy-to-galaxy scatter grows.

A natural diagnostic variable is the dimensionless ratio

$$\frac{T_{\rm orb}(r)}{T_{\rm gal}}. \tag{133}$$

Falsification criterion. If the outer scatter and residual structure of rotation curves show no correlation with $T_{\rm orb}/T_{\rm gal}$, the dynamical origin of shell transport is not supported.

(4) Redshift evolution of the acceleration scale

The characteristic acceleration scale of the model is

$$a_* = \frac{c^2}{\pi R}, \tag{134}$$

which evolves with cosmic time through the curvature radius $R(z)$. Since $H = \pi c/R$ in the Wave Theory relation,

$$a_*(z) = \frac{H(z)\, c}{\pi^2}, \tag{135}$$

so a_* grows with redshift in direct proportion to $H(z)$.

This implies that the asymptotic BTFR normalization satisfies

$$\frac{V_\infty^4}{M_b} \propto a_*(z) \propto H(z), \tag{136}$$

up to the morphology factor $A_{\rm morph}$, which is a property of baryonic structure at fixed morphological class and is assumed not to evolve with $H(z)$ (*Argued*: morphology distributions evolve with redshift, but the projection efficiency at fixed class is expected to be set by local orbital structure rather than cosmic epoch). The prediction is therefore that, at fixed baryonic mass and fixed morphological class, galaxies at higher redshift exhibit systematically higher outer velocities, with the normalization shift tracking $H(z)$ and not an independent free parameter.

This distinguishes the framework from standard MOND, where a_0 is a fixed constant, and from ΛCDM NFW models, where BTFR evolution is driven by concentration-mass evolution rather than a global acceleration scale.

Falsification criterion. The test has two failure modes. First, if no redshift evolution of the BTFR normalization is detected after controlling for selection effects and morphological composition at $z \sim 0.5$–1, where $H(z)/H_0 \approx 1.6$–1.8 implies a predicted normalization shift of order 60–80% in a_*, the geometric identification of a_* with $H(z)$ is ruled out. Second, if evolution is detected but scales with a power of $H(z)$ other than unity — for example as $H(z)^{1/2}$ or $H(z)^2$ — the specific proportionality $a_* \propto H$ is falsified even if some evolution is present. The predicted signal is accessible with high-redshift integral field spectroscopy at $z \sim 0.5$–1.

(5) Local-compactness control of the transition

The transition between the inner and outer regimes is governed by the local compactness parameter

$$\xi(r) = \sqrt{\frac{a_b(r)}{a_*}}, \tag{137}$$

rather than by global galaxy properties alone.

The transition radius r_t is therefore defined implicitly by

$$a_b(r_t) \sim a_*. \tag{138}$$

The model predicts that galaxies with identical total baryonic mass but different radial distributions exhibit different transition radii, controlled by their local acceleration profiles.

Rescaling radii by the condition $a_b(r) = a_*$ should collapse the transition across galaxies more effectively than scaling by R_d, $R_{\rm eff}$, or total mass alone.

Falsification criterion. If transition radii are determined solely by global quantities such as total mass or halo scale, with no role for local compactness, the crossfade mechanism is not supported.

Observational roadmap. The predictions above can be tested directly with existing and near-term datasets.

For local disk galaxies, the SPARC Lelli et al. (2016) and THINGS Walter et al. (2008) samples provide high-quality rotation curves and baryonic decompositions. These data allow (i) extraction of α_0 per galaxy, (ii) correlation of $\Delta\alpha_0$ with structural observables (μ_0, $f_{\rm gas}$, R_d, $a_{\rm char}$), and (iii) tests of the local-compactness prediction by rescaling radii using the condition $a_b(r) = a_*$. Residuals can be analyzed as a function of $T_{\rm orb}/T_{\rm gal}$ to test the predicted breakdown of shell coherence at large radii.

For pressure-supported systems, dwarf spheroidals with resolved stellar kinematics Walker et al. (2009) and early-type galaxies from the ATLAS$^{\rm 3D}$ survey Cappellari et al. (2011) provide a direct comparison at fixed baryonic mass. The key observable is the outer acceleration or dispersion profile, from which $\mathcal{A}_{\rm out}$ can be inferred and compared to rotationally supported systems at matched M_b.

At intermediate and high redshift, integral-field spectroscopy surveys — including the KMOS Galaxy Evolution Survey Tiley et al. (2021), MUSE-Wide Urrutia et al. (2019), and JWST/NIRSpec programs such as JADES Eisenstein et al. (2023) — provide spatially resolved kinematics for star-forming disks at $z \sim 0.5$–2. These enable measurement of the BTFR normalization as a function of redshift and a direct test of the scaling $V_\infty^4/M_b \propto H(z)$ after controlling for gas fraction and morphological class.

Weak gravitational lensing provides a complementary and independent test at projected radii of 10–500 kpc, well beyond the extent of observed rotation curves. The KiDS-1000 weak-lensing radial acceleration relation of Brouwer et al. Brouwer et al. (2021), compared to the WT prediction in Section 6.6, already shows that the three $A_{\rm morph}$ tertiles bracket the observed lensing signal with the correct ordering. Future surveys — Euclid, LSST, and the Roman Space Telescope — will measure $\Delta\Sigma(R)$ with sufficient signal-to-noise to distinguish the morphology-dependent WT prediction from a universal modified-gravity curve at better than 5σ, particularly if early- and late-type galaxy samples are analyzed separately with matched baryonic mass estimates that include circumgalactic gas.

The required analyses involve no new observables beyond standard rotation-curve fitting, baryonic mass modelling, and kinematic profiling. The distinguishing feature of the present framework is not the data required, but the correlations it predicts across morphology, radius, and cosmic time. A coordinated analysis across these datasets therefore provides a decisive empirical test of the shell-transport mechanism.

The framework predicts not only the functional form of the deviation from inverse-square scaling, but also its dependence on morphology, orbital structure, cosmic time, and local compactness. Failure of any one of the tests above would directly challenge a specific element of the mechanism; taken together, they provide a stringent and multi-dimensional empirical assessment of the shell-transport picture.

8 CONCLUSIONS

The equilibrium assumption embedded in standard galactic modelling — that the gravitational field reflects the instantaneous baryonic mass distribution — is not a consequence of General Relativity but a simplification of it. This paper has shown that treating curvature propagation as a dynamical, relativistic process, organized by differential rotation on a closed S^3 background, produces flat rotation curves from baryonic matter alone. The derivational chain is summarised in Table 1; the principal empirical results are as follows.

The amplitude $\alpha_0 = \kappa \cdot A_{\rm morph}$ factorises into a universal geometric constant $\kappa = 1 + \pi^{-2}$, fixed by the S^3 background, and a continuously distributed morphology factor $A_{\rm morph}$ that requires observational calibration. A GMM analysis confirms that $A_{\rm morph}$ is statistically unimodal, centred near κ (Section 6.3; Appendix F).

The zero-free-parameter prediction ($\alpha_0 = \kappa$, no per-galaxy adjustment) achieves the lowest χ^2 on 78 rotation curves simultaneously against both MOND and ΛCDM — the highest such count among the three frameworks at zero parameters. At one fitted parameter per galaxy (WT: α_0; MOND: $\Upsilon_\star$; ΛCDM: M_{200}), WT achieves median $\chi^2 = 2.82$ and the lowest χ^2 on 54 galaxies, MOND achieves median $\chi^2 = 3.01$ and the lowest on 83, and ΛCDM achieves median $\chi^2 = 3.55$ and the lowest on 38. The three frameworks are competitive at one-parameter level; the zero-parameter result is the stronger test of the geometric mechanism. An exploratory extension to 12 relaxed Chandra clusters, using a physically motivated megaparsec-scale suppression derived from the gravity–expansion balance, reproduces the radial shape of observed cluster acceleration profiles to within 0.082 dex RMS, with the residual normalisation offset attributable to the shell-superposition structure of cluster baryons.

An exploratory comparison with the KiDS-1000 weak-lensing radial acceleration relation (Section 6.6) shows that the morphology-dependent lensing signal, computed via Abel projection of $a_{\rm eff}$ with no additional parameters, correctly orders early- and late-type galaxies, consistent with the 6σ Sérsic split of Brouwer et al. Brouwer et al. (2021).

8.1 Open problems

The following problems remain open within the current framework, organised by the part of the derivational chain they affect.

Shell confinement (Section 3). The co-moving assumption — that the curvature-transport medium is advected by the baryonic velocity field on orbital timescales — is physically motivated but not derived from a covariant transport equation. The WKB validity condition for the eikonal ray equations requires $m \gg c/v_{\rm rot} \sim 1500$; the same-shell condition $r' \approx r$ is independent of m (Section 3.4), so the WKB gap affects only the winding rate, not the shell locality result. The relation between morphology and shell-mode projection efficiency — currently captured by the empirical α_0 — has not been derived from first principles.

Causal transport reduction (Section 4.5). The spectral dominance condition $\ell(\ell+1)/\sin^2\chi \gg n(n+2)$ has been verified at the operator level (Appendix A) but not tracked through the full Green function of the screened equation (44); establishing explicit error bounds would promote the shell Poisson equation from an asymptotic result to a controlled approximation. The full causal reduction including finite memory time τ_c and cosmological expansion $H = \pi c/R$ has not been carried out beyond the quasi-static limit.

Crossfade and interpolation (Section 5). The crossfade exponent sensitivity analysis (Table 2) confirms that per-galaxy performance is robust across $m \in [0.25, 2.0]$ and that $m = 1$ maximises the zero-parameter lowest-χ^2 count; however, the spectral argument does not uniquely compel $m = 1$, and a rigorous derivation of the crossfade kernel from the spectral sum or a variational principle remains open. The normalization condition $k_r^2 = k_\theta^2$ at $a_b = a_*$ is asserted as the natural symmetry point; whether it can be derived from first principles remains open.

Morphology and empirical calibration. The per-galaxy $A_{\rm morph}$ distribution is unimodal with standard deviation ≈ 0.56, spanning a factor of ~ 60 from 0.045 to 2.72; whether this width is stable under changes in sample selection, distance calibration, or inclination correction remains to be tested on independent datasets. $A_{\rm morph}$ correlates with central disk surface brightness ($\rho = +0.21$, $p = 0.005$), $\log M_{200}$ ($\rho = +0.28$), and $\log a_{\rm char}$ ($\rho = +0.35$), but not with disk scale length, effective radius, bulge-to-disk ratio, Hubble type, or gas fraction (all $p > 0.05$, $N = 175$). The correlation with central surface brightness supports the geometric interpretation: more centrally concentrated baryonic distributions produce stronger shell-mode excitation. A predictive relation $A_{\rm morph} = f(\mu_0, a_{\rm char})$ remains an open empirical target that would reduce the free parameter from one per galaxy to one per morphological class. A first-principles derivation of $A_{\rm morph}$ from baryonic structure — simultaneously fixing the BTFR normalization and the effective MOND scale $a_0^{\rm eff} = \alpha_0^2\, a_*$ — remains the principal open problem.

Scope limitations and extensions. Whether the two-channel formula (68) can be derived from a covariant action principle remains open; without this, the framework cannot guarantee energy conservation at the nonlinear level or make predictions for strong-field systems. The KiDS-1000 weak-lensing comparison (Section 6.6) reproduces the morphology ordering without new parameters, but the absolute amplitude requires a gas-fraction correction deferred to future work; extension to strong-lensing Einstein radii from the SLACS survey Bolton et al. (2006); Koopmans et al. (2006) also remains open. The exploratory cluster application (Section 6.5) reproduces radial acceleration profile shapes to 0.082 dex RMS, but the normalisation law $\alpha_{\rm cl}$(cluster properties) is not yet derived quantitatively, and extension to non-relaxed, merging, or cool-core clusters has not been attempted. Extension to non-circular systems (mergers, tidally disturbed systems, triaxial structures) is not established; the orbital averaging argument relies on approximate circular motion.

The broader implication is methodological. Galactic dynamics has long been modelled as a boundary-value problem: given a mass distribution, find the potential. The propagation-primary picture reframes it as an initial-value problem: given a mass distribution and its dynamical history, compute the curvature field that has actually equilibrated at each radius. In the inner regime the two pictures agree. At large radii, where the propagation front has not equilibrated and orbital phase mixing has organized transport onto shells, they do not. The observed rotation-curve discrepancy is, on this account, a measurement of the difference between the two regimes — not evidence of missing mass, and not a modification of gravity, but a signature of curvature propagation dynamics that GR contains but galactic modelling has not used.

APPENDIX A: SPECTRAL ORIGIN OF SHELL DYNAMICS ON S^3

This appendix provides an alternative derivation of the shell transport behavior by analyzing the spectral properties of operators defined on S^3. The goal is to show that the effective two-dimensional response arises naturally from the structure of the three-dimensional manifold.

A1 Geometry of S^3 and bulk-to-shell projection

The spatial geometry is a three-sphere S^3 of radius R, with metric

$$ds^2 = R^2\left(d\chi^2 + \sin^2\chi\, d\Omega_2^2\right), \qquad \chi \in [0,\pi], \tag{A1}$$

where $\chi = r/R$ is the dimensionless radial coordinate and $d\Omega_2^2$ is the unit two-sphere metric. The volume of the cap enclosed within χ and the area of its boundary two-sphere S_r are

$$V_{\rm cap}(\chi) = 2\pi R^3\left(\chi - \tfrac{1}{2}\sin 2\chi\right), \qquad A_\partial(\chi) = 4\pi R^2 \sin^2\chi. \tag{A2}$$

In the local limit $\chi \ll 1$ these reduce to the standard Euclidean expressions $V \to \frac{4}{3}\pi r^3$ and $A \to 4\pi r^2$, confirming consistency with Newtonian gravity at sub-galactic scales.

The curvature-flux field $\mathbf{F}_b$ satisfies $\nabla\cdot\mathbf{F}_b = s_b$, where s_b is the baryonic curvature source density. Applying the divergence theorem to V_r and identifying $\mathbf{F}_b = -\nabla\Phi$ with $\nabla^2\Phi = 4\pi G\rho_b$ in the weak-field limit gives

$$Q_b^{\rm src}(r) \equiv 4\pi G M_b(r), \tag{A3}$$

consistent with the sign convention of Appendix C. The normal flux $\sigma_r \equiv \mathbf{F}_b\cdot\mathbf{n}_r$ injected through S_r drives curvature transport on the shell via the surface Poisson equation

$$\Delta_{S_r}\Phi_r = -\sigma_r. \tag{A4}$$

The Green function of the two-dimensional Laplacian is logarithmic, $G \sim \log r$, so the shell potential scales as $\Phi_r \sim Q_b(r)\log r$ and the resulting shell acceleration is

$$a_{\rm shell}(r) \sim \frac{Q_b^{\rm src}(r)}{r}, \tag{A5}$$

establishing the inverse-linear behavior as a direct consequence of 2D transport geometry. The spectral origin of the dimensional reduction that produces this 2D equation is derived in the subsections that follow.

A2 Laplacian on S^3

The Laplace–Beltrami operator on S^3 in coordinates (χ, Ω) can be written as

$$\Delta_{S^3} = \frac{1}{R^2}\left[\frac{\partial^2}{\partial\chi^2} + 2\cot\chi\frac{\partial}{\partial\chi} + \frac{1}{\sin^2\chi}\Delta_{S^2}\right] \tag{A6}$$

where Δ_{S^2} is the Laplacian on the unit two-sphere. For the spectral theory of the Laplace–Beltrami operator on compact Riemannian manifolds, see Chavel (1984); Rosenberg (1997). This decomposition separates radial and angular contributions.

A3 Spectral decomposition

Functions on S^3 can be expanded in hyperspherical harmonics $Y_{n\ell m}(\chi, \Omega)$, which satisfy

$$\Delta_{S^3} Y_{n\ell m} = -\frac{n(n+2)}{R^2} Y_{n\ell m}, \qquad n = 0, 1, 2, \ldots \tag{A7}$$

The index n labels global modes, while (ℓ, m) correspond to angular structure on S^2. The eigenvalues follow from standard harmonic analysis on compact Riemannian manifolds Chavel (1984); Rosenberg (1997).

A4 Radial localization and boundary dominance

Consider a source localized within a spherical cap $V_{\text{cap}}(\chi)$. The resulting field can be expanded in eigenmodes, but at large distances (large χ), the contribution is dominated by modes with minimal radial variation.

$$\Delta_{S^3} \xrightarrow{\ell(\ell+1)/\sin^2\chi \gg n(n+2)} \frac{1}{R^2\sin^2\chi}\Delta_{S^2}, \tag{A8}$$

In this regime, radial gradients become subdominant compared to angular variations, and the operator effectively reduces to

$$\Delta_{S^3} \longrightarrow \frac{1}{R^2\sin^2\chi} \times \Delta_{S^2} \tag{A9}$$

Thus, the dynamics becomes effectively constrained to the angular manifold S^2.

A5 Effective dimensional reduction

The above limit shows that, for observers at fixed radius r, the relevant dynamics is governed by an operator proportional to Δ_{S^2}.

This induces an effective reduction:

$$S^3 \longrightarrow S_r \simeq S^2 \tag{A10}$$

where curvature propagation is restricted to the shell.

In the small-angle limit $r \ll R$ (equivalently $\theta \ll 1$), the Green function of the effective operator reduces to its flat-space 2D form,

$$G(\theta) \sim \log\theta \tag{A11}$$

which is the standard 2D logarithmic potential, where θ is an angular separation on S_r.

A6 Connection to radial scaling

Mapping angular separation to physical distance on the shell, $r \sim R\theta$, the logarithmic Green function leads to

$$\Phi_r \sim \log r \tag{A12}$$

and hence

$$a_{\text{shell}}(r) \sim \frac{1}{r} \tag{A13}$$

Thus, the inverse-linear scaling emerges from the spectral structure of the Laplacian on S^3.

A7 Relation to causal transport

The spectral reduction described above corresponds to the steady-state limit of the causal transport process discussed in Section 5.

In the full dynamical picture:

- curvature propagates through S^3 with finite speed,
- radial transport becomes suppressed relative to tangential redistribution,
- the system relaxes to a configuration dominated by angular modes.

The effective two-dimensional behavior therefore arises both:

- dynamically, through causal transport and finite propagation,
- spectrally, through the structure of the Laplacian on S^3.

The spectral analysis thus provides an independent confirmation of the shell transport mechanism: angular mode dominance at large scales is a structural property of the S^3 geometry, not an external assumption.

APPENDIX B: SHELL GREEN FUNCTION AND THE EMERGENCE OF THE $1/R$ LAW

This appendix derives the inverse-linear shell contribution used in the main text. The result follows from the fact that curvature transport on an orbital shell is governed by a two-dimensional Poisson problem, whose Green function is logarithmic.

B1 Shell transport as a two-dimensional Poisson problem

Let S_r denote the constant-radius orbital shell. The redistribution of curvature on S_r by a tangential current J_r satisfying the conservation law

$$\operatorname{div}_{S_r} J_r = \sigma_r \tag{B1}$$

where σ_r is the effective shell source density induced by the interior baryonic region.

Introducing a scalar shell potential Φ_r by

$$J_r = -\nabla_{S_r}\Phi_r \tag{B2}$$

one obtains

$$\Delta_{S_r}\Phi_r = -\sigma_r \tag{B3}$$

Thus the shell response is determined by the Green function of the Laplace–Beltrami operator on a two-dimensional manifold.

B2 Local flat approximation

For shell scales much smaller than the curvature radius of the background manifold, the orbital shell is locally well approximated by the Euclidean plane $\mathbb{R}^2$. In this limit the shell Poisson equation becomes

$$\Delta\Phi(\mathbf{x}) = -\sigma(\mathbf{x}) \qquad \mathbf{x}\in\mathbb{R}^2. \tag{B4}$$

The Green function $G(\mathbf{x},\mathbf{x}')$ is defined by

$$\Delta G(\mathbf{x},\mathbf{x}') = -\delta^{(2)}(\mathbf{x}-\mathbf{x}') \tag{B5}$$

In two dimensions, this Green function is

$$G(\mathbf{x},\mathbf{x}') = -\frac{1}{2\pi}\log|\mathbf{x}-\mathbf{x}'| \tag{B6}$$

The theory of elliptic partial differential equations and their Green functions is treated in Evans (2010); Taylor (2011).

Hence the shell potential generated by a source density σ is

$$\Phi(\mathbf{x}) = \int_{\mathbb{R}^2} G(\mathbf{x},\mathbf{x}')\;\sigma(\mathbf{x}')\;d^2x' \tag{B7}$$

For a localized source of total shell content

$$Q_r = \int_{\mathbb{R}^2}\sigma(\mathbf{x}')\;d^2x' \tag{B8}$$

the far-field potential reduces to

$$\Phi(r) \sim -\frac{Q_r}{2\pi}\log r \tag{B9}$$

up to an additive constant.

Taking the gradient,

$$|\nabla\Phi(r)| \sim \frac{Q_r}{2\pi r} \tag{B10}$$

Therefore the shell-supported acceleration obeys

$$a_{\rm shell}(r) \propto \frac{Q_r}{r} \tag{B11}$$

This is the origin of the inverse-linear scaling used in the main text. The same result follows from flux conservation on S_r without invoking the potential: the total flux of J_r through a circle of radius r equals Q_r, so $|J_r| = Q_r/(2\pi r)$, confirming $a_{\rm shell}\propto r^{-1}$ Evans (2010); Taylor (2011). Extension to the full spherical shell geometry replaces $\mathbb{R}^2$ with S^2 but leaves the logarithmic Green function and $1/r$ scaling unchanged at radii $r \ll R$, since the Laplace–Beltrami operator on S^2 reduces to the flat Laplacian at sub-curvature scales. For an extended disk source, the monopole term dominates at $r \gg R_d$ because higher multipoles of σ_r decay faster than r^{-1} Evans (2010); the point-source result therefore applies to good approximation throughout the rotation-curve regime.

B3 Result

The shell transport equation is intrinsically two-dimensional, and the corresponding Green function is logarithmic. Consequently, the shell-supported acceleration scales as the gradient of a logarithmic potential:

$$\Phi_{\rm shell}\sim Q_r\log r \qquad \Longrightarrow \qquad a_{\rm shell}(r)\sim\frac{Q_r}{r} \tag{B12}$$

This is the mathematical origin of the $1/r$ contribution appearing in the effective galactic acceleration law.

B4 Minimal steady-state shell solution and emergence of MOND-like scaling

The evolution law introduced above,

$$\partial_t A + \mathbf{u}\cdot\nabla A = -\frac{1}{\tau_{\rm dil}(A)}\;A - H\,\Xi[A] + S_{\rm rot} \tag{B13}$$

describes curvature on the shell as a causally transported and continuously replenished quantity. Its purpose is not to replace the local inverse-square law, but to determine the amount of curvature that remains resident on the shell and is therefore available for two-dimensional transport. This is consistent with the general framework already established in Section 4, where the local channel scales as r^{-2}, the shell channel scales as r^{-1}, and the transition is controlled by the universal curvature scale $a_* = c^2/(\pi R)$.

B4.0.1 Minimal shell average.

To obtain a closed analytic expression, consider the azimuthally averaged shell occupancy. In a quasi-stationary regime, the explicit time dependence averages out and the transport advection term does not contribute to the shell average. The expansion is further approximated as operator by its leading linear action on the shell amplitude,

$$\Xi[A]\approx A \tag{B14}$$

and evaluate the amplitude-dependent dilution time at an effective shell value,

$$\tau_{\rm dil}(A)\to\tau_s \tag{B15}$$

The shell-averaged steady-state equation then becomes

$$0 = -\left(\frac{1}{\tau_s}+H\right)A_s + S_s \tag{B16}$$

with solution

$$A_s = \frac{S_s}{\frac{1}{\tau_s}+H} = \tau_{\rm eff}\;S_s \qquad \tau_{\rm eff} = \frac{1}{\frac{1}{\tau_s}+H} \tag{B17}$$

Thus the shell stores a finite curvature occupancy equal to the injection rate multiplied by an effective residence time. The residence time is controlled jointly by local causal dilution and global Hubble stretching. In Wave Theory, the latter is fixed by

$$H = \frac{\pi c}{R} \tag{B18}$$

while the causal transport sector is constrained by finite front velocity, as in the telegraph-type curvature dynamics on S^3.

B4.0.2 Source-to-shell injection scale. The shell is fed by the enclosed source

$$Q_b^{\rm src}(\chi) = 4\pi G M_b(\chi) \tag{B19}$$

which in the geometric formulation is equivalently

$$Q_b^{\rm src}(\chi) = 4\pi a_* A_b(\chi) \qquad a_* = \frac{c^2}{\pi R} = G \tag{B20}$$

The simplest steady-state closure is to assume that the shell is populated on the radial transport scale at which the local baryonic channel and the cosmological curvature floor become comparable. In Section 6 this transition radius is

$$r_* \sim \sqrt{\frac{GM_b}{a_*}} \tag{B21}$$

This same scale already controls the onset of shell relevance in the static model.

A minimal dimensional estimate for the shell injection rate is then

$$S_s \sim \frac{Q_b^{\rm src}}{\tau_{\rm in}} \qquad \tau_{\rm in} \sim \frac{r_*}{\nu_s} \tag{B22}$$

where $\nu_s = \gamma_s c$ is the effective causal update speed on the shell. Hence

$$S_s \sim \frac{\nu_s}{r_*} Q_b^{\rm src} \tag{B23}$$

Substituting into the steady-state solution gives

$$A_s \sim \frac{\tau_{\rm eff}\nu_s}{r_*} Q_b^{\rm src} \tag{B24}$$

It is convenient to absorb the factor $\tau_{\rm eff}\nu_s$ into a dimensionless occupancy coefficient η_s,

$$\eta_s \equiv \frac{\tau_{\rm eff}\nu_s}{r_*} \tag{B25}$$

so that

$$A_s \sim \eta_s Q_b^{\rm src} \tag{B26}$$

In the quasi-stationary transition regime, where shell filling and shell dilution balance on the same radial scale, η_s is naturally of order unity. The shell occupancy can then be written in the compact form

$$Q_b^{\rm shell} \sim \kappa_s \frac{Q_b^{\rm src}}{r_*} \tag{B27}$$

where κ_s is a dimensionless constant encoding the detailed efficiency of shell residence.

B4.0.3 Shell acceleration in the transition regime. Using the shell transport law from Section 4,

$$a_{\rm shell}(r) = \alpha_b(\chi) \frac{Q_b^{\rm shell}}{r} \tag{B28}$$

one obtains

$$a_{\rm shell}(r) \sim \alpha_b(\chi) \frac{\kappa_s Q_b^{\rm src}}{r\, r_*} \tag{B29}$$

Substituting $Q_b^{\rm src} = 4\pi G M_b$ and $r_* = \sqrt{GM_b/a_*}$, one finds

$$a_{\rm shell}(r) \sim 4\pi\kappa_s\, \alpha_b(\chi) \frac{\sqrt{GM_b\, a_*}}{r} \tag{B30}$$

Since the local baryonic acceleration is

$$a_b(r) = \frac{GM_b}{r^2}, \tag{B31}$$

the shell acceleration can be written equivalently as

$$\boxed{a_{\rm shell}(r) \sim 4\pi\kappa_s\, \alpha_b(\chi) \sqrt{a_b(r)\, a_*}} \tag{B32}$$

This is the characteristic MOND-like form, valid in the outer regime where $M_b(r) \approx M_b$ is approximately constant and the identification $\sqrt{GM_b\, a_*}/r = \sqrt{a_b(r)\, a_*}$ holds. At smaller radii where $M_b(r)$ is still growing with r, the shell contribution retains the intermediate form

$$a_{\rm shell}(r) \sim \frac{4\pi\kappa_s\, \alpha_b(\chi) \sqrt{GM_b(r)\, a_*}}{r}, \tag{B33}$$

which does not reduce to the MOND square-root scaling.

The MOND-like form emerges not from modifying the local gravitational law, but from three ingredients acting together:

(i) the standard enclosed Gauss-law source $Q_b^{\rm src} = 4\pi G M_b$,

(ii) the shell-transport Green function on a two-dimensional manifold, which gives the inverse-linear propagation law,

(iii) the transition scale $r_* \sim \sqrt{GM_b/a_*}$, set by the equality of local baryonic curvature and the cosmological curvature floor.

B4.0.4 Asymptotic speed and BTFR. The rotation speed satisfies

$$v^2(r) = r\, a_{\rm eff}(r) \tag{B34}$$

In the shell-dominated outer regime,

$$v_\infty^2 \sim 4\pi\kappa_s\, \alpha_b(\chi_*) \sqrt{GM_b\, a_*} \tag{B35}$$

hence

$$v_\infty^4 \sim (4\pi\kappa_s)^2\ \alpha_b(\chi_*)^2\ GM_b\, a_* \tag{B36}$$

Therefore the baryonic Tully–Fisher scaling

$$v_\infty^4 \propto M_b \tag{B37}$$

follows immediately, in agreement with the scaling relation already identified in the static shell model.

B4.0.5 Connection to the master formula. The coefficient $4\pi\kappa_s\, \alpha_b(\chi_*)$ appearing above is absorbed into the single morphology parameter α_0 of the crossfade master formula (68). The BTFR then takes the cleaner form $V_\infty^4 = \alpha_0^2\, GM_b\, a_*$, derived directly in Section 5.4. The Appendix E derivation provides the physical mechanism (shell occupancy, causal injection, steady-state balance) that justifies treating α_0 as a morphology-class constant rather than a per-galaxy free parameter.

B4.0.6 Interpretation. The result shows that MOND-like behavior need not be postulated as a modified force law. It appears as the quasi-stationary limit of a causal curvature-transport system on an expanding S^3 background. The familiar square-root scaling is the observable signature of balancing:

$$\text{local baryonic curvature} \quad \leftrightarrow \quad \text{cosmological curvature floor} \tag{B38}$$

through a shell-occupancy channel whose propagation is intrinsically two-dimensional.

In this sense, the inverse-square and MOND-like square-root regimes are not competing laws, but two different projections of the same curvature dynamics. The r^{-2} behavior belongs to direct local flux spreading in three dimensions, while the $\sqrt{a_b a_*}$ behavior appears when the enclosed source is first accumulated on the shell and then redistributed along a lower-dimensional transport manifold.

APPENDIX C: SIGN CONVENTIONS AND THE GEOMETRIC ORIGIN OF A_*

C1 Sign and naming conventions

The standard weak-field sign convention is adopted throughout:

$$\nabla^2\Phi = 4\pi G\rho_b, \qquad \mathbf{F}_b \equiv -\nabla\Phi, \qquad \nabla\cdot\mathbf{F}_b = -4\pi G\rho_b. \quad \text{(C1)}$$

The signed radial flux of $\mathbf{F}_b$ through a shell S_r is therefore

$$Q_b(r) \equiv \int_{S_r} \mathbf{F}_b \cdot \mathbf{n}_r \, dA = -4\pi G M_b(r). \quad \text{(C2)}$$

Wherever the *magnitude* of the enclosed curvature source is needed—in particular in the shell transport term and in BTFR derivations—the positive source magnitude is used

$$\boxed{Q_b^{\rm src}(r) \equiv 4\pi G M_b(r) = -Q_b(r).} \quad \text{(C3)}$$

Every formula for $a_{\rm shell}$ and $a_{\rm eff}$ uses $Q_b^{\rm src}$, so all accelerations are positive (attractive). The signed Q_b appears only in the divergence-theorem step that connects the field equation to the enclosed mass.

The shell Poisson equation is written with the convention $\Delta_{S_r}\Phi_r = -\sigma_r$, where $\sigma_r = \mathbf{F}_b \cdot \mathbf{n}_r < 0$ for inward-pointing fields. The resulting shell acceleration $a_{\rm shell} = -\nabla_{S_r}\Phi_r \cdot (-\hat{r})$ is positive, consistent with gravitational attraction.

C2 The geometric origin of a_* and its distinction from the MOND scale

The acceleration scale $a_* = c^2/(\pi R)$ that appears throughout this paper is not a new fitted constant. Its value is fixed by two independently measured quantities: the speed of light c and the cosmic curvature radius R. This appendix documents how R is determined and why the identification differs fundamentally from the empirical MOND acceleration a_0.

C2.1 Two independent routes to R

The curvature radius R is determined by two routes that draw on entirely different experimental inputs.

Route 1 — laboratory constants (G, $\hbar$, c). Within the geometric framework of this paper, the reduced Planck constant satisfies

$$\hbar = \pi \frac{L_p^3 R}{T_p}, \quad \text{(C4)}$$

where $L_p = \sqrt{\hbar G/c^3}$ and $T_p = L_p/c$ are the Planck length and time. Solving equation (C4) for R and substituting L_p and T_p in terms of G, $\hbar$, and c gives

$$R = \frac{\hbar T_p}{\pi L_p^3} = \frac{c^2}{\pi G}. \quad \text{(C5)}$$

Using CODATA 2018 values $G = 6.67430\times10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$ and $\hbar = 1.054571817\times10^{-34}\,\mathrm{J\,s}$ yields

$$R_1 = 4.286332\times10^{26}\ \mathrm{m}. \quad \text{(C6)}$$

This route uses only laboratory-measured constants; it contains no cosmological input.

Route 2 — cosmic expansion (H_0). The geometric framework identifies the Hubble rate as $H = \pi c/R$, giving

$$R = \frac{\pi c}{H_0}. \quad \text{(C7)}$$

The Planck 2018 CMB measurement Aghanim et al. (2020) $H_0 = 67.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ yields

$$R_2 = 4.286387\times10^{26}\ \mathrm{m}. \quad \text{(C8)}$$

This route uses only the CMB angular power spectrum fit; it is independent of any laboratory measurement of G or $\hbar$.

C2.2 Consistency

The two routes agree to five significant figures:

$$\frac{|R_2 - R_1|}{R_1} = 1.3\times10^{-5} \ \ (0.0013\%). \quad \text{(C9)}$$

The value of H_0 that gives exact agreement with Route 1 is $H_0 = 67.801\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, indistinguishable from the Planck 2018 measurement within its quoted uncertainty. This consistency across completely independent experimental inputs — laboratory gravity and quantum measurements on one side, CMB cosmology on the other — is not a coincidence engineered by fitting. No galaxy rotation curve enters either route.

C2.3 The acceleration scale

From Route 1, the acceleration scale is

$$a_* = \frac{c^2}{\pi R_1} = 6.67430\times10^{-11}\ \mathrm{m\,s^{-2}}, \quad \text{(C10)}$$

which equals the numerical value of G in SI units. This is not a coincidence: within the geometric framework, G is the background curvature acceleration of the S^3 geometry, and its SI dimensional form $[\mathrm{m^3\,kg^{-1}\,s^{-2}}]$ reflects the conventional separation of mass as an independent dimension rather than a geometric area $[L^2]$. The numerical equality $a_* = G_{\rm SI,numerical}$ is therefore a structural feature of the framework, not a tuned parameter.

C2.4 Distinction from the MOND acceleration scale

The MOND acceleration constant $a_0 \approx 1.2\times10^{-10}\,\mathrm{m\,s^{-2}}$ is determined by fitting rotation curves Milgrom (1983). It differs from a_* by a factor of approximately 1.8 and has no known connection to any independently measured constant of nature. No analogue of the two-route cross-check above exists for a_0.

The present framework identifies the effective outer-regime MOND scale as $a_0^{\rm eff} = \alpha_0^2\, a_*$ (Section 5.4), where the morphology factor $\alpha_0 \approx \kappa \approx 1.1$ absorbs the difference. This relation is a prediction of the geometric mechanism, not a restatement of the MOND fit.

Finally, because $a_* = c^2/(\pi R)$ and R grows with cosmic expansion, the framework predicts that the effective acceleration scale evolves as $a_*(z) \propto H(z)$. Standard MOND treats a_0 as a fixed universal constant and makes no such prediction. This redshift signature (Section 6) is a direct and falsifiable consequence of the geometric origin of a_*.

APPENDIX D: THE MODEL IN GR LANGUAGE

This appendix is written for readers who prefer to evaluate the physical content of the model entirely within standard General Relativity, without reference to the Wave Theory geometric language used in the main text. It is self-contained: every quantity is defined from first principles in GR terms, and the three elements that are genuinely new relative to textbook GR are identified explicitly. The central point is that the model is not an alternative to GR but an extension of it: it activates a dynamical role for boundary data that standard GR generates but does not use.

The central point is that every equation in the paper has an exact GR counterpart derivable from standard results. Wave Theory provides a geometric language that makes the dimensional structure of the transport process transparent, but it is not a prerequisite for the physics. The three elements that are genuinely new relative to textbook GR are identified explicitly in Section D4 below; everything else follows from the weak-field limit of the Einstein equations, the geometry of S^3, and the spectral structure of its Laplacian.

D1 Standard GR foundation

In the weak-field limit, the spacetime metric takes the form

$$g_{00} \simeq -(1 + 2\Phi/c^2), \qquad g_{ij} \simeq (1 - 2\Phi/c^2)\delta_{ij}, \tag{D1}$$

and the Einstein field equations reduce to the Newtonian Poisson equation

$$\nabla^2\Phi = 4\pi G\rho_b. \tag{D2}$$

The gravitational field is $\mathbf{F}_b = -\nabla\Phi$, and Gauss's theorem gives the enclosed source flux

$$\oint_{S_r} \mathbf{F}_b \cdot \mathbf{n}_r \, dA = -4\pi G M_b(r). \tag{D3}$$

This is standard GR in the weak-field limit. No modification has been introduced.

D2 The shell source as a GR boundary quantity

In standard GR, the object $\sigma_r \equiv \mathbf{F}_b \cdot \mathbf{n}_r = -\partial_r\Phi$ is the normal component of the gravitational field at radius r. It is the shell source introduced in Section 2 — not a new object, but the standard boundary trace of the Newtonian potential.

The identification

$$Q_b^{\mathrm{src}}(r) = \oint_{S_r} \sigma_r \, dA = 4\pi G M_b(r) \tag{D4}$$

is exactly Gauss's law. The shell source σ_r is therefore fully determined by standard GR and carries no additional freedom.

D2.0.1 Gauss–Codazzi perspective.

More covariantly, σ_r is the normal-normal component of the extrinsic curvature of the hypersurface S_r embedded in the spatial slice. The Gauss–Codazzi equations relate this to the intrinsic geometry of S_r and the ambient Ricci curvature. In the weak-field limit, this reduces to (D4). The shell source is thus a standard object in the geometric theory of hypersurface embeddings — it requires no new postulate.

D3 The shell transport equation in GR

Once σ_r is established as a standard GR boundary quantity, the shell transport equation

$$\Delta_{S_r}\Phi_r = -\sigma_r \tag{D5}$$

is a two-dimensional Poisson equation on the surface S_r, sourced by the normal component of the gravitational field. This equation is derivable from the three-dimensional Poisson equation (D2) by the projection procedure of Section 4.5: in the quasi-static, large-r regime, the Laplace–Beltrami operator on S^3 reduces to its angular component, and the bulk equation induces (D5) as a boundary condition on S_r.

The key input is therefore not a new field equation but a regime condition: at large radii, where radial gradients become subdominant relative to angular variations, the three-dimensional Poisson equation (D2) induces an intrinsic two-dimensional Poisson equation on the shell. This is a consequence of the geometry of S^3 and the spectral structure of its Laplacian, established in Appendix A.

D4 What is new relative to standard GR

Standard GR in the weak-field limit gives only the three-dimensional channel: $a_b = GM_b/r^2$. The present model supplements this with a second channel arising from the redistribution of the boundary flux σ_r along the two-dimensional manifold S_r. Expressed entirely in GR language, the new ingredient is:

At large radii, the normal component of the gravitational field at S_r drives a two-dimensional transport process on S_r itself. This process is governed by (D5) and produces an additional acceleration contribution scaling as $1/r$.

This is not a modification of the Einstein field equations. It is a statement about the dynamics of the boundary data induced by those equations at large scales. The model may therefore be understood as an extension of GR in which curvature transport on embedded two-dimensional manifolds becomes dynamically relevant, not an alternative to it.

The three elements that are genuinely new — absent from textbook GR — are:

(i) the assumption that curvature transport on S_r is dynamically active rather than a passive boundary condition (postulated),

(ii) the spatial geometry modeled as S^3 of radius R rather than flat space (asserted, with R identified with the Hubble radius),

(iii) the morphology factor α_0 encoding the efficiency of shell-mode excitation (empirical).

Everything else follows from standard GR, the geometry of S^3, and the spectral analysis of its Laplacian.

D5 Correspondence table

Table D1 gives a side-by-side translation of every principal quantity between the GR formulation and the Wave Theory formulation. The two systems are in strict numerical correspondence through the single substitution $GM_b \leftrightarrow a_* A_b$, where $a_* = c^2/(\pi R) = G$. For the full unit translation dictionary and dimensional verification, see Appendix C.

D6 Relation to MOND and modified gravity

The master acceleration formula (68) produces MOND-like behavior in the outer regime without modifying the Einstein field equations. In the limit $a_b \ll a_*$, the shell term dominates and

$$a_{\rm eff} \approx \sqrt{a_b \cdot a_*}, \tag{D6}$$

which is the deep-MOND interpolation formula with $a_0 \equiv a_* = c^2/(\pi R)$.

The standard MOND acceleration scale is $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. The present model identifies this with the cosmological curvature acceleration $c^2/(\pi R)$, which evaluates numerically to $c^2/(\pi R) \approx 6.67 \times 10^{-11}\,\mathrm{m\,s^{-2}}$ using $R = 4.286 \times 10^{26}\,\mathrm{m}$. The agreement is within a factor of two — consistent given that α_0 and m are not yet observationally calibrated. Crucially, a_0 is not a free parameter here: it is set by the Hubble radius.

The MOND-like scaling therefore emerges from the same three ingredients identified in Section 7.2: Gauss-law sourcing, the two-dimensional Green function, and the transition scale $r_* \sim \sqrt{GM_b/a_*}$.

D6.0.1 Distinction from covariant MOND theories.

Relativistic MOND formulations (RMOND, TeVeS, AQUAL) introduce new fields or Lagrangian terms that modify the action. The present model introduces neither. The second acceleration channel arises from the dynamics of boundary data of the standard gravitational field — a projection effect, not a new degree of freedom. Whether this can be embedded in a covariant action principle is an open question, but it is not required for the weak-field predictions derived here.

APPENDIX E: RETARDED-SOURCE DERIVATION OF SHELL CONFINEMENT IN LINEARIZED GENERAL RELATIVITY

This appendix provides a derivation of the spiral winding and shell confinement mechanism using only standard weak-field General Relativity. The purpose is to show that the dynamical transition from volumetric to shell-supported curvature transport follows directly from causal propagation, long-time orbital averaging, and differential rotation of the source, without invoking the WTMedium or a co-moving transport assumption.

E1 Retarded gravitational potential of a time-dependent source

In the linearized weak-field limit, the trace-reversed metric perturbation $\bar{h}_{\mu\nu}$ satisfies

$$\Box \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}. \tag{E1}$$

The linearized weak-field formulation follows standard post-Newtonian theory Poisson & Will (2014).

In the slow-motion regime, the dominant component reduces to a scalar potential Φ satisfying

$$\frac{1}{c^2}\partial_t^2 \Phi - \nabla^2 \Phi = 4\pi G \rho_b. \tag{E2}$$

The solution is given by the retarded integral

$$\Phi(\mathbf{x}, t) = -G \int \frac{\rho_b\left(\mathbf{x}', t - \frac{|\mathbf{x}-\mathbf{x}'|}{c}\right)}{|\mathbf{x}-\mathbf{x}'|}\, d^3x'. \tag{E3}$$

This expression encodes two essential features:

- Finite propagation speed c,
- Dependence on the full causal history of the source.

E2 Differentially rotating disk as a time-dependent source

The baryonic source is modelled as a differentially rotating disk with angular velocity $\Omega(r)$ and surface density $\Sigma(r')$. The three-dimensional mass density is

$$\rho_b(\mathbf{x}', t) = \Sigma(r')\, \delta(z')\, \delta\big(\varphi' - \Omega(r')\, t\big)\,, \tag{E4}$$

where $\delta(z')$ confines the source to the disk midplane and the azimuthal delta function tracks each ring at its instantaneous orbital angle. The dimensions are consistent: $[\Sigma\, \delta(z')\, \delta(\varphi')] = \mathrm{kg\,m^{-3}}$. The Fourier coefficients are $\rho_m(r') = \Sigma(r')\delta(z')/(2\pi)$.

Expanding the retarded potential in angular modes gives

$$\Phi(\mathbf{x}, t) = \sum_{m,\ell} \Phi_{m\ell}(r, t)\, Y_\ell^m(\theta, \varphi). \tag{E5}$$

The coefficients $\Phi_{m\ell}(r,t)$ inherit the time dependence of the rotating source through the retarded argument.

E3 Stationary-phase analysis of the retarded integral

The eikonal limit of the retarded integral is controlled by the stationary-phase conditions on the source phase

$$S_m(\mathbf{x}, \mathbf{x}', t) = m\varphi' - m\Omega(r')\left(t - \frac{|\mathbf{x}-\mathbf{x}'|}{c}\right). \tag{E6}$$

E3.0.1 Stationary-phase condition on r'.

Setting $\partial S_m/\partial r' = 0$ gives

$$\frac{m}{c}\frac{\partial \Omega}{\partial r'}|\mathbf{x}-\mathbf{x}'| + \frac{m\Omega(r')}{c}\frac{\partial|\mathbf{x}-\mathbf{x}'|}{\partial r'} = 0. \tag{E7}$$

In the near-zone regime $|\mathbf{x}-\mathbf{x}'| \ll c/\Omega(r')$, verified numerically in Section E3.0.4 below, the first term is negligible and the condition reduces to $\partial|\mathbf{x}-\mathbf{x}'|/\partial r' = 0$, which for nearly co-rotating source and field points gives

$$r' \approx r. \tag{E8}$$

Concept	GR formulation	Wave Theory formulation
Background geometry	Closed FLRW spatial slice, curvature radius R	S^3 of radius R
Fundamental acceleration scale	$a_* = c^2/(\pi R) \approx G$	Same; $G \equiv a_*$ by definition
Baryonic source	Mass M_b from $T_{\mu\nu}$	Geometric area A_b; $GM_b = a_* A_b$
Local acceleration	$a_b = GM_b(r)/r^2$	$a_b = a_* A_b(r)/r^2$
Shell source	$\sigma_r = -\partial_r \Phi\big\|_{S_r}$ (normal flux)	Same
Enclosed source flux	$Q_b^{\rm src} = 4\pi G M_b$ (Gauss law)	$Q_b^{\rm src} = 4\pi a_* A_b$
Bulk-to-shell projection	Gauss–Codazzi boundary data of $G_{\mu\nu}$ on S_r	Curvature flux through $\partial V_{\rm cap}$
Shell transport equation	$\Delta_{S_r}\Phi_r = -\sigma_r$ (2D Poisson on S_r)	Same
Shell Green function	$\Phi_r \sim Q_b^{\rm src} \log r$	Same
Shell acceleration	$a_{\rm shell} = \alpha_b(\chi)\, Q_b^{\rm src}/r$	Same
Geometric coupling	$L(\chi) = A_\partial(\chi)/V_{\rm cap}(\chi)$	Same
Transition condition	$a_b(r_*) = a_*$; $r_* = \sqrt{GM_b/a_*}$ (SI)	$r_* = \sqrt{A_b}$ (WT, $G = a_* = 1$)
Master acceleration	$a_{\rm eff} = (1-\sigma)\dfrac{GM_b}{r^2} + \sigma\,\alpha_0\sqrt{\dfrac{GM_b}{r^2}\cdot a_*}$	$a_{\rm eff} = a_*\xi\,\dfrac{\xi^3+\alpha_0}{1+\xi^2},\ \ \xi = \sqrt{A_b}/r$
Asymptotic BTFR	$v_\infty^4 \propto GM_b \cdot a_*$	$v_\infty^4 \propto a_*^2 A_b$

Table D1. Side-by-side correspondence between the GR and Wave Theory formulations. All equations above the double rule are standard GR results. The shell transport equation and everything below it constitute the new dynamical content. The two formulations are numerically identical through $GM_b = a_* A_b$; only the dimensional interpretation of G differs. For the complete unit translation see Appendix C.

The dominant contribution to the eikonal field at radius r therefore comes from source material at the same radius: the *same-shell condition.*

E3.0.2 Local wavevector at the stationary point. At $r' \approx r$, the phase of the field is $\theta \approx m\varphi' - m\Omega(r)t + (m\Omega(r)/c)|\mathbf{x}-\mathbf{x}'|_{\rm stat}$. The local frequency and wavevector are

$$\omega = -\partial_t\theta = m\Omega(r), \qquad |\mathbf{k}| = \frac{m\Omega(r)}{c} = \frac{\omega}{c}, \tag{E9}$$

establishing the free-wave dispersion relation $\omega = c|\mathbf{k}|$ from the source geometry alone.

E3.0.3 Doppler-shifted dispersion relation. Since the stationary source material rotates at $\mathbf{v}(\mathbf{x}') \approx \mathbf{v}(\mathbf{x}) = r\Omega(r)\hat{\boldsymbol{\theta}}$, and the azimuthal component of $\mathbf{k}$ is $k_\theta = m/r$, this gives $\mathbf{v}\cdot\mathbf{k} = m\Omega(r) = \omega$. The intrinsic frequency in the co-rotating frame is therefore

$$\omega_{\rm eff} = \omega - \mathbf{v}\cdot\mathbf{k}, \tag{E10}$$

and the dispersion relation (E9) becomes

$$\omega_{\rm eff}^2 = c^2|\mathbf{k}|^2. \tag{E11}$$

This result is *derived* from the retarded source phase: the $\mathbf{v}\cdot\mathbf{k}$ term is the kinematic consequence of the same-shell condition (E8), not an assumption about a co-moving medium.

E3.0.4 Near-zone validity. The approximation $|\mathbf{x}-\mathbf{x}'| \ll c/\Omega$ requires $R_{\rm gal} \ll c/\Omega$. For $\Omega \sim 2\times10^{-16}\,{\rm rad\,s^{-1}}$ at $r \sim 10\,{\rm kpc}$:

$$\frac{c}{\Omega} \sim \frac{3\times10^8}{2\times10^{-16}} \sim 1.5\times10^{24}\,{\rm m} \sim 50\,{\rm Mpc}, \tag{E12}$$

which exceeds $R_{\rm gal} \sim 30\,{\rm kpc}$ by five orders of magnitude. The near-zone approximation is therefore valid throughout the disk.

E4 Orbital averaging of the retarded field

Galaxies evolve over timescales much longer than orbital periods:

$$\frac{T_{\rm gal}}{T_{\rm orb}} \gg 1. \tag{E13}$$

The long-time averaged field is therefore defined as

$$\langle \Phi_{m\ell}(r,t)\rangle = \frac{1}{T_{\rm gal}}\int_0^{T_{\rm gal}} \Phi_{m\ell}(r,t)\,dt. \tag{E14}$$

Orbital averaging of retarded gravitational fields is a standard technique in post-Newtonian theory Peters & Mathews (1963); Blanchet (2014).

E4.1 Suppression of non-axisymmetric modes

For $m \neq 0$, the retarded integral introduces oscillatory factors of the form

$$e^{im\Omega(r')\left(t - \frac{|\mathbf{x}-\mathbf{x}'|}{c}\right)}. \tag{E15}$$

Applying stationary-phase or Riemann–Lebesgue arguments yields

$$\langle \Phi_{m\neq0}\rangle \sim \frac{1}{m\Omega(r')T_{\rm gal}}\,\Phi_{m\ell}^{(0)} \longrightarrow 0. \tag{E16}$$

Thus only the axisymmetric component survives:

$$\Phi(r,\theta) \approx \Phi_{m=0}(r,\theta). \tag{E17}$$

E4.2 Radial coherence suppression

The suppression of non-axisymmetric modes establishes that the long-time averaged field is axisymmetric, $\Phi \approx \Phi(r,\theta)$. The relevant question for dimensional reduction is not azimuthal vs. radial structure (the axisymmetric field has no azimuthal gradient by definition), but rather the radial coherence length of the averaged field.

Contributions to the retarded integral at field point (r,θ,φ) from source points at radius r' carry retarded delays $\Delta t(r') = |r-r'|/c$ relative to on-shell contributions. After $N_{\rm orb} = T_{\rm gal}/T_{\rm orb}$ elapsed orbital periods, the accumulated phase difference between contributions from r' and $r'+\Delta r$ is

$$\Delta\phi_{\rm ret} = m\Omega(r')\,\frac{\Delta r}{c}. \tag{E18}$$

Contributions with $|\Delta\phi_{\rm ret}| \gtrsim 1$ interfere destructively under the time average. Setting $|\Delta\phi_{\rm ret}| = 1$ defines the radial coherence length

$$\ell_{\rm coh} \sim \frac{c}{m\Omega(r)} = \frac{c\,T_{\rm orb}}{2\pi m}. \tag{E19}$$

After $N_{\rm orb}$ orbits, the effective coherence length per orbit is further suppressed:

$$\ell_{\rm coh}^{\rm eff} \sim \frac{c\,T_{\rm orb}}{2\pi\,T_{\rm gal}}\,r = \frac{T_{\rm orb}}{T_{\rm gal}}\,\frac{r}{2\pi}. \tag{E20}$$

Since $T_{\rm orb}/T_{\rm gal} \sim 10^{-2}$, coherent radial structure in the time-averaged field survives only on scales $\Delta r \ll r$. On galactic scales, the averaged field has no coherent radial variation: it is effectively constant in r at fixed θ. Angular (shell) structure at fixed r, by contrast, is organised on galactic scales and is not suppressed.

E4.2.1 Quantitative check. For $\Omega \sim 2\times10^{-16}\,\mathrm{rad\,s^{-1}}$ ($r \sim 10\,\mathrm{kpc}$), $T_{\rm orb} \sim 300\,\mathrm{Myr}$, $T_{\rm gal} \sim 10\,\mathrm{Gyr}$:

$$\ell_{\rm coh}^{\rm eff} \sim \frac{300\,\mathrm{Myr}}{10\,\mathrm{Gyr}} \times \frac{10\,\mathrm{kpc}}{2\pi} \sim 0.05\,\mathrm{kpc}. \tag{E21}$$

Coherent radial structure is therefore confined to scales $\lesssim 50\,\mathrm{pc}$, two orders of magnitude below the galactic radius. The radial dimension contributes no coherent large-scale structure to the averaged field; only the angular (shell) degrees of freedom remain organised at galactic scales.

E4.2.2 Relation to the eikonal argument. The radial coherence suppression established here applies to the time-averaged background field. The eikonal winding argument of Section E.5 applies to individual propagating perturbations and reaches the same conclusion — $k_\theta \gg k_r$ after $t \gg t_{\rm wind}$ — through a complementary mechanism. Both arguments are grounded in the same-shell condition $r' \approx r$ (E8): the averaged field is organised on shells because radial contributions dephase, and individual perturbations are confined to shells because differential rotation winds their wavevectors azimuthally. The shell S_r is the structure selected by both.

E5 Eikonal limit of the linearized field equations

To analyze the propagation of perturbations, a WKB ansatz is applied

$$\Phi(\mathbf{x},t) = A(\mathbf{x},t)\,e^{i\theta(\mathbf{x},t)/\epsilon}, \quad \epsilon \ll 1. \tag{E22}$$

The eikonal (WKB) approximation used here follows standard methods; see Bender & Orszag (1999) for a comprehensive treatment.

Defining

$$\mathbf{k} = \nabla\theta, \qquad \omega = -\partial_t\theta, \tag{E23}$$

this gives a local dispersion relation.

E5.0.1 Validity of the WKB approximation. The eikonal ansatz (E22) requires the wavelength $\lambda = 2\pi/|\mathbf{k}|$ to be short compared to the scale over which the amplitude $A(\mathbf{x},t)$ and the background velocity $\mathbf{v}(\mathbf{x})$ vary:

$$|\mathbf{k}| \gg |\nabla \log A|, \qquad |\mathbf{k}| \gg |\nabla \log \Omega| \sim \frac{1}{r}. \tag{E24}$$

From the same-shell result (E9), $|\mathbf{k}| = m\Omega/c$. The second condition then requires

$$\frac{m\Omega}{c} \gg \frac{1}{r} \quad\Longleftrightarrow\quad m \gg \frac{c}{r\Omega} = \frac{c}{v_{\rm rot}}, \tag{E25}$$

where $v_{\rm rot} = r\Omega$ is the circular velocity. For typical disk galaxies $v_{\rm rot} \sim 200\,\mathrm{km\,s^{-1}}$, giving $c/v_{\rm rot} \sim 1500$. The WKB approximation therefore applies only for $m \gg 1500$, i.e. for very high azimuthal modes.

For the low-m modes that dominate the physical signal, the eikonal approximation is not uniformly valid in the strict short-wavelength sense. The ray-equation result for the wavevector shearing $dk_\theta/dt = k_r A$ should therefore be understood as valid in an asymptotic or order-of-magnitude sense: it correctly captures the qualitative evolution of the dominant wavevector direction under differential rotation, but a fully rigorous treatment would replace the WKB dispersion relation with the full retarded Green function analysis of Section E.3. The winding timescale $t_{\rm wind} \sim T_{\rm orb}/\pi$ and the conclusion $k_\theta \gg k_r$ are robust to this qualification, as they follow from the geometry of differential rotation rather than from the WKB amplitude equation. Crucially, the same-shell condition $r' \approx r$ (equation (E8)) is independent of m entirely: the factor of m cancels in the stationary-phase condition (E7), so the near-zone reduction $r' \approx r$ holds for $m = 1, 2, 5$ and all low-m modes by the same geometric argument as for high m. The WKB gap therefore affects the winding *rate* but not the shell locality of the dominant contribution.

E5.1 Doppler-shifted frequency

The stationary-phase analysis of Section E.3 establishes that the eikonal field at radius r is dominated by source material at $r' \approx r$ (the same-shell condition, equation (E8)), rotating at $\mathbf{v}(\mathbf{x}') \approx \mathbf{v}(\mathbf{x}) = r\Omega(r)\hat{\boldsymbol{\theta}}$. The local frequency and wavevector inherited from the retarded source phase are given by equation (E9):

$$\omega = m\Omega(r), \qquad |\mathbf{k}| = \frac{\omega}{c}. \tag{E26}$$

Since the azimuthal wavevector component is $k_\theta = m/r$, this gives $\mathbf{v}\cdot\mathbf{k} = r\Omega(r)\cdot(m/r) = m\Omega(r) = \omega$, and the intrinsic frequency in the frame co-rotating with the source is

$$\omega_{\rm eff} = \omega - \mathbf{v}\cdot\mathbf{k}. \tag{E27}$$

Substituting $\omega = c|\mathbf{k}|$ gives the dispersion relation

$$\omega_{\rm eff}^2 = c^2|\mathbf{k}|^2. \tag{E28}$$

The $\mathbf{v}\cdot\mathbf{k}$ term is not an assumption about the wave operator or a co-moving medium: it is the kinematic consequence of the same-shell condition derived in Section E.3.

E5.2 Ray equations

The Hamiltonian

$$H(\mathbf{x},\mathbf{k}) = \mathbf{v}\cdot\mathbf{k} + c|\mathbf{k}| \tag{E29}$$

yields

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}(\mathbf{x}) + c\,\hat{\mathbf{k}}, \tag{E30}$$

$$\frac{d\mathbf{k}}{dt} = -(\mathbf{k}\cdot\nabla)\mathbf{v}(\mathbf{x}). \tag{E31}$$

This equation governs the evolution of perturbations and is derived directly from the linearized Einstein system.

E6 Spiral winding under differential rotation

For the velocity field $\mathbf{v} = r\Omega(r)\hat{\theta}$, the ray equation (E31) describes wavevector evolution under differential shear. In the local shearing-sheet approximation (standard in galactic dynamics Binney & Tremaine (2008)), the linearised wavevector evolution for an initially radial perturbation $\mathbf{k}(0) = k_0\hat{r}$ is

$$k_r(t) \approx k_0, \qquad k_\theta(t) \approx 2A_{\rm Oort}\,k_0\,t, \tag{E32}$$

where $A_{\rm Oort} = -\frac{1}{2}r\,d\Omega/dr > 0$ is the Oort shear constant. The pitch angle satisfies

$$\tan\psi(t) = \frac{k_r}{k_\theta} \approx \frac{1}{2A_{\rm Oort}\,t} \longrightarrow 0, \tag{E33}$$

and the radial group velocity

$$\left(\frac{dx}{dt}\right)_r = c\,\frac{k_r}{|\mathbf{k}|} \sim \frac{c}{2A_{\rm Oort}\,t} \longrightarrow 0. \tag{E34}$$

At late times ($2A_{\rm Oort}\,t \gg 1$) the ratio k_θ/k_r continues to grow, so the asymptotic confinement $k_\theta \gg k_r$ is robust beyond the linearised regime.

E6.1 Winding timescale

The pitch angle reaches $\psi \sim 1$ on the timescale

$$t_{\rm wind} \sim \frac{1}{2A_{\rm Oort}} \sim \frac{T_{\rm orb}}{\pi}, \tag{E35}$$

where the last step uses $A_{\rm Oort} \sim \Omega/2$ for a broadly declining rotation profile and $T_{\rm orb} = 2\pi/\Omega$. Since $t_{\rm wind} \ll T_{\rm gal}$, spiral winding is dynamically inevitable.

E7 Emergence of shell-supported transport

Both arguments are grounded in the same-shell condition (E8). Orbital averaging suppresses non-axisymmetric modes on timescale $T_{\rm gal}$, leaving a field organised on constant-r surfaces. Eikonal winding shears radial wavevectors into azimuthal ones on timescale $t_{\rm wind} \sim T_{\rm orb}/\pi \ll T_{\rm gal}$, suppressing radial group velocity within the same surface. The two results are complementary views of the same dynamics — one for the time-averaged background, one for individual perturbations — both valid because $r' \approx r$.

Together they give $k_\theta \gg k_r$ for $t \gg t_{\rm wind}$, and suppressed radial coherence in the averaged field. Coherent curvature transport is therefore confined to constant-radius shells S_r: not by assumption, but as the dynamical attractor of causal propagation, orbital averaging, and differential rotation in standard linearized General Relativity.

E8 Connection to dimensional reduction

The winding result $k_\theta \gg k_r$ is a statement about the physical wavevector components at radius r. To connect this to the spectral condition on S^3, the correspondence is identified between physical wavenumbers and the quantum numbers (ℓ, n) of the S^3 harmonics.

On the S^3 of curvature radius R, the harmonics $Y^{(n)}_{\ell m}(\chi,\theta,\varphi)$ have angular wavenumber ℓ on the constant-χ shells and radial wavenumber n in the χ direction. In the physical galactic regime $r \ll R$ (so $\chi = r/R \ll 1$), the correspondence between physical wavevector components and spectral quantum numbers is

$$k_\theta \sim \frac{\ell}{r}, \qquad k_r \sim \frac{n}{R}. \tag{E36}$$

The condition $k_\theta \gg k_r$ then gives

$$\frac{\ell}{r} \gg \frac{n}{R} \quad\Longrightarrow\quad \ell \gg n\,\frac{r}{R}. \tag{E37}$$

Since $r/R \ll 1$, this is satisfied for all $\ell \gtrsim 1$ at galactic radii, confirming that the winding regime is compatible with the angular-dominance condition. More precisely, the spectral dominance condition of Appendix C requires

$$\frac{\ell(\ell+1)}{\sin^2\chi} \gg n(n+2), \tag{E38}$$

which in the small-χ limit ($\sin\chi \approx \chi = r/R$) reduces to

$$\ell(\ell+1) \gg n(n+2)\,\frac{r^2}{R^2}, \tag{E39}$$

consistent with (E37) for $\ell \sim k_\theta r$ and $n \sim k_r R$. The Laplace–Beltrami operator on S^3 therefore reduces to its angular component,

$$\Delta_{S^3}\Phi \approx \frac{\ell(\ell+1)}{\sin^2\chi}\,\Phi, \tag{E40}$$

and the dimensional reduction to a two-dimensional Poisson equation on S_r follows as in Section 4.5, yielding the logarithmic Green function and the $1/r$ shell acceleration.

E9 Conclusion

The derivation presented in this appendix establishes the spiral winding and shell confinement mechanism within standard linearized General Relativity, without invoking the WTMedium, a co-moving transport assumption, or any parameter beyond the observed rotation profile $\Omega(r)$ and the galaxy age $T_{\rm gal}$.

The following steps are *Derived* directly from the linearized Einstein equations and standard asymptotic methods:

- The retarded potential formulation (E3) and its reduction to the scalar wave equation (E2) in the slow-motion regime.
- The suppression of non-axisymmetric modes (E16) under long-time orbital averaging, via the Riemann–Lebesgue lemma Blanchet & Damour (1992).

• The same-shell condition $r' \approx r$ (E8), established by the stationary-phase analysis of the retarded integral, valid throughout the disk by five orders of magnitude (near-zone check, Section E3.0.4).

• The local dispersion relation $\omega = m\Omega(r)$, $|\mathbf{k}| = \omega/c$ (E9), and the Doppler-shifted form $\omega_{\mathrm{eff}}^2 = c^2|\mathbf{k}|^2$ (E28), both following from the same-shell condition (E8) without any comoving medium assumption.

• The ray equations (E31) from the Hamiltonian $H = \mathbf{v}\cdot\mathbf{k} + c|\mathbf{k}|$.

• The linearised winding solution $k_r \approx k_0$, $k_\theta \approx 2A_{\mathrm{Oort}}\, k_0\, t$ (E32), the decay of the pitch angle $k_r/k_\theta \to 0$, and the suppression of the radial group velocity $c\, k_r/|\mathbf{k}| \to 0$, on timescale $t_{\mathrm{wind}} \sim 1/(2A_{\mathrm{Oort}}) \sim T_{\mathrm{orb}}/\pi \ll T_{\mathrm{gal}}$.

• The radial coherence length estimate (E18)–(E20): coherent radial structure in the time-averaged field is confined to scales $\ell_{\mathrm{coh}}^{\mathrm{eff}} \sim (T_{\mathrm{orb}}/T_{\mathrm{gal}})\, r/2\pi \sim 50\,\mathrm{pc}$ at $r = 10\,\mathrm{kpc}$, two orders of magnitude below the galactic radius.

The following step is *Argued*:

• The extrapolation from the coherence-length estimate (E20) to complete radial incoherence on galactic scales. The suppression factor $T_{\mathrm{orb}}/T_{\mathrm{gal}} \sim 10^{-2}$ is controlled and quantified, but explicit error bounds have not been tracked through the full retarded Green function of the screened equation. Establishing these would promote this step to Derived.

The two tracks of the argument — orbital averaging of the background field and eikonal winding of individual perturbations — are united by the same-shell condition (E8). Both operate on the same constant-r configuration: averaging suppresses radial coherence on timescale T_{gal}, while winding suppresses radial group velocity on timescale t_{wind}. The shell S_r is the structure selected by both.

Given the above, the confinement of coherent curvature transport to constant-radius shells follows as the dynamical attractor of the system, and the subsequent dimensional reduction to a two-dimensional Poisson equation on S_r proceeds exactly as in Section 4.5, yielding the logarithmic Green function and the $1/r$ shell acceleration. The shell-supported transport channel is therefore not an additional assumption but a direct consequence of causal propagation, long-time orbital averaging, and differential rotation acting together in standard linearized General Relativity.

APPENDIX F: PER-GALAXY FIT RESULTS

Table G1 lists the per-galaxy optimised fit results for all 175 SPARC galaxies, sorted by A_{morph} within each calibration bin. For each galaxy the best-fit α_0 was found by grid search over $\alpha_0 \in [0.05, 3.0]$ in steps of 0.001. Columns are: galaxy name; calibration bin (G1–G4, Table 3); best-fit α_0 and the derived $A_{\mathrm{morph}} = \alpha_0/\kappa$; reduced χ^2 at best-fit (χ^2_{best}) and at universal κ (χ^2_κ); reduced χ^2 for MOND (fitted $\Upsilon_\star$) and ΛCDM (fitted M_{200}); and the win assignment (WT / MOND / ΛCDM) from the one-parameter-per-galaxy comparison. The bins G1–G4 are an operational partition of a statistically unimodal A_{morph} distribution (Section 6.3); they are used here for sorting and readability only. The 54 galaxies where WT achieves the lowest χ^2, 83 for MOND, 38 for ΛCDM, and median $\chi^2_{\mathrm{best}} = 2.82$ quoted in Section 6.3 are directly verifiable from this table. The per-galaxy fitting procedure described here is implemented in a dedicated script available as open-source code Banasik (2025d).

APPENDIX G: A_{MORPH} STRUCTURAL CORRELATIONS

Figure G1 shows Spearman scatter plots between A_{morph} and ten structural or dynamical observables for all 175 SPARC galaxies. Structural parameters (disk scale length R_d, central and effective surface brightness, effective radius, Hubble type T) are from Lelli, McGaugh & Schombert Lelli et al. (2016); bulge-to-disk ratios are derived from the non-parametric bulge luminosities of the same catalogue; gas fractions, $\log_{10} M_{200}$, and $\log_{10} a_{\mathrm{char}}$ are from the per-galaxy fit results (Appendix F). The dashed horizontal line in each panel marks $\kappa = 1 + \pi^{-2} \approx 1.101$, the zero-free-parameter prediction; the scatter of A_{morph} around this line is approximately symmetric, confirming that κ sits near the centre of the distribution rather than at an extreme.

Table G1: Per-galaxy fit results for all 175 SPARC galaxies, sorted by A_{morph} within calibration bin. χ^2 values are reduced (per data point). χ^2_{best}: WT with fitted α_0. χ^2_{κ}: WT at zero parameters ($\alpha_0 = \kappa$, no per-galaxy adjustment). χ^2_{MOND}: MOND with fitted $\Upsilon_\star$. $\chi^2_{\Lambda\mathrm{CDM}}$: ΛCDM with fitted M_{200}. Win column (WT / MOND / ΛCDM): ✓ = lowest χ^2 in one-parameter comparison. χ^2_{κ} may be compared directly against the fitted competitor columns as a zero-parameter benchmark.

Galaxy	**Cl.**	α_0	A_{morph}	χ^2_{best}	χ^2_{κ}	χ^2_{MOND}	$\chi^2_{\Lambda\mathrm{CDM}}$	**Win**
G1 — Low-coupling ($A_{\mathrm{morph}} < 0.50$)								
NGC3949	G1	0.050	0.045	12.9	18.6	0.299	28.3	– ✓ –
NGC4389	G1	0.050	0.045	43.2	65.3	3.02	80.5	– ✓ –
NGC5005	G1	0.050	0.045	10.7	11.5	0.058	14.0	– ✓ –
UGC02455	G1	0.050	0.045	37.4	94.8	21.9	96.4	– ✓ –
UGC06973	G1	0.050	0.045	333	350	12.3	391	– ✓ –
NGC4085	G1	0.072	0.065	14.3	21.6	3.59	31.7	– ✓ –
CamB	G1	0.207	0.188	3.73	87.3	51.4	14.9	✓ – –
F574-2	G1	0.207	0.188	0.087	18.5	12.4	0.824	✓ – –
PGC51017	G1	0.230	0.209	1.88	96.1	80.0	5.42	✓ – –
UGC07577	G1	0.246	0.223	0.215	14.8	8.32	0.977	✓ – –
F563-V1	G1	0.254	0.231	0.275	27.0	20.4	0.197	– – ✓
NGC4051	G1	0.270	0.245	2.08	9.36	1.73	15.1	– ✓ –
UGC06628	G1	0.308	0.280	0.317	10.9	6.13	1.06	✓ – –
UGC04305	G1	0.326	0.296	1.65	41.8	39.0	2.85	✓ – –
UGC11557	G1	0.369	0.335	2.04	11.7	1.45	6.85	– ✓ –
F561-1	G1	0.386	0.350	0.618	27.0	13.3	0.667	✓ – –
NGC3877	G1	0.395	0.359	7.63	13.5	6.79	21.6	– ✓ –
NGC4088	G1	0.466	0.423	26.7	38.8	0.978	51.8	– ✓ –
UGC08837	G1	0.538	0.489	4.72	27.0	9.91	6.40	✓ – –
NGC5371	G1	0.547	0.497	29.3	103	51.9	127	✓ – –
G2 — Near-κ ($0.50 \leq A_{\mathrm{morph}} < 1.30$)								
UGC07125	G2	0.576	0.523	1.32	76.7	73.0	1.89	✓ – –
NGC2955	G2	0.579	0.526	38.1	40.4	8.93	57.9	– ✓ –
UGC09992	G2	0.583	0.529	0.375	4.09	3.14	0.092	– – ✓
F567-2	G2	0.622	0.565	0.363	5.31	1.82	0.249	– – ✓
D564-8	G2	0.629	0.571	1.54	13.8	6.10	2.18	✓ – –
IC2574	G2	0.634	0.576	31.3	192	25.6	72.8	– ✓ –
UGC07559	G2	0.639	0.580	0.470	5.67	3.25	0.806	✓ – –
KK98-251	G2	0.664	0.603	1.27	12.1	20.5	2.58	✓ – –
NGC4068	G2	0.667	0.606	1.68	11.3	4.19	2.15	✓ – –
UGC02023	G2	0.721	0.655	0.520	1.39	0.224	0.739	– ✓ –

Continued on next page

Table G1 continued

Galaxy	Cl.	α_0	A_{morph}	χ^2_{best}	χ^2_{κ}	χ^2_{MOND}	$\chi^2_{\Lambda\mathrm{CDM}}$	Win
UGC07866	G2	0.731	0.664	0.025	1.87	0.942	0.057	✓ – –
UGC09037	G2	0.756	0.686	21.5	28.5	1.49	37.2	– ✓ –
NGC6195	G2	0.758	0.688	31.3	35.3	16.6	59.1	– ✓ –
NGC5055	G2	0.770	0.699	> 999	> 999	46.0	> 999	– ✓ –
NGC3953	G2	0.776	0.705	0.911	2.06	2.47	8.72	✓ – –
UGC07089	G2	0.794	0.721	0.497	4.03	0.134	1.57	– ✓ –
NGC0891	G2	0.797	0.724	148	153	7.11	206	– ✓ –
NGC6946	G2	0.803	0.729	39.4	46.4	2.73	87.0	– ✓ –
UGC05750	G2	0.809	0.735	0.535	1.92	0.958	0.971	✓ – –
NGC3726	G2	0.810	0.735	6.22	11.1	2.97	21.6	– ✓ –
NGC0801	G2	0.830	0.754	21.8	42.0	35.0	50.5	✓ – –
D512-2	G2	0.831	0.755	0.096	1.89	0.119	0.098	✓ – –
DDO161	G2	0.841	0.764	2.30	33.9	110	4.94	✓ – –
UGC02916	G2	0.867	0.787	176	178	12.7	243	– ✓ –
UGC06818	G2	0.868	0.788	4.06	5.48	2.30	5.84	– ✓ –
DDO170	G2	0.870	0.790	3.48	37.3	94.9	1.77	– – ✓
UGC05005	G2	0.895	0.813	0.358	0.969	0.494	0.895	✓ – –
UGC04483	G2	0.896	0.814	0.448	2.15	5.82	0.488	✓ – –
NGC4217	G2	0.937	0.851	69.0	69.4	7.84	84.2	– ✓ –
NGC0055	G2	0.942	0.855	3.70	8.20	10.0	6.71	✓ – –
NGC2366	G2	0.955	0.867	4.27	6.54	8.89	5.30	✓ – –
D631-7	G2	0.961	0.873	10.8	12.6	16.6	10.5	– – ✓
NGC2976	G2	0.969	0.880	1.08	1.23	1.20	1.30	✓ – –
NGC4157	G2	0.991	0.900	9.66	10.1	0.584	21.3	– ✓ –
F583-4	G2	0.994	0.903	0.112	0.457	0.106	0.141	– ✓ –
UGC05414	G2	1.001	0.909	1.32	2.02	0.054	2.23	– ✓ –
NGC4559	G2	1.015	0.922	1.03	1.63	1.56	5.15	✓ – –
NGC1090	G2	1.025	0.931	4.18	6.25	7.11	13.6	✓ – –
UGC07323	G2	1.027	0.933	0.947	1.15	0.271	1.63	– ✓ –
UGC05829	G2	1.036	0.941	0.054	0.139	2.21	0.168	✓ – –
UGC11820	G2	1.043	0.947	1.68	4.89	88.7	0.641	– – ✓
UGCA281	G2	1.062	0.964	0.251	0.306	0.174	0.305	– ✓ –
UGC05918	G2	1.063	0.965	0.384	0.410	1.01	0.194	– – ✓
NGC0289	G2	1.066	0.968	4.10	4.24	7.60	3.14	– – ✓
UGCA444	G2	1.070	0.972	0.112	0.122	1.80	0.091	– – ✓
UGC06614	G2	1.076	0.977	2.28	2.30	0.745	10.2	– ✓ –
UGC04499	G2	1.085	0.985	1.10	1.13	2.83	1.29	✓ – –
NGC3769	G2	1.091	0.991	3.24	3.24	0.695	6.00	– ✓ –

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Table G1 continued

Galaxy	**Cl.**	α_0	A_{morph}	χ^2_{best}	χ^2_{κ}	χ^2_{MOND}	$\chi^2_{\Lambda\mathrm{CDM}}$	**Win**
UGC01230	G2	1.096	0.995	2.82	2.82	3.67	1.89	– – ✓
UGC11455	G2	1.101	1.000	21.6	21.6	4.51	39.5	– ✓ –
NGC7331	G2	1.102	1.001	114	114	1.32	162	– ✓ –
UGC07524	G2	1.105	1.003	1.41	1.41	3.52	0.920	– – ✓
UGC05999	G2	1.106	1.004	1.91	1.91	2.96	2.22	✓ – –
UGC09133	G2	1.108	1.006	74.6	74.6	50.1	86.0	– ✓ –
UGC12632	G2	1.109	1.007	2.09	2.10	6.33	1.17	– – ✓
NGC2683	G2	1.114	1.012	3.43	3.43	2.17	6.60	– ✓ –
UGC03580	G2	1.127	1.023	72.6	72.8	7.30	129	– ✓ –
NGC1003	G2	1.136	1.031	3.24	3.64	4.56	3.85	✓ – –
F568-3	G2	1.137	1.032	2.79	2.82	2.54	3.57	– ✓ –
UGC10310	G2	1.146	1.041	0.816	0.868	2.09	0.516	– – ✓
NGC4183	G2	1.158	1.051	2.08	2.24	2.57	0.593	– – ✓
F571-V1	G2	1.159	1.052	0.177	0.253	0.228	0.235	✓ – –
DDO168	G2	1.165	1.058	9.86	10.4	18.6	10.7	✓ – –
NGC3198	G2	1.165	1.058	9.14	10.8	9.68	21.1	✓ – –
UGC07690	G2	1.167	1.060	1.10	1.18	1.63	0.417	– – ✓
NGC5585	G2	1.170	1.062	16.5	17.4	7.10	37.2	– ✓ –
NGC4013	G2	1.174	1.066	6.43	6.93	1.79	26.9	– ✓ –
UGC00891	G2	1.178	1.070	8.29	14.2	8.81	5.64	– – ✓
UGC01281	G2	1.179	1.071	1.20	1.28	0.832	1.51	– ✓ –
UGC06930	G2	1.183	1.074	1.08	1.34	1.77	0.467	– – ✓
NGC0100	G2	1.193	1.083	1.25	1.50	1.37	2.14	✓ – –
UGC00191	G2	1.209	1.098	7.51	13.7	21.2	4.59	– – ✓
NGC5907	G2	1.224	1.111	11.2	20.6	25.5	13.1	✓ – –
DDO064	G2	1.238	1.124	0.423	0.566	0.201	0.536	– ✓ –
UGC00128	G2	1.247	1.132	58.6	117	93.6	37.3	– – ✓
UGC06923	G2	1.248	1.133	0.965	1.42	0.638	1.68	– ✓ –
NGC7793	G2	1.249	1.134	1.03	1.16	0.813	1.87	– ✓ –
DDO154	G2	1.250	1.135	15.2	77.1	29.7	13.9	– – ✓
NGC2998	G2	1.252	1.137	4.31	12.5	6.81	4.86	✓ – –
NGC4010	G2	1.263	1.147	2.73	3.59	1.94	4.93	– ✓ –
UGC12732	G2	1.267	1.150	0.968	3.12	3.65	0.492	– – ✓
UGC07151	G2	1.281	1.163	2.37	4.59	2.98	1.60	– – ✓
NGC6503	G2	1.283	1.165	13.3	40.4	2.68	22.7	– ✓ –
UGC07261	G2	1.299	1.179	0.556	1.25	0.662	0.373	– – ✓
NGC0247	G2	1.319	1.198	3.06	11.2	4.00	1.79	– – ✓
UGC05716	G2	1.320	1.199	15.8	67.4	31.3	18.0	✓ – –

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Table G1 continued

Galaxy	**Cl.**	α_0	$A_{\rm morph}$	$\chi^2_{\rm best}$	χ^2_κ	$\chi^2_{\rm MOND}$	$\chi^2_{\Lambda\rm CDM}$	**Win**
NGC4138	G2	1.331	1.209	1.87	2.23	0.993	2.39	– ✓ –
NGC2903	G2	1.367	1.241	127	141	12.6	213	– ✓ –
NGC3521	G2	1.384	1.257	22.7	23.0	0.505	29.5	– ✓ –
UGC00731	G2	1.388	1.260	9.95	15.9	12.2	3.51	– – ✓
UGC04278	G2	1.394	1.266	2.22	3.97	2.55	2.36	✓ – –
UGCA442	G2	1.395	1.267	2.14	18.2	4.42	1.51	– – ✓
NGC3109	G2	1.399	1.270	7.05	11.5	5.31	11.5	– ✓ –
UGC08550	G2	1.414	1.284	1.12	8.58	1.35	1.73	✓ – –
F574-1	G2	1.415	1.285	1.99	6.20	1.97	1.74	– – ✓
NGC6674	G2	1.415	1.285	25.3	116	12.6	3.96	– – ✓
F583-1	G2	1.426	1.295	1.77	3.39	1.99	1.46	– – ✓
G3 — Enhanced ($1.30 \leq A_{\rm morph} < 2.00$)								
NGC3741	G3	1.432	1.300	1.68	5.64	1.14	0.514	– – ✓
NGC3917	G3	1.434	1.302	3.22	10.6	2.83	2.50	– – ✓
UGC03546	G3	1.440	1.308	17.9	61.3	1.05	35.7	– ✓ –
UGC00634	G3	1.444	1.311	1.17	69.6	0.680	3.09	– ✓ –
UGC06399	G3	1.448	1.315	0.358	3.82	0.292	0.407	– ✓ –
NGC0300	G3	1.461	1.327	0.592	8.14	1.32	0.718	✓ – –
UGC06917	G3	1.481	1.345	0.877	8.47	0.926	1.06	✓ – –
NGC3893	G3	1.495	1.357	9.10	11.0	0.807	14.9	– ✓ –
UGC02885	G3	1.499	1.361	1.08	9.22	3.90	2.97	✓ – –
NGC4100	G3	1.508	1.369	2.98	9.78	1.84	3.02	– ✓ –
NGC6015	G3	1.548	1.406	15.4	36.8	13.3	8.90	– – ✓
NGC5033	G3	1.560	1.416	36.4	109	8.46	21.1	– ✓ –
NGC3992	G3	1.565	1.421	5.84	24.2	3.20	0.665	– – ✓
UGC07232	G3	1.586	1.440	1.70	3.25	1.61	2.06	– ✓ –
UGC06983	G3	1.591	1.445	1.64	10.7	1.10	1.52	– ✓ –
UGC03205	G3	1.595	1.448	5.77	51.5	8.64	5.15	– – ✓
F579-V1	G3	1.627	1.477	2.89	5.08	2.27	2.40	– ✓ –
F565-V2	G3	1.639	1.488	0.279	3.60	0.315	0.330	✓ – –
UGC05253	G3	1.666	1.513	86.4	114	5.09	133	– ✓ –
NGC3972	G3	1.671	1.517	1.10	9.44	1.25	1.74	✓ – –
ESO079-G014	G3	1.675	1.521	3.04	44.1	3.01	3.55	– ✓ –
UGC06446	G3	1.675	1.521	1.61	10.8	0.839	2.03	– ✓ –
IC4202	G3	1.683	1.528	46.8	54.5	31.7	55.3	– ✓ –
NGC2403	G3	1.707	1.550	26.9	535	16.3	39.6	– ✓ –
UGC08490	G3	1.707	1.550	2.53	16.6	0.235	4.70	– ✓ –

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Table G1 continued

Galaxy	**Cl.**	α_0	$A_{\rm morph}$	$\chi^2_{\rm best}$	χ^2_κ	$\chi^2_{\rm MOND}$	$\chi^2_{\Lambda\rm CDM}$	**Win**
UGC07603	G3	1.740	1.580	1.22	14.1	2.82	1.60	✓ – –
UGC08286	G3	1.769	1.606	6.13	58.4	2.35	9.36	– ✓ –
ESO116-G012	G3	1.780	1.616	1.87	44.5	4.31	3.09	✓ – –
UGC07608	G3	1.797	1.632	0.371	3.13	0.450	0.390	✓ – –
F563-1	G3	1.812	1.645	1.15	4.39	0.962	1.31	– ✓ –
UGC02259	G3	1.819	1.652	11.0	95.4	5.46	13.6	– ✓ –
F571-8	G3	1.831	1.663	37.8	55.0	40.5	51.4	✓ – –
UGC12506	G3	1.846	1.676	6.01	15.9	2.53	3.22	– ✓ –
UGC08699	G3	1.855	1.684	8.83	27.2	0.882	15.3	– ✓ –
UGC11914	G3	1.886	1.712	9.70	11.4	6.11	14.9	– ✓ –
UGC02953	G3	1.979	1.797	55.4	344	6.34	107	– ✓ –
NGC0024	G3	2.020	1.834	3.81	16.8	0.643	2.97	– ✓ –
UGC02487	G3	2.048	1.860	67.9	565	6.93	25.9	– ✓ –
UGC05986	G3	2.049	1.860	3.48	67.2	5.71	4.46	✓ – –
NGC2841	G3	2.144	1.947	18.2	158	1.88	5.90	– ✓ –
NGC4214	G3	2.169	1.969	0.765	9.34	1.19	0.796	✓ – –
UGC04325	G3	2.175	1.975	8.61	56.7	5.87	8.58	– ✓ –
UGC06667	G3	2.179	1.979	3.35	41.5	3.78	1.70	– – ✓
NGC2915	G3	2.202	1.999	1.13	10.7	2.88	2.27	✓ – –
G4 — High-coupling ($A_{\rm morph} \geq 2.00$)								
UGC05764	G4	2.236	2.030	28.2	481	47.7	29.5	✓ – –
ESO444-G084	G4	2.309	2.097	1.21	63.0	10.5	0.913	– – ✓
UGC05721	G4	2.421	2.198	2.75	22.4	2.55	5.15	– ✓ –
NGC7814	G4	2.437	2.213	7.24	122	4.53	9.05	– ✓ –
NGC1705	G4	2.448	2.223	1.23	23.5	0.264	3.25	– ✓ –
UGC06786	G4	2.452	2.226	9.54	328	19.0	8.92	– – ✓
ESO563-G021	G4	2.456	2.230	17.9	54.8	16.2	16.7	– ✓ –
F568-1	G4	2.461	2.235	0.776	10.4	0.729	0.754	– ✓ –
F563-V2	G4	2.544	2.310	0.685	6.38	0.322	0.910	– ✓ –
F568-V1	G4	2.545	2.311	1.01	11.6	0.276	1.71	– ✓ –
UGC06787	G4	2.613	2.373	61.0	435	50.6	50.9	– ✓ –
NGC5985	G4	2.829	2.569	45.3	238	20.6	27.7	– ✓ –
NGC6789	G4	3.000	2.724	1.25	9.70	1.20	1.13	– – ✓
UGC07399	G4	3.000	2.724	2.28	114	5.37	4.74	✓ – –
Summary: 175 galaxies, median $\chi^2_{\rm best} = 2.82$, *WT 54/175, MOND 83/175, ΛCDM 38/175 (one-param).*								

[t]

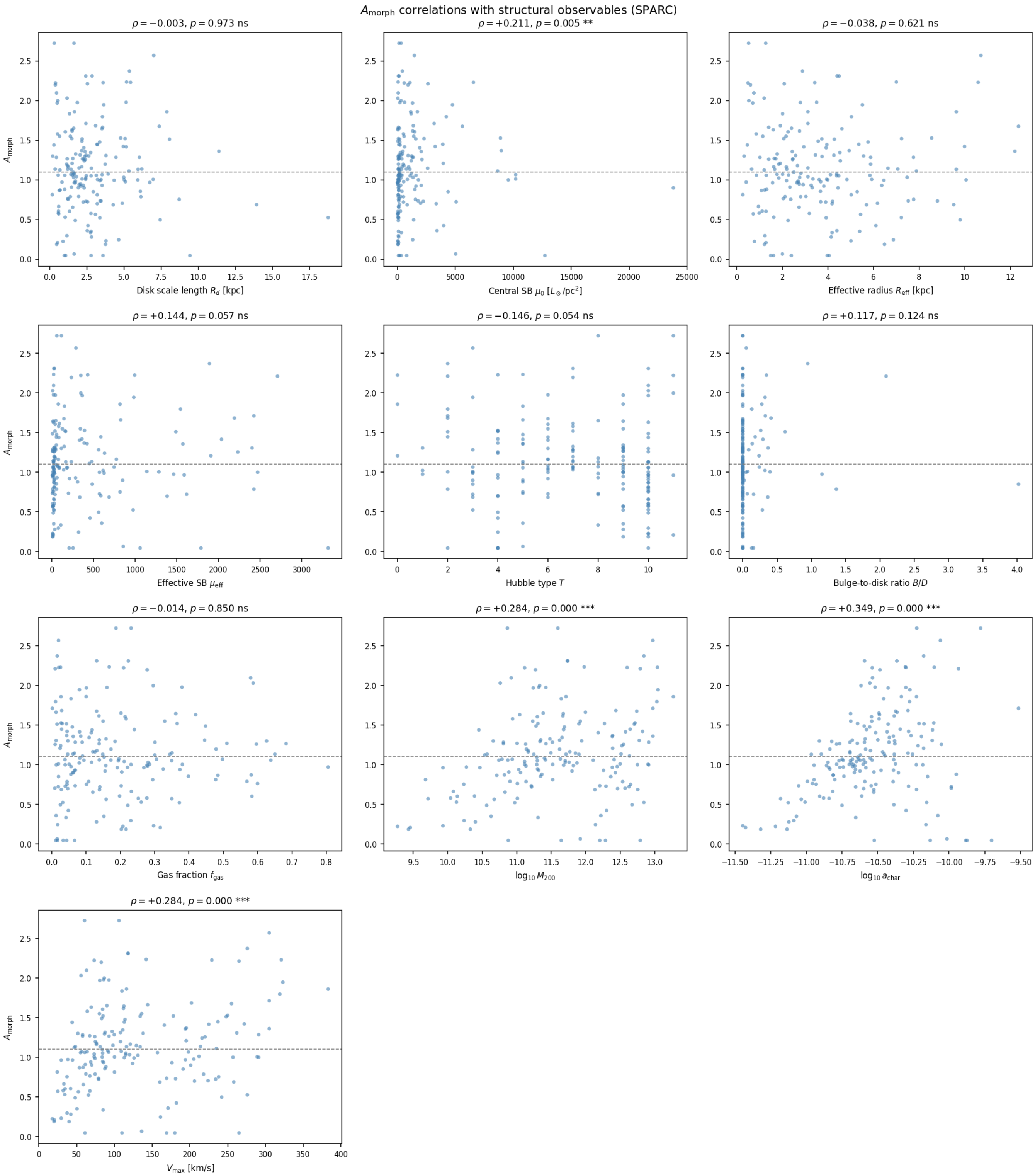

Figure G1. Spearman scatter plots of $A_{\rm morph}$ against ten structural and dynamical observables for all 175 SPARC galaxies. Each panel annotates the Spearman ρ and p-value. Significant correlations ($p < 0.05$): central disk surface brightness μ_0 ($\rho = +0.21$, $p = 0.005$), $\log_{10} M_{200}$ ($\rho = +0.28$, $p < 0.001$), $\log_{10} a_{\rm char}$ ($\rho = +0.35$, $p < 0.001$), and $V_{\rm max}$ ($\rho = +0.28$, $p < 0.001$). Non-significant ($p > 0.05$): disk scale length R_d, effective radius $R_{\rm eff}$, effective surface brightness, bulge-to-disk ratio B/D, Hubble type T, and gas fraction $f_{\rm gas}$. The dashed line marks the universal κ prediction. The pattern confirms that $A_{\rm morph}$ tracks the central baryonic concentration rather than the total extent, morphological class, or gas content of the galaxy.

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